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二維材料於純量與向量渦旋光激發下之激子躍遷理論研究
Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

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二維材料於純量與向量渦旋光激發下之激子躍遷理論研究

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摘　要

二維過渡金屬硫族化物（transition-metal dichalcogenides, TMDs）因其直接能隙與強烈激子效應，成為具潛力的光電應用材料。透過電子-電洞交換能，使此材料的亮激子態產生混合，形成激子於質心動量空間中，躍遷偶極矩具有不同的指向。因此，利用具有結構的光場激發此類激子成為重要研究方向。由兩個 Laguerre-Gaussian 光束疊加而成的向量渦旋光束（vector vortex beams, VVBs），透過光場中的自旋-軌道交互作用（spin-orbit interaction），VVB 的縱向分量會形成特徵性的方向角幾何結構，並與激子在 COM 動量空間中的躍遷速率分布具有對應關係 [1]。基於此關聯，本論文研究更廣義的 VVB 組態及其與激子的交互作用。

對於具有相同軌道角動量（orbital angular momentum, OAM）大小的 VVB，其總角動量（total angular momentum, TAM）差與 OAM 差必然為偶數，導致激子躍遷速率分布保持對稱。透過拓展至具有不同 OAM 大小的 VVB，本論文展示 TAM 差與 OAM 差分別決定縱向與橫向分量在方向角地幾何結構，並分別表現在光強度與偏振分布。當組成光束間維持足夠的空間重疊時，此決定機制也適用在擁有不同 OAM 大小地 VVB。特別在奇數 TAM 差或 OAM 差時，可實現有限動量激子的方向選擇性激發。

最後，本論文建立保留激子內部波函數的激子波包。我們發現，對於谷混合激子（valley-mixed excitons），來自激子內部結構的相位會轉化為不同谷分量間的相對相位，並在決定激子波包結構中扮演關鍵角色，因此仍需進一步數值計算以完整描述其波包結構。

關鍵字：渦漩光、角動量、自旋-軌道交互作用、亮與灰激子、躍遷速率、波包、二維過度金屬二流族化合物

Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

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Abstract

Two-dimensional transition-metal dichalcogenides (TMDs) are promising materials for optoelectronic applications due to their direct band gaps and strong excitonic effects. Through the electron–hole exchange interaction, the valley-mixed excitonic branches possess different transition dipole orientations in the center-of-mass (COM) momentum space. Therefore, utilizing structured optical fields to excite these excitonic states has become an important research direction. Vector vortex beams (VVBs), formed by the superposition of two Laguerre–Gaussian beams, provide a versatile platform for tailoring structured light. Through spin–orbit interaction in the optical field, the longitudinal component of VVBs exhibits characteristic angular geometry, which corresponds to exciton transition-rate distributions in COM momentum space [1]. Motivated by this connection, this thesis investigates generalized VVB configurations and their interaction with excitons.

For VVBs composed of modes with equal orbital angular momentum (OAM) magnitudes, the total angular momentum (TAM) difference and OAM difference are necessarily even integers, resulting in symmetric exciton transition-rate distributions. By extending to VVBs with unequal OAM magnitudes, we demonstrate that the TAM difference and OAM difference independently determine the angular geometry of the longitudinal and transverse components, respectively, manifested as intensity and polarization distributions. This governing mechanism remains valid for VVBs with unequal OAM magnitudes as long as sufficient spatial overlap between the constituent beams is maintained. In particular, odd TAM or OAM differences enable directional-selective excitation of finite-momentum excitons.

Finally, we construct exciton wave packets while retaining the internal exciton wave function. We find that, for valley-mixed excitons, the phase contribution originating from the internal exciton structure becomes a relative phase between different valley components and plays a crucial role in determining the exciton wave-packet structure. Therefore, further numerical calculations are required for a complete description of the wave-packet structure.

Keywords: vortex beams, angular momentum, spin-orbit interaction, bright and gray exciton, transition rate, wave packet, two dimensional transition metal dichalcogenides

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Chapter 1 Introduction

1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer

Conventional semiconductors such as diamond, silicon [2], and aluminum arsenide are not suitable for the selective initialization, manipulation, and readout of valley states. This is because they possess multiple inequivalent valleys in their band structures that do not couple distinctly to external optical fields. The situation changed with the discovery of graphene in 2004 [3], which triggered rapid progress in research on two-dimensional (2D) materials. Graphene has a hexagonal honeycomb lattice that places two inequivalent valleys at the K and K' points of the hexagonal Brillouin zone. Nevertheless, the absence of a bandgap in graphene allows electrons to be thermally excited between valleys, making valley polarization unstable at room temperature.

In contrast, transition-metal dichalcogenide (TMD) monolayers share a similar honeycomb lattice but possess a finite direct bandgap (see Figure 1.1) on the order of $\sim$ 1–2 eV, which lies in the visible to near-infrared spectral range. A TMD monolayer consists of two group-VIA chalcogen atoms, such as sulfur (S) or selenium (Se), with a group-VIB transition-metal atom, such as molybdenum (Mo) or tungsten (W), located between them.

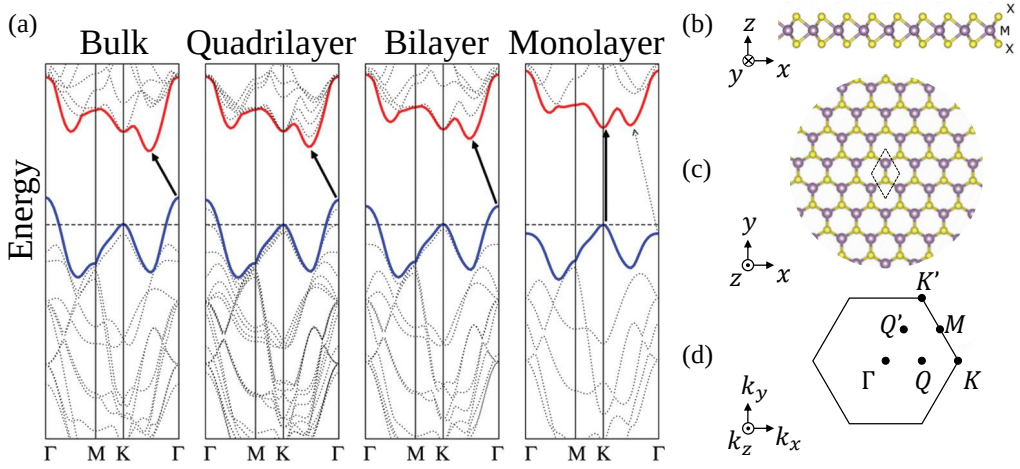


Figure 1.1 : (a) Calculated band structures of MoS_2 with different layers, reproduced from Ref. [4]. (b) Side view and (c) top view of transition-metal dichalcogenide (TMD) monolayer with the unit cell framed by the dashed line, reproduced from Ref. [5]. (d) The first Brillouin zone of TMD monolayer in the recipical space and the selected high symetry points

Under optical excitation, electrons in a TMD monolayer can be promoted from the valence

band to the conduction band near the K and K' valleys, leaving holes in the valence band. Due to the strong Coulomb interaction in atomically thin materials with weak dielectric screening, the electron–hole pair can bind together to form a quasiparticle known as an exciton. Figure 1.2 shows that center-of-mass (COM) momentum of exciton is related to momentum of light $\boldsymbol{Q}_c$ with incident angle θ through $\boldsymbol{Q} = \boldsymbol{Q}_c \sin\theta$, which gives the idea of the light cone.

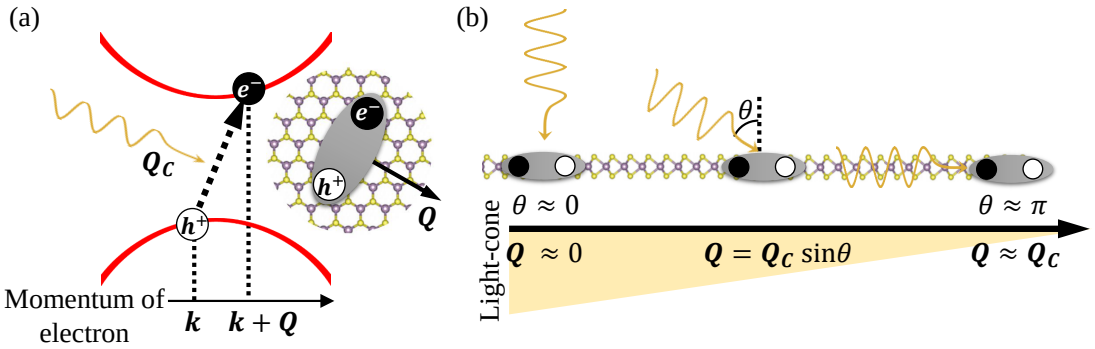


Figure 1.2：(a) Illustration of a photon carrying momentum $\boldsymbol{Q}_c$ exciting (leaving) an electron (hole) with crystal momentum $\boldsymbol{k} + \boldsymbol{Q}$ $(-\boldsymbol{k})$, forming an exciton with center-of-mass (COM) momentum $\boldsymbol{Q}$. (b) Schematic of ideal of light cone and relation between COM momentum $\boldsymbol{Q}$ and $\boldsymbol{Q}_c$

Moreover, the lack of inversion symmetry in the lattice, together with strong spin–orbit coupling from the heavy transition-metal atoms, gives rise to pronounced spin splitting of the electronic bands. At the K/K' valleys, this splitting is typically on the order of a few meV in the conduction band and several hundreds of meV in the valence band, as illustrated in Figure 1.3(a).

The same symmetry considerations also lead to valley-dependent optical selection rules. As shown in Figure 1.3(b), excitons can be classified according to both momentum and spin selection rules. Only excitons with center-of-mass (COM) momentum $\boldsymbol{Q}$ within the light cone,

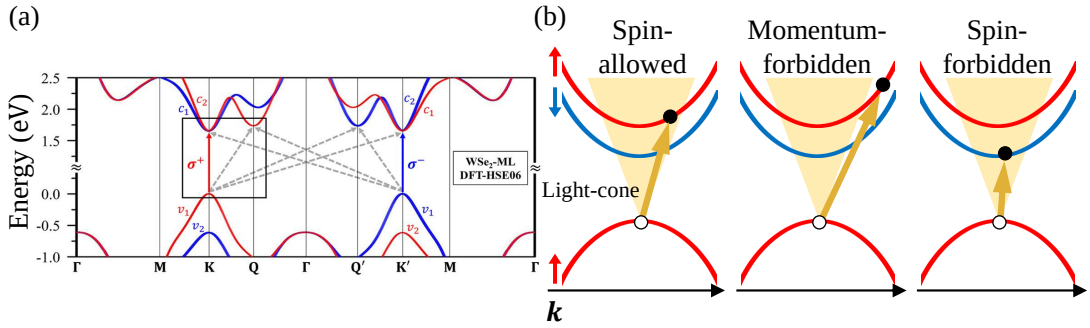


Figure 1.3：(a) Calculated band structure of WSe_2, showing spin-up (red) and spin-down (blue) states, together with a schematic illustration of valley-selective optical transitions at the K and K' valleys, which couple to right- and left-circularly polarized light, respectively, reproduced from Ref. [6]. (b) Schematic configurations of excitonic states, including spin-allowed bright excitons (BXs), momentum-forbidden dark excitons (DXs), and spin-forbidden DXs.

i.e., $|\boldsymbol{Q}| \lesssim Q_c$, and with antiparallel spin configurations can recombine radiatively and are classified as bright excitons (BXs). In contrast, excitons that lie outside the light cone or violate

the spin selection rules remain optically inactive and are therefore classified as dark excitons (DXs).

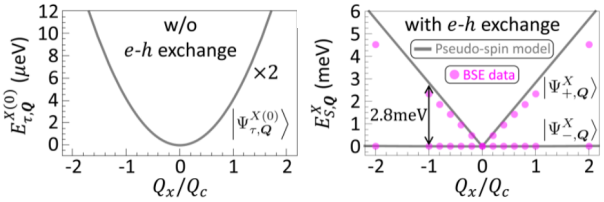


Figure 1.4 : The lowest valley-degenerate (left) and valley-split (right) exciton bands in MoS_2 along the Q_x direction, reproduced from Ref. [7].

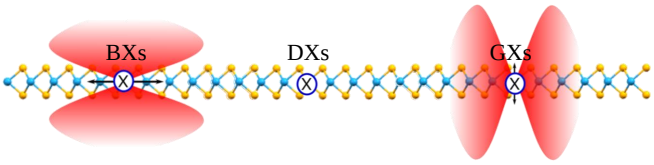


Figure 1.5 : Illustration of the radiation patterns of BXs, DXs, and GXs in neutral WSe_2, along with the corresponding optical excitation indicated by arrows (reproduced from Ref. [8]).

The finer band structure further enriches the classification of quasiparticles. In particular, the lowest bright exciton band, which is initially twofold degenerate, splits into two linearly dispersing branches when electron–hole exchange interactions are included within a pseudospin model. These branches are commonly referred to as valley-mixed excitons [7].

In W-based TMDs, an additional excitonic state known as the gray exciton (GX) can also emerge. As shown in Figure 1.5, GXs emit light polarized perpendicular to the two-dimensional plane (out-of-plane direction). Compared with BXs and DXs, GXs exhibit relatively long lifetimes—typically an order of magnitude longer than BXs (BX $\sim$ 10 ps) [8]—while retaining weak but finite optical activity. These properties make GXs particularly useful for studying exciton dynamics and light–matter interactions, and highlight the rich excitonic physics in two-dimensional TMDs.

1.2 Spin and Orbital Angular Momentum (SAM and OAM) of Laguerre–Gaussian (LG) beam

Following the theoretical work of Poynting in 1909 [9] and the experimental demonstration by Beth in 1936 [10], it has been established that each photon in circularly polarized light with finite dimensions [11–13] carries a spin angular momentum (SAM) of $S = \sigma\hbar = \pm\hbar$ along the propagation direction, where σ denotes the photon helicity. In Beth's experiment, the torque exerted on a suspended half-wave plate by circularly polarized light was measured, demonstrating that spin angular momentum can be transferred to a birefringent medium during propagation (see Figure 1.6(a)).

In 1992, Allen *et al.* showed that a laser beam in a Laguerre–Gaussian (LG) mode possesses a well-defined orbital angular momentum (OAM), also directed along the propagation axis, with $L = \ell\hbar$ per photon, where $\ell \in \mathbb{Z}$ is the azimuthal mode index, commonly referred to as the topological charge [11]. In their experiment, an input Hermite–Gaussian mode, which carries zero OAM, was converted into an LG mode with nonzero OAM. This mode conversion subsequently induced a measurable torque on cylindrical optical elements (see Figure 1.6(b, c)),

establishing that light can also carry OAM associated with the spatial structure of its wavefront. The helical phase structure associated with a quantized amount of $\ell\hbar$ of OAM is illustrated in Figure 1.7, while intensified charge-coupled device (ICCD) images show the characteristic doughnut-shaped intensity distribution.

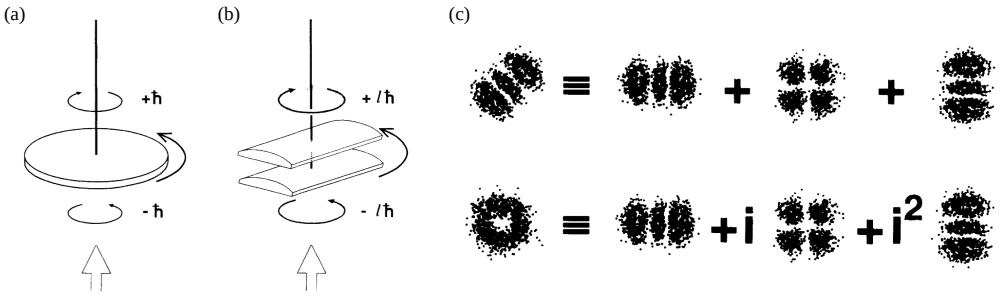


Figure 1.6：(a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre－Gaussian (LG) mode carrying OAM $\ell\hbar$ per photon into a mode with $-\ell\hbar$ per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero ℓ (left side of bottom row) by introducing a phase shift when converting Hermite−Gaussian modes with zero ℓ (left side of top row). All reproduced from Ref. [11].

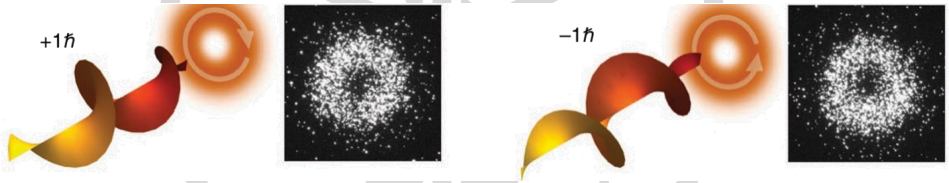


Figure 1.7：Illustration of twisted light with a helical wavefront, where each photon carries $\ell\hbar$ of orbital angular momentum. The intensified charge-coupled device camera images show the characteristic doughnut-shaped intensity profile. Reproduced from Ref. [14].

Since then, a wide range of applications of light carrying OAM has been developed. Early studies focused on angular momentum transfer from light to matter, where objects experience mechanical torque due to OAM transfer, resulting in rotational motion [15–17]. Later developments have extended to optical communications [18–20], quantum key distribution [21, 22], astronomy [23], data storage [24, 25], enhanced imaging sensitivity [26, 27], and optical tweezers [15, 28]. These advances motivate investigation of how structured light carrying orbital angular momentum interacts with electronic excitations in low-dimensional materials.

1.3 SAM and OAM coupling in LG beam

In free space, coupling between SAM and OAM has been shown to occur in non-paraxial (tightly focused) light. Along the propagation direction, SAM can be partially converted into

OAM as the focusing angle increases [29]. This effect can be understood by considering the reduction of the spin component along the beam axis induced by a focusing lens (see Figure 1.8).

The initial spin s_0 is aligned with the propagation direction. After focusing, the spin is no longer parallel to the beam axis; instead, the final spin s_1 is tilted, so only its longitudinal component s_z contributes to the beam' s spin angular momentum. Since the maximum spin angular momentum is $s\hbar$ per photon, the total spin is reduced. However, total angular momentum is conserved during focusing. Consequently, this reduction in SAM is compensated by orbital angular momentum generation, leading to a transfer from SAM to OAM [30].

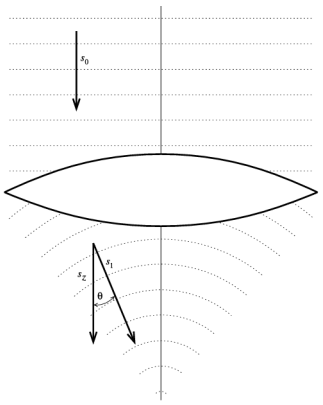


Figure 1.8：Illustration of a beam focused by a lens, showing the variation of SAM before and after the lens along the propagation direction. Reproduced from Ref. [30].

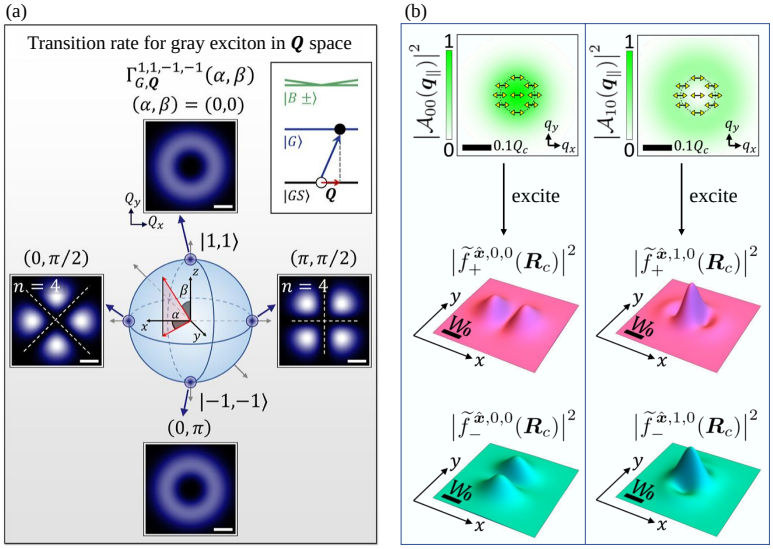


Figure 1.9：(a) Transition-rate distribution for gray excitons in momentum space, reproduced from Ref. [1]. (b) Distribution of the incident light and the excited BX wave-packet states, reproduced from Ref. [7].

However, it should be emphasized that the simple proportionality between spin and orbital angular momentum with the optical helicity σ and topological charge ℓ holds strictly only within the paraxial approximation [31, 32], which often neglects the longitudinal component. However, it is precisely this longitudinal component that can couple to GXs. More importantly, this neglected component already contains two distinct types of spin–orbit interaction (SOI).

In 2024, Oscar Javier Gomez Sanchez *et al.* demonstrated that a type of SOI originating from the phase of the longitudinal component induces momentum-dependent transition rates of GXs through the superposition of two twisted-light beams, forming the vector vortex beams (VVBs) (see Figure 1.9(a)) [1]. However, the combined contribution of SAM and OAM was not fully clarified in their formulation. Meanwhile, in 2022, Guan-Hao Peng *et al.* investigated the photoexcitation of valley-mixed BXs by LG beams within the pseudospin model, suggesting a connection to excitonic topology [7] (see Figure 1.9(b)). Nevertheless, the global phase of the valley-mixed excitons may still influence the excited wave-packet profile.

In this work, we extend the studies of Oscar Javier Gomez Sanchez *et al.* and Guan-Hao Peng *et al.* to investigate the interplay between spin and orbital angular momentum of light in determining transition rates and wave-packet distributions of excitonic states within SI units.

Chapter 2 SAM and OAM interaction (SOI) of LG Beam

2.1 LG Beams : SOI effect

Although electromagnetic fields are the directly observable physical quantities, structured light beams such as Laguerre–Gaussian (LG) beams are often formulated in terms of the scalar and vector potentials. However, the potentials possess gauge freedom, leading to different forms of the wave equations under different gauge conditions. Importantly, the derivation of LG beams relies on the paraxial approximation, whose implementation is not completely independent of the chosen gauge.

In this section, we examine how gauge choice affects the formulation of wave equations. We begin by introducing the wave equations under different gauge conditions, and then analyze the compatibility of the paraxial approximation. This discussion clarifies why certain gauges are more suitable than others for describing paraxial optical fields.

2.1.1 Gauge Freedom in Wave Equations

Maxwell's equations describe how the **electric field** $\boldsymbol{E}(\boldsymbol{r},t)$ and the **magnetic field** $\boldsymbol{B}(\boldsymbol{r},t)$ evolve in space and time. These equations form the fundamental basis for describing electromagnetic waves, including Laguerre–Gaussian beams. In **free space**, the charge density ρ and current density **J** vanish. In this case, Maxwell's equations reduce to four homogeneous differential equations involving the vacuum permittivity ϵ_0, the vacuum permeability μ_0, and the speed of light in vacuum $c = 1/\sqrt{\mu_0\epsilon_0}$.

Within this electrodynamics framework, Maxwell's equations can be rewritten as two coupled wave equations for the **scalar and vector potentials**, $\phi(\boldsymbol{r},t)$ and $\boldsymbol{A}(\boldsymbol{r},t)$, derived in appendix A.1 as

$$\nabla^2\phi(\boldsymbol{r},t) + \frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}(\boldsymbol{r},t)\right) = 0, \tag{2.1}$$

$$\nabla^2\boldsymbol{A}(\boldsymbol{r},t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}(\boldsymbol{r},t) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}(\boldsymbol{r},t) + \frac{1}{c^2}\frac{\partial}{\partial t}\phi(\boldsymbol{r},t)\right) = 0. \tag{2.2}$$

These equations contain the same physical information as the original Maxwell equations, but are expressed in terms of the scalar and vector potentials.

Further simplification of Eqs. (2.1) and (2.2) can be achieved by exploiting the gauge freedom of the potentials. Below, we introduce two commonly used gauges: the **Coulomb gauge**

and the **Lorenz gauge**. Superscripts C and L are used to distinguish quantities defined in the Coulomb and Lorenz gauges, respectively. The gauge conditions are given by

$$\text{Coulomb gauge condition}: \quad \boldsymbol{\nabla} \cdot \boldsymbol{A}^C(\boldsymbol{r}, t) = 0, \tag{2.3}$$

$$\text{Lorenz gauge condition}: \quad \boldsymbol{\nabla} \cdot \boldsymbol{A}^L(\boldsymbol{r}, t) = -\frac{1}{c^2}\frac{\partial}{\partial t}\phi^L(\boldsymbol{r}, t). \tag{2.4}$$

Both gauges simplify the wave equations. In the Coulomb gauge, they reduce to

$$\nabla^2 \phi^C(\boldsymbol{r}, t) = 0, \text{and} \tag{2.5}$$

$$\nabla^2 \boldsymbol{A}^C(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}^C(\boldsymbol{r}, t) = \frac{1}{c^2}\frac{\partial}{\partial t}\boldsymbol{\nabla}\phi^C(\boldsymbol{r}, t) = 0, \tag{2.6}$$

where the last equality follows from Eq. (2.5). In the Lorenz gauge, Eqs. (2.1) and (2.2) become

$$\nabla^2 \phi^L(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi^L(\boldsymbol{r}, t) = 0, \text{and} \tag{2.7}$$

$$\nabla^2 \boldsymbol{A}^L(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}^L(\boldsymbol{r}, t) = 0. \tag{2.8}$$

Here, we introduce the **monochromatic wave** ansatz, in which the electromagnetic fields and potentials are expressed as $\boldsymbol{E}(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{E}(\boldsymbol{r})e^{-i\omega t}\right]$, $\boldsymbol{A}^C(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^C(\boldsymbol{r})e^{-i\omega t}\right]$, $\boldsymbol{A}^L(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\right]$, and $\phi^L(\boldsymbol{r}, t) = \text{Re}\left[\phi^L(\boldsymbol{r})e^{-i\omega t}\right]$, where ω is the angular frequency of light. Substituting these expressions into Eq. (A.1.6) from Maxwell's equations and applying the corresponding gauge conditions leads to

$$\boldsymbol{E}(\boldsymbol{r}) = i\omega \boldsymbol{A}^C(\boldsymbol{r}) = i\omega\left(\boldsymbol{A}^L(\boldsymbol{r}) + \frac{\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \boldsymbol{A}^L(\boldsymbol{r})\right)}{q_0^2}\right). \tag{2.9}$$

Here $q_0 = \omega/c$ denotes the wavenumber of light in free space. Under the monochromatic-wave ansatz, electric field $\boldsymbol{E}(\boldsymbol{r}, t)$ and magnetic field $\boldsymbol{B}(\boldsymbol{r}, t)$ are related to potentials through Eq. (2.9) and Eq. (A.1.5)

2.1.2 Incompatibility of Gauge Freedom with the Paraxial Approximation

In principle, the vector potential formulation of LG beams can be derived starting from either Eq. (2.6) or Eq. (2.8). However, within Coulomb gauge, the directional constraint imposed on the transversely polarized wave is ultimately incompatible with the standard paraxial Helmholtz formulation. For clarity, this incompatibility will be discussed in detail after deriving the paraxial Helmholtz equation within the Lorenz gauge framework [33].

The derivation starts from the scalar Helmholtz equation, which is obtained from Eq. (2.8) by imposing both the monochromatic ($\boldsymbol{A}^L(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\right]$) and transverse wave ansätze ($\boldsymbol{A}^L(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}} A^L(\boldsymbol{r})$, where we condsider polarization unit vector $\hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\varepsilon}}_\sigma = \hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}}$, $\sigma = \pm 1$ as right/left circularly polarized)(see Sec. A.2):

$$\nabla^2 A^L(\boldsymbol{r}) + q_0^2 A^L(\boldsymbol{r}) = 0. \tag{2.10}$$

The **paraxial approximation** [33–35] assumes that the rays have only small inclinations with respect to the optical axis in geometric optics. In wave optics, this corresponds to an angular spectrum dominated by plane-wave components whose wavevectors are close to the propagation direction. Here, we choose the propagation direction along the $\boldsymbol{z}$ axis, such that the dominant component of the wavevector $\boldsymbol{q}_0$ lies along $\hat{\boldsymbol{z}}$.

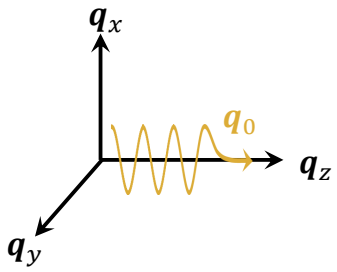


Figure 2.1：Illustration of a dominant component of wavevector $\boldsymbol{q}_0$ lies along $\hat{\boldsymbol{z}}$

Under this approximation, we introduce the **paraxial wave** ansatz, $A^L(\boldsymbol{r}) = U(\boldsymbol{r})e^{iq_0 z}$, and substitute it into Eq. (2.10), which yields

$$\nabla_\parallel^2 U(\boldsymbol{r}) + \frac{\partial^2}{\partial z^2}U(\boldsymbol{r}) + 2iq_0\frac{\partial}{\partial z}U(\boldsymbol{r}) = 0. \tag{2.11}$$

Here, $\nabla_\parallel^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ denotes the transverse Laplacian. The paraxial wave ansatz assumes that the **envelope function** $U(\boldsymbol{r})$ **varies slowly along the $\boldsymbol{z}$ direction** compared to the rapid phase variation of $e^{iq_0 z}$. Consequently, the paraxial approximation implies the following relations:

$$\left|\frac{\partial^2}{\partial z^2}U(\boldsymbol{r})\right| \ll \left|\nabla_\parallel^2 U(\boldsymbol{r})\right|, \tag{2.12}$$

$$\left|\frac{\partial^2}{\partial z^2}U(\boldsymbol{r})\right| \ll q_0\left|\frac{\partial}{\partial z}U(\boldsymbol{r})\right|. \tag{2.13}$$

Neglecting the second-order derivative with respect to z then leads to the paraxial Helmholtz equation

$$\nabla_\parallel^2 U(\boldsymbol{r}) + 2iq_0\frac{\partial}{\partial z}U(\boldsymbol{r}) \approx 0. \tag{2.14}$$

We now clarify why a transversely polarized light ($\hat{\boldsymbol{\varepsilon}}$ confined to the x–y plane) is not suitable in this derivation under the Coulomb gauge. For a transversely polarized wave, the Coulomb gauge condition gives

$$\boldsymbol{\nabla}\cdot\boldsymbol{A}^C(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\cdot\boldsymbol{\nabla}_\parallel A^C(\boldsymbol{r}) \tag{2.15}$$

$$= \left(\frac{\partial}{\partial x}\hat{\boldsymbol{x}} + i\sigma\frac{\partial}{\partial y}\hat{\boldsymbol{y}}\right)A^C(\boldsymbol{r}) = 0 \tag{2.16}$$

where $\boldsymbol{\nabla}_\parallel = \frac{\partial}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}\hat{\boldsymbol{y}}$. This condition imposes a directional constraint in the transverse plane on the vector potential and results in the transverse Laplacian of the potential becoming zero.

$$\nabla_\parallel^2 A^C(\boldsymbol{r}) = \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)A^C(\boldsymbol{r}) = 0, \tag{2.17}$$

which in turn is incompatible with the approximation in Eq. (2.12) [33]. In contrast, a transversely polarized light does not have such constraint within Lorenz gauge. For Coulomb gauge, a nonzero longitudinal (z) component of the vector potential should be introduced, going be-

yond the simple transverse polarization. Therefore, the Lorenz gauge provides a more natural and consistent framework for deriving the paraxial Helmholtz equation and Laguerre–Gaussian beam solutions.

2.1.3 Solutions of LG Beams

Within the paraxial approximation, the optical helicity σ and the topological charge ℓ are directly proportional to the spin and orbital angular momentum of light, respectively [31, 32]. This correspondence provides a convenient framework for analyzing the angular momentum properties of LG beams.

Moreover, when the vector potential formulation of LG beams is expressed in the Coulomb gauge, an SOI can emerge through the additional longitudinal component of the field, giving rise to terms proportional to $\sigma\ell$ and $\sigma + \ell$. Adopting the Coulomb gauge is further justified by the theory of light–matter interaction, where the vector potential is conventionally formulated within both the Coulomb gauge and the angular spectrum representation. The angular-spectrum representation is particularly useful here because it allows us to describe the vector potential in terms of individual plane-wave components with different in-plane wavevectors $\boldsymbol{q}_\parallel$, which can be directly connected to the center-of-mass momentum $\boldsymbol{Q}$ of the excitons.

In the following section, we first introduce the LG beams solutions in both real-space and angular spectrum representations. We then perform a gauge transformation to reveal the SOI.

Real space representation Under the **long Rayleigh-range** approximation [36][1], the LG beam solutions of the paraxial Helmholtz equation, Eq. (2.14), can be expressed in cylindrical coordinates as

$$\boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma A^L_{\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma \, u_{\ell,p}(\boldsymbol{r}_\parallel)\, e^{iq_0 z}, \tag{2.18}$$

where the position vector is written as $\boldsymbol{r} = (\rho, \phi, z) = (\boldsymbol{r}_\parallel, z)$. Here $\boldsymbol{r}_\parallel$ denotes the in-plane position vector consisting of the radial coordinate ρ and the azimuthal angle ϕ. The integer $\ell \in \mathbb{Z}$ is the **topological charge**, and the integer $p \geq 0$ is the radial mode index. The vector $\boldsymbol{q}_0 = q_0\hat{\boldsymbol{z}}$ denotes the wavevector along the propagation direction. The envelope function is expressed as

$$u_{\ell,p}(\boldsymbol{r}_\parallel) = f_{\ell,p}(\rho)\, e^{i\ell\phi}, \tag{2.19}$$

where the radial function is $f_{\ell,p}(\rho) = A_0^{\mathrm{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}}\, L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right)$, which is given by [1, 7, 36] and A_0^{LG} is a constant amplitude. Here the generalized Laguerre polynomial is defined as $L_p^{|\ell|}(x) = \sum_{m=0}^{p}(-1)^m \frac{(|\ell|+p)!}{(p-m)!(|\ell|+m)!m!} x^m$, with $x = \frac{2\rho^2}{W_0^2}$. The parameter W_0 denotes the beam waist of the LG beam, while the cylindrical coordinates are related to the Cartesian

[1] For a light beam, the Rayleigh range is the propagation distance from the beam waist to the point where the beam cross-sectional area doubles (see Figure 2.2). It is given by $z_R = \frac{1}{2} q_0 W_0^2$, where W_0 is the beam waist. When the Rayleigh range is much larger than the characteristic propagation distance of interest, diffraction-induced beam spreading becomes negligible. In this limit, the beam waist can be treated as approximately constant, and the envelope function may be regarded as independent of z. This simplification is referred to as the long Rayleigh-range approximation.

coordinates through $\rho = \sqrt{x^2 + y^2}$ and $\phi = \tan^{-1}(y/x)$. The geometric meaning of the beam waist and the Rayleigh range is illustrated in Figure 2.2.

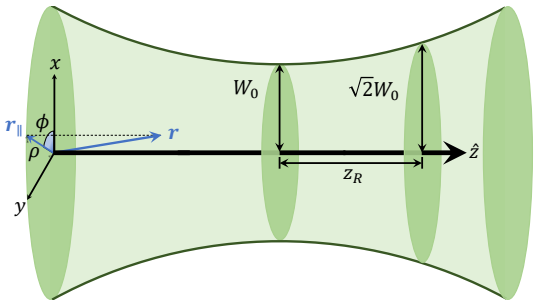


Figure 2.2 : Illustration of the evolution of the beam radius $W(z)$ during propagation. The minimum beam radius W_0 occurs at the beam waist, and the Rayleigh range z_R is defined as the distance from the waist at which $W(z_R) = \sqrt{2}W_0$.

Focusing on the minimum cross-section area of the beam ($z = 0$), Figure 2.3 shows that as ℓ and/or p increase, $u_{\ell,p}(\boldsymbol{r}_\parallel)$ develops a broader doughnut-shaped intensity profile, accompanied by an increasing number of radial nodes along the radial direction. The phase varies continuously with the azimuthal angle ϕ, reflecting the helical wavefront associated with orbital angular momentum. In particular, the phase winds by $2\pi\ell$ around the beam center. The beam waist is chosen as $W_0 = 1.5$ μm, which sets the spatial scale of the beam and determines the radial extent of the intensity distribution.

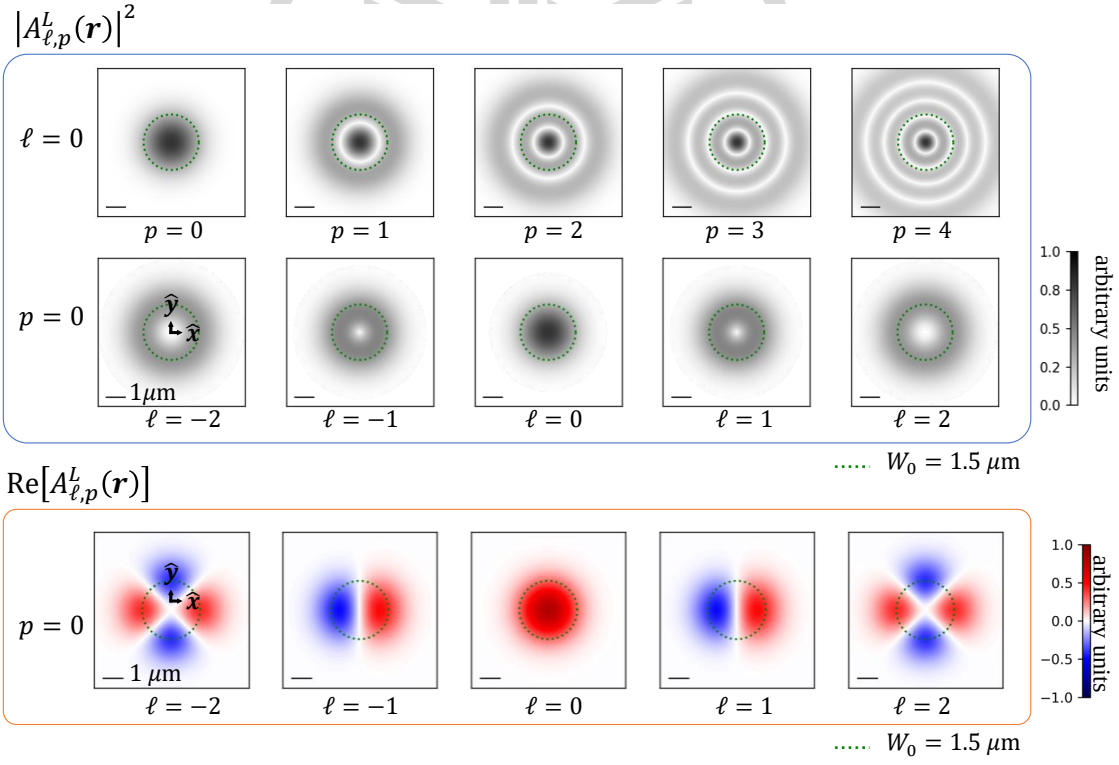


Figure 2.3 : Squared magnitude (intensity) and real part distribution of $A^L_{\ell,p}(\boldsymbol{r})$ at $z = 0$. The green dotted line denotes the beam waist W_0. The scale bar indicates the length of 1 μm.

Comparison between the Paraxial and Long-Rayleigh approximations In the derivation, both the paraxial and long Rayleigh-range approximations are employed. Although both approximations constrain beam propagation, they impose different physical constraints and therefore

cannot be directly compared solely in terms of the propagation angle. The paraxial approximation restricts the propagation angle of the constituent plane waves [37], whereas the long Rayleigh-range approximation requires the beam radius to vary negligibly over the propagation distance of interest :

$$\text{Paraxial approximation :} \quad \theta \equiv \sin^{-1}\left(\frac{q_{\parallel}}{q_0}\right) < 0.5 \text{ rad} \approx 30^{\circ}, \tag{2.20}$$

$$\text{Long Rayleigh-range approximation :} \quad z \ll z_R, \tag{2.21}$$

where $q_{\parallel} = \sqrt{q_x^2 + q_y^2}$ is the magnitude of the transverse component of the wavevector, and the Rayleigh range is $z_R = \frac{1}{2} q_0 W_0^2$.

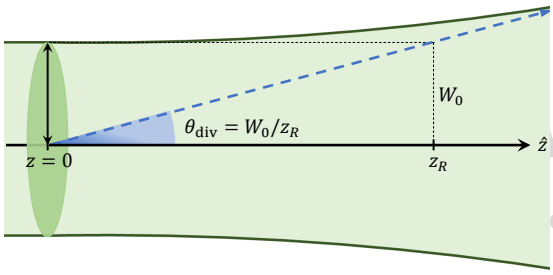


Figure 2.4 : Illustration of the beam-radius evolution and far-field divergence angle θ_{div}.

To facilitate the comparison, the long Rayleigh-range condition can be related to the far-field divergence angle [35], defined as

$$\theta_{\text{div}} \equiv \lim_{z\to\infty} \frac{dW(z)}{dz} = \frac{W_0}{z_R}, \tag{2.22}$$

where the beam radius is $W(z) = W_0\sqrt{1 + (z/z_R)^2}$. Equation (2.22) shows that a smaller far-field divergence angle corresponds to a slower increase in the beam radius, allowing the beam to remain approximately collimated over a longer propagation distance (see Figure 2.4).

Although θ and θ_{div} describe different physical quantities, they are related through the angular distribution of the constituent plane waves. For a qualitative comparison, we may approximately identify the characteristic propagation angle θ with the far-field divergence angle θ_{div}, i.e., $\theta_{\text{div}} \approx \theta$. Under this rough approximation, the two criteria can be expressed in terms of the same angular quantity:

$$\text{Paraxial approximation :} \quad \theta \leq 30^{\circ}, \tag{2.23}$$

$$\text{Long Rayleigh-range approximation :} \quad \theta \approx \theta_{\text{div}} \ll \frac{W_0}{z}, \tag{2.24}$$

where the first condition limits the propagation angle of the constituent plane waves, while the second expresses the requirement that the transverse spreading over the propagation distance z remain much smaller than the beam waist.

Angular spectrum representation To derive the vector potential of LG beams in the angular spectrum representation, the potential is expanded **in a plane-wave basis** as

$$\boldsymbol{A}(\boldsymbol{r}) = \sum_{\boldsymbol{q}} \boldsymbol{\mathcal{A}}(\boldsymbol{q}) e^{i\boldsymbol{q}\cdot\boldsymbol{r}}, \tag{2.25}$$

where $\boldsymbol{\mathcal{A}}(\boldsymbol{q})$ is the complex amplitude associated with each plane-wave component $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$. The LG beams in Eq. (2.18) can be written in this form through a defined **two-dimensional Fourier transformation** and its corresponding inverse transformation,

$$u_{\ell,p}(\boldsymbol{r}_\parallel) = \frac{1}{(2\pi)^2}\int \mathrm{d}^2\boldsymbol{q}_\parallel\, F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel)\, e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}, \quad \longleftrightarrow \quad F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel) = \int \mathrm{d}^2\boldsymbol{r}_\parallel\, u_{\ell,p}(\boldsymbol{r}_\parallel)\, e^{-i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}. \tag{2.26}$$

For applications to two-dimensional materials, the in-plane wavevector $\boldsymbol{q}_\parallel$ can be approximately identified with a reciprocal-space vector. Under this approximation, Eq. (2.26) can be discretized, and Eq. (2.25) becomes

$$\boldsymbol{\mathcal{A}}^L_{\sigma,\ell,p}(\boldsymbol{r}) = \sum_{\boldsymbol{q}_\parallel} \boldsymbol{\mathcal{A}}^L_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel} e^{iq_{0z}z}. \tag{2.27}$$

Here the in-plane wavevector is written as $\boldsymbol{q}_\parallel = (q_\parallel, \phi_{\boldsymbol{q}_\parallel})$. The corresponding complex amplitude can be obtained analytically, as summarized in Table 2.1.

Table 2.1：Real space and angular spectrum representations of LG beams

Real space	**Angular spectrum**
Vector potential	
$\boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma A^L_{\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\, u_{\ell,p}(\boldsymbol{r}_\parallel)\, e^{iq_0 z}$	$\boldsymbol{\mathcal{A}}^L_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) = \hat{\boldsymbol{\varepsilon}}_\sigma \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel) = \hat{\boldsymbol{\varepsilon}}_\sigma \dfrac{1}{\Omega} F^{\mathrm{LG}}_{\lvert\ell\rvert,p}(\boldsymbol{q}_\parallel)$
Envelope function	
$u_{\ell,p}(\boldsymbol{r}_\parallel) = f_{\ell,p}(\rho)e^{i\ell\phi}$	$\dfrac{1}{\Omega}F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel) = \tilde{F}_{\lvert\ell\rvert,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}_\parallel}}$
Solutions at $p=0$	
$f_{\ell,p=0}(\rho) = \dfrac{A^{\mathrm{LG}}_0}{\sqrt{(\lvert\ell\rvert)!\pi/2}} \left(\dfrac{\sqrt{2}\rho}{W_0}\right)^{\lvert\ell\rvert} \exp\left(-\dfrac{\rho^2}{W_0^2}\right)$	$\tilde{F}_{\lvert\ell\rvert,p=0}(q_\parallel) = \dfrac{A^{\mathrm{LG}}_0 W_0^2\,(-i)^{\lvert\ell\rvert}\,\pi}{\Omega\sqrt{(\lvert\ell\rvert)!\pi/2}} \left(\dfrac{q_\parallel W_0}{\sqrt{2}}\right)^{\lvert\ell\rvert} \exp\left(-\dfrac{q_\parallel^2 W_0^2}{4}\right)$

where $\tilde{F}_{|\ell|,p}(q_\parallel) = \frac{(-i)^{|\ell|}}{\Omega} 2\pi\, \mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)]$, and $\mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)] = \int_0^\infty \mathrm{d}\rho\ \rho\, f_{\ell,p}(\rho)\, J_{|\ell|}(q_\parallel\rho)$ is the **Hankel transform** of order $|\ell|$, where $J_{|\ell|}$ is the **Bessel function of the first kind**. Here Ω denotes the area of the two-dimensional system

Figure 2.5 shows that $\mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel)$ exhibits an intensity profile analogous to that of $A^L_{\ell,p}(\boldsymbol{r})$, reflecting the characteristic doughnut-shaped structure in angular spectrum. This correspondence follows directly from the Fourier transform relation between two representations of the LG beam. The doughnut-shaped distribution in angular spectrum indicates that the dominant in-plane wavevectors lie within a finite range determined by the beam waist W_0.

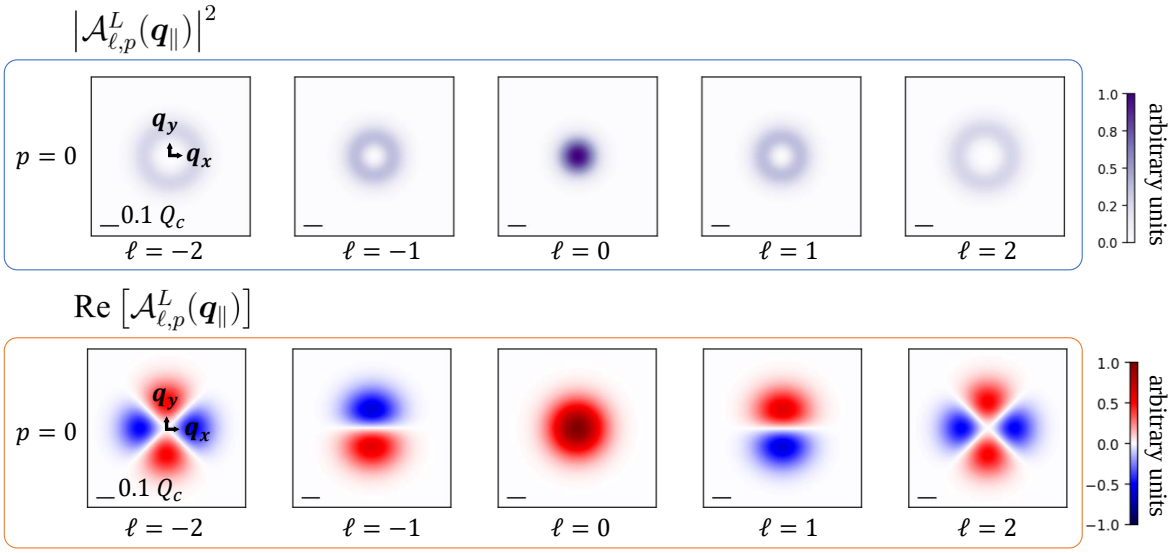


Figure 2.5：Squared magnitude (intensity) and real part distribution of $\mathcal{A}^L_{\ell,p=0}(\boldsymbol{q}_\parallel)$. The scale bar indicates the considered 0.1 times light cone range for WSe_2-monolayers, $Q_c = 8.6\ \mu\mathrm{m}^{-1}$.

2.1.4 Longitudinal Component of LG Beam : SOI effect

In the previous section, Eq. (2.9) was introduced to relate the vector potentials defined in the two gauges to the physical fields. Using this relation, we can perform gauge transformation by rewritten Eq. (2.9) as

$$\boldsymbol{A}^C_{\sigma,\ell,p}(\boldsymbol{r}) = \boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r}) + \frac{\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r})\right)}{q_0^2}. \tag{2.28}$$

To explicitly incorporate the SAM of light, the polarization unit vector is defined as $\hat{\boldsymbol{\varepsilon}} \equiv \hat{\boldsymbol{\varepsilon}}_\sigma = \frac{1}{\sqrt{2}}(\hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}})$, where $\sigma = \pm 1$ denotes the spin of the photon. From now on we omit the superscript C to simplify the notations and the vector potential in real space under the Coulomb gauge is obtained as

$$\boldsymbol{A}_{\sigma,\ell,p}(\boldsymbol{r}) \equiv \boldsymbol{A}^C_{\sigma,\ell,p}(\boldsymbol{r}) = \tilde{A}^{\parallel}_{\ell,p}(\boldsymbol{r})\hat{\boldsymbol{\varepsilon}} + \tilde{A}^z_{\sigma,\ell,p}(\boldsymbol{r})\hat{\boldsymbol{z}}. \tag{2.29}$$

For LG beams in the fundamental radial mode $p = 0$

$$\begin{aligned} \tilde{A}^{\parallel}_{\ell,p}(\boldsymbol{r}) &\approx f_{\ell,p=0}(\rho)e^{i\ell\phi}e^{iq_0 z} \\ \tilde{A}^z_{\sigma,\ell,p}(\boldsymbol{r}) &\approx \frac{i}{\sqrt{2}q_0\rho}\left(\left(|\ell| + \sigma\ell\right) - \frac{2\rho^2}{W_0^2}\right) f_{\ell,p=0}(\rho)e^{iJ\phi}e^{iq_0 z} \end{aligned} \tag{2.30}$$

where

$$J \equiv \sigma + \ell$$

is the total angular momentum (TAM). In general, the second term in Eq. (2.28) introduces corrections to all spatial components of the vector potential. However, under the paraxial approximation, these corrections are suppressed. Therefore, we retain only the longitudinal (z) component as the leading correction.

The longitudinal component in Eq. (2.30) reveals two distinct types of SOI: a $\sigma\ell$ dependence in the amplitude and a J dependence in the phase. Figure 2.6 shows that the magnitude (absolute value) at $z = 0$ of this longitudinal component is significantly weaker than that of the transverse components. The $\sigma\ell$ dependence manifests as an asymmetry between cases with the same ℓ but opposite spin σ.

It is worth noting that the longitudinal term is weighted by a factor proportional to $(q_0 W_0^2)^{-1}$, which is inversely related to the Rayleigh range ($\frac{1}{2} q_0 W_0^2$) [34] . This indicates that a smaller beam waist W_0 leads to a stronger longitudinal component, corresponding to a shorter Rayleigh range and tighter focusing, as shown in the bottom row of Figure 2.6. The beam waist W_0 is chosen to be $1.5\ \mu$m and $0.5\ \mu$m, representing a typical paraxial beam and a case at the boundary of the paraxial regime, respectively [38–40]. This demonstrates that the longitudinal component, although small, remains finite even within the paraxial regime.

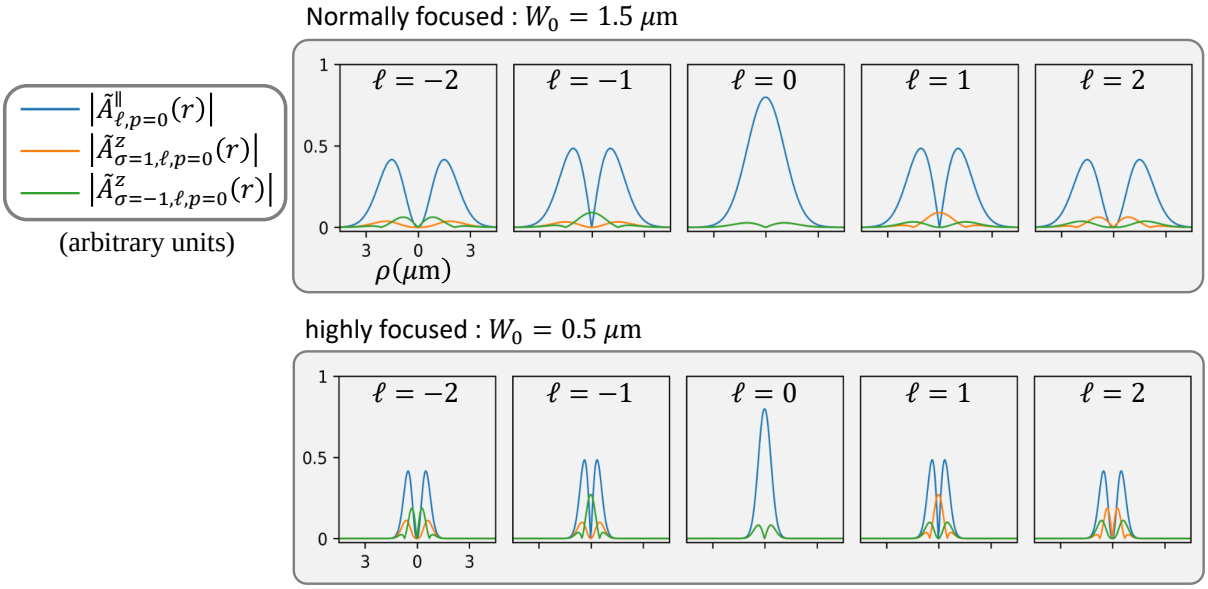


Figure 2.6 : Magnitude (absolute value) of $\boldsymbol{A}_{\sigma,\ell,p=0}(\boldsymbol{r})$ at $z = 0$ for different components and spin. The $\sigma\ell$ term leads to asymmetry between cases with the same ℓ in the longitudinal component (a wavelength of 532 nm is assumed).

As for the $\sigma + \ell$ dependence in the phase term, it is retained in the angular spectrum representation and will be discussed below. By combining Eq. (2.28) with the two-dimensional Fourier transformation defined in Coulomb gauge, vector potential can be written as

$$\boldsymbol{A}_{\sigma,\ell,p}(\boldsymbol{r}) = \frac{1}{(2\pi)^2} \int \mathrm{d}^2 \boldsymbol{q}_{\parallel}\, \boldsymbol{\mathcal{A}}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel}) e^{i\boldsymbol{q}_{\parallel}\cdot\boldsymbol{r}_{\parallel}} e^{iq_0 z}. \tag{2.31}$$

Since we primarily focus on the fundamental radial mode $p = 0$, from now on we omit the subscript p to simplify the notations. The complex amplitude under Coulomb gauge in the angular spectrum representation is then written as

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) = \boldsymbol{\mathcal{A}}_{\sigma,\ell,p=0}(\boldsymbol{q}_{\parallel}) = \tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{\varepsilon}}_{\sigma} + \tilde{\mathcal{A}}^{z}_{\sigma,\ell}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{z}}. \tag{2.32}$$

The corresponding complex amplitude in the angular spectrum representation is obtained as

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) = \left[\hat{\boldsymbol{\varepsilon}}_\sigma - \left(\frac{|\boldsymbol{q}|}{q_0}\right)^2 (\hat{\boldsymbol{\varepsilon}}_\sigma \cdot \hat{\boldsymbol{q}}_\parallel)\hat{\boldsymbol{q}}\right] \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel) \tag{2.33}$$

$$= \left[\hat{\boldsymbol{\varepsilon}}_\sigma - \left(\frac{|\boldsymbol{q}|}{q_0}\right)^2 \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\sigma\phi_{\boldsymbol{q}_\parallel}} \hat{\boldsymbol{q}}\right] \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel) \tag{2.34}$$

$$\approx \left[\hat{\boldsymbol{\varepsilon}}_\sigma - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\sigma\phi_{\boldsymbol{q}_\parallel}} \hat{\boldsymbol{z}}\right] \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel), \tag{2.35}$$

where the final approximation follows from the paraxial condition $\cos\theta_{\boldsymbol{q}} = q_0/|\boldsymbol{q}| \approx 1$. The wavevector unit vector is written as $\hat{\boldsymbol{q}} = \sin\theta_{\boldsymbol{q}}\left(\cos\phi_{\boldsymbol{q}_\parallel}\hat{\boldsymbol{x}} + \sin\phi_{\boldsymbol{q}_\parallel}\hat{\boldsymbol{y}}\right) + \cos\theta_{\boldsymbol{q}}\hat{\boldsymbol{z}} \approx \hat{\boldsymbol{z}}$, and $\sin\theta_{\boldsymbol{q}} = \frac{q_\parallel}{\sqrt{q_\parallel^2+q_0^2}}$. Substituting the complex amplitude under Lorenz gauge derived previously, each component of the complex amplitude takes the form

$$\begin{aligned}
\tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_\parallel) &\equiv \tilde{\mathcal{A}}^{\parallel}_{\ell,p=0}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|}(q_\parallel) e^{i\ell\phi_{\boldsymbol{q}_\parallel}} \\
\tilde{\mathcal{A}}^{z}_{\sigma,\ell}(\boldsymbol{q}_\parallel) &\equiv \tilde{\mathcal{A}}^{z}_{\sigma,\ell,p=0}(\boldsymbol{q}_\parallel) \approx -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(q_\parallel) e^{iJ\phi_{\boldsymbol{q}_\parallel}},
\end{aligned} \tag{2.36}$$

where $\tilde{F}_{|\ell|}(q_\parallel) = \tilde{F}_{|\ell|,p=0}(q_\parallel)$. This expression shows that the SOI is encoded in the phase of the longitudinal component through the combined dependence $(\sigma+\ell)$, corresponding to TAM. This behavior difference is illustrated in Figure 2.7 and Figure 2.8. However, this phase structure is generally difficult to detext experimentally, as it is averaged out in intensity-based measurements. To access this TAM-dependent phase, interference between two LG beams can be employed, forming vector vortex beams (VVBs), which will be discussed in the following section.

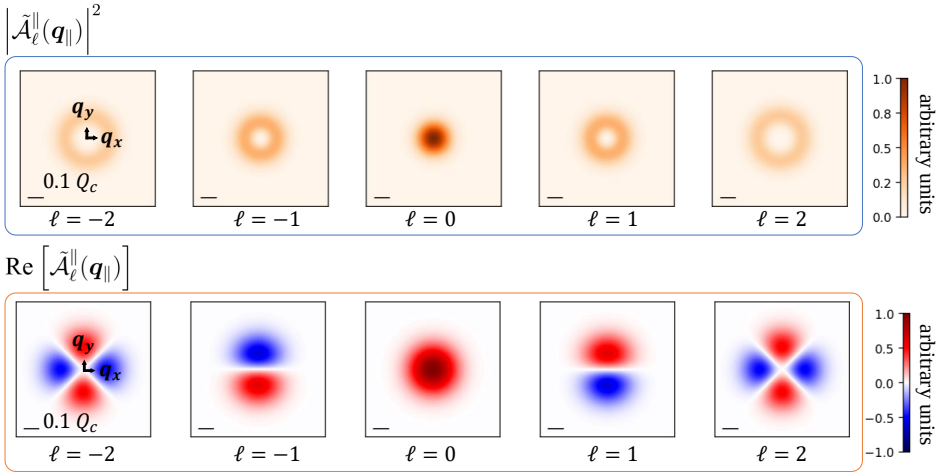


Figure 2.7：Squared magnitude (intensity) and real part distribution of $\tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_\parallel)$. The scale bar indicates 0.1 times the light-cone radius for WSe_2 monolayers, $Q_c = 8.6\ \mu\text{m}^{-1}$.

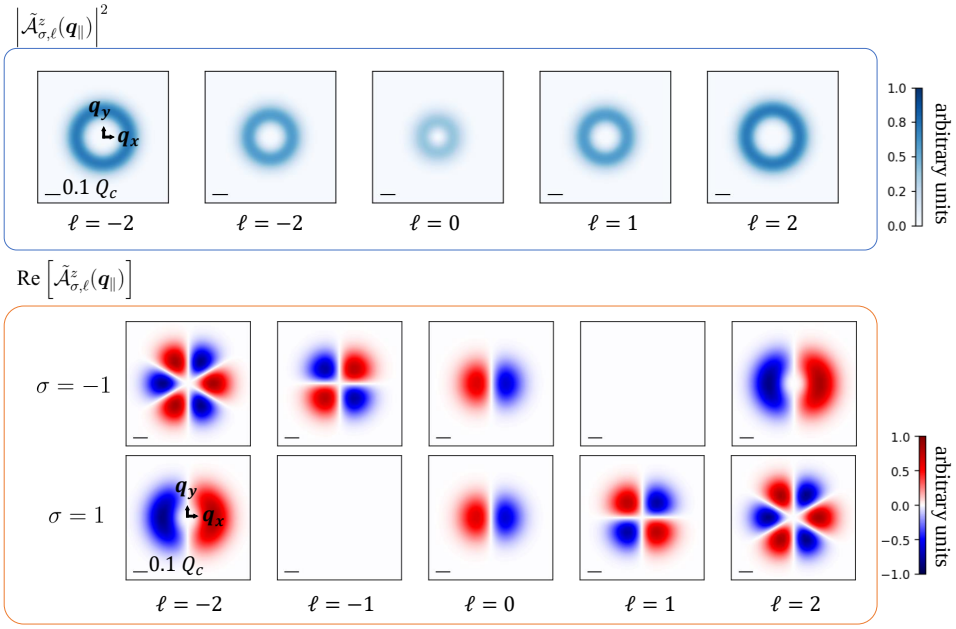


Figure 2.8：Squared magnitude (intensity) and real part distribution of $\tilde{\mathcal{A}}^{z}_{\sigma,\ell}(\boldsymbol{q}_\parallel)$. The scale bar indicates 0.1 times the light-cone radius for WSe_2 monolayers, $Q_c = 8.6\ \mu\text{m}^{-1}$.

2.2 Vector Vortex Beams(VVBs) : Wave Beating

Subject to decoherence, the information of TAM encoded in the phase is likely to be lost when calculating optical transition rates, as pointed out by Oscar Javier Gomez Sanchez *et al.* in Ref. [1]. To retain and control this information, they introduced vector vortex beams (VVBs), which are constructed by superimposing two twisted light beams , which in general, carry angular momentum of the same magnitude but opposite signs. Thus, a VVB carries vectorial polarization that varies across the transverse plane. In contrast, a scalar vortex beam, which consists of a single twisted light beam, carries uniform polarization. The VVB state is expressed as a superposition through

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) = \cos\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma',\ell'}(\boldsymbol{q}_\parallel). \tag{2.37}$$

The parameter α controls the relative phase between the two beams, while β determines their relative weight in the superposition. Throughout this thesis, the radial mode index is set to $p = 0$, as the primary focus remains on the interplay between SAM and OAM. σ' and ℓ' denote the second LG beam and its different combination of SAM and OAM.

Similar to the previous section, a VVB state contains both transverse and longitudinal components, which can be expressed as

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) = \tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) + \tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta), \tag{2.38}$$

2.2.1 Analogy between 1D Wave Beating and the Longitudinal Component

Here we review wave mixing to obtain an intuitive physical picture of how Vector Vortex Beams (VVBs) behave. To begin with, let us consider two signals and their additive mixing :

$$\text{Signal } 1(\phi,t) = \text{Re}\left[\tilde{A}e^{i(k\phi-\omega t)}\right], \qquad \text{Signal } 2(\phi,t) = \text{Re}\left[\tilde{A}'e^{i(k'\phi-\omega t+\alpha)}\right], \quad \text{and}$$
$$\text{Sum} \equiv \text{Re}\left[\tilde{A}e^{i(k\phi-\omega t)} + \tilde{A}'e^{i(k'\phi-\omega t+\alpha)}\right].$$

Below, we examine how the resulting wave behaves under different conditions.

1. Standing wave $\tilde{A} = \tilde{A}' = A$ and $k' = -k$:

$$\text{Sum} = \text{Re}\Big[\underbrace{2A\cos\left(\frac{\Delta k}{2}\phi + \frac{c}{2}\right)}_{\text{Envelope}}\underbrace{e^{i\left(-\omega t+\frac{c}{2}\right)}}_{\text{Carrier}}\Big], \tag{2.39}$$

where the number of nodes is determined by the difference in wave numbers

$$\Delta k \equiv k' - k.$$

An example for standing wave is shown in Figure 2.9.

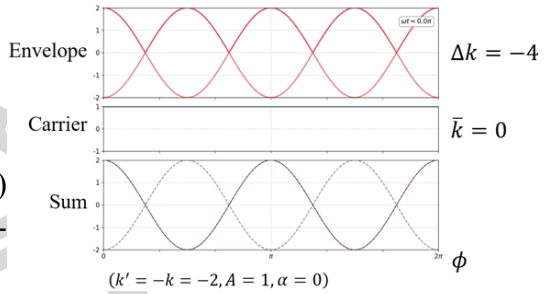


Figure 2.9: Example of a standing wave with $\Delta k = -4$ at $t = 0$.

2. Beating wave$\tilde{A} = \tilde{A}' = A$ and $k' \neq k$:

$$\text{Sum} = \text{Re}\Big[\underbrace{2A\cos\left(\frac{\Delta k}{2}\phi + \frac{\alpha}{2}\right)}_{\text{Envelope}}\underbrace{e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}\right)}}_{\text{Carrier}}\Big], \tag{2.40}$$

where the carrier starts to move along the ϕ direction (The field is rotating) when

$$\bar{k} \equiv \tfrac{k'+k}{2}$$

is non-zero. An example for beating wave is shown in Figure 2.10.

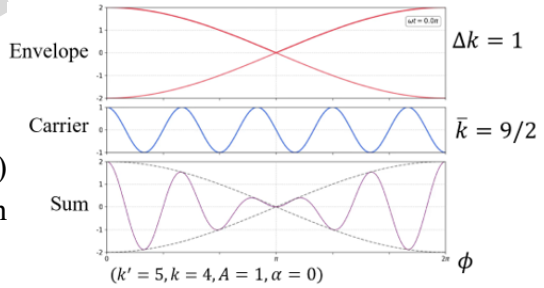


Figure 2.10: Example of a beating wave with $\bar{k} = 9/2$ at $t = 0$

To draw an analogy to VVBs, we consider mismatched amplitudes that remain similar by assuming

$$|\tilde{A}| + |\tilde{A}'| \gg |\tilde{A}| - |\tilde{A}'|$$

3. Real amplitudes $\tilde{A} \neq \tilde{A}' \in \mathbb{R}$ and $k' \neq -k$

$$\begin{aligned} \text{Sum} &= \text{Re}\left\{\left[(\tilde{A}+\tilde{A}')\cos\left(\tfrac{\Delta k}{2}\phi+\tfrac{\alpha}{2}\right) - i(\tilde{A}-\tilde{A}')\sin\left(\tfrac{\Delta k}{2}\phi+\tfrac{\alpha}{2}\right)\right] e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}\right)}\right\} \\ &\approx \text{Re}\left\{\underbrace{(\tilde{A}+\tilde{A}')\cos\left(\tfrac{\Delta k}{2}\phi+\tfrac{\alpha}{2}\right)}_{\text{Envelope}}\underbrace{e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}\right)}}_{\text{Carrier}}\right\} \end{aligned} \tag{2.41}$$

Under the approximation, the formula is the same but we need to keep in mind that the node becomes non-zero, which corresponds to the non-zero term $(\tilde{A} - \tilde{A}')$. An example is shown in the left panel in Figure 2.11

4. Complex amplitudes $\tilde{A} \neq \tilde{A}' \in \mathbb{C}$ and $k' \neq -k$:

$$\begin{aligned} \text{Sum} &= \text{Re}\left\{\left[(|\tilde{A}|+|\tilde{A}'|)\cos\left(\tfrac{\Delta k}{2}\phi+\tfrac{\alpha}{2}\right) - i(|\tilde{A}|-|\tilde{A}'|)\sin\left(\tfrac{\Delta k}{2}\phi+\tfrac{\alpha}{2}\right)\right] e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}+\frac{1}{2}(\text{Arg}[\tilde{A}']+\text{Arg}[\tilde{A}])\right)}\right\} \\ &\approx \text{Re}\{\underbrace{(|\tilde{A}|+|\tilde{A}'|)\cos\left(\tfrac{\Delta k}{2}\phi+\tfrac{\alpha'}{2}\right)}_{\text{Envelope}}\underbrace{e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}+\frac{1}{2}(\text{Arg}[\tilde{A}']+\text{Arg}[\tilde{A}])\right)}}_{\text{Carrier}}\} \end{aligned} \tag{2.42}$$

The relative phase now includes contributions from the complex amplitudes such that

$$\alpha' \equiv \alpha + \text{Arg}[\tilde{A}'] - \text{Arg}[\tilde{A}].$$

The result shows that the envelope (intensity) is

$$\text{rotated (counter-clockwise)by } -\frac{\alpha'}{\Delta k}.$$

The rotation (shift) can be seen in the example in the right panel of Figure 2.11.

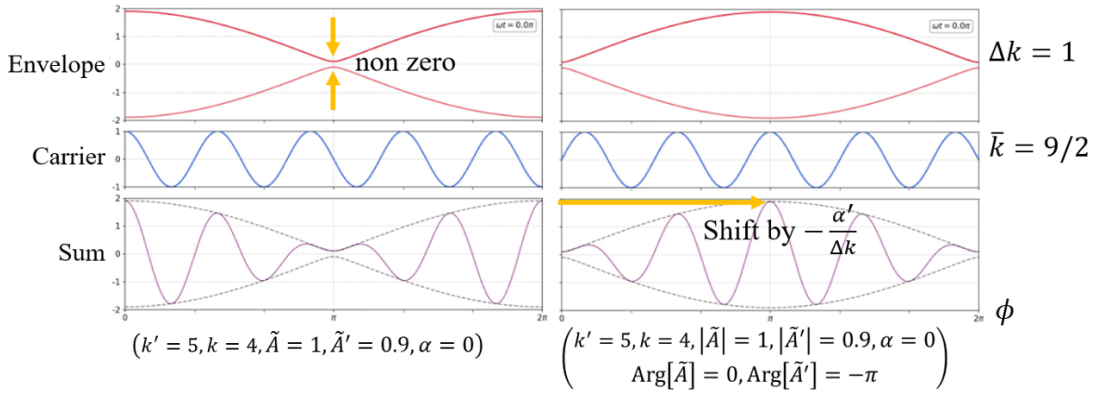


Figure 2.11 : Examples for mismatched amplitudes. The left(right) panel shows the case for real(complex) amplitude, where $-\frac{\alpha'}{\Delta k} = \pi$ for the complex amplitude case.

By analogy with VVBs, under the approximation $G^{+}_{|\ell|,|\ell'|}(q_\parallel;\beta) \gg G^{-}_{|\ell|,|\ell'|}(q_\parallel;\beta) \approx 0$, the longitudinal component can be expressed as

$$\begin{aligned}\tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel,t;\alpha,\beta) &= \left[\cos\left(\tfrac{\beta}{2}\right)\tfrac{-\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|}(q_\parallel)e^{i\left(J\phi_{\boldsymbol{q}_\parallel}-\omega t\right)} + \sin\left(\tfrac{\beta}{2}\right)\tfrac{-\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell'|}(q_\parallel)e^{i\left(J'\phi_{\boldsymbol{q}_\parallel}-\omega t+\alpha\right)}\right]\hat{\boldsymbol{z}} \\ &\approx \underbrace{G^{+}_{|\ell|,|\ell'|}(q_\parallel;\beta)\cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)}_{\text{Envelope}} \times \underbrace{e^{i\left(\bar{J}\phi_{\boldsymbol{q}_\parallel}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}}_{\text{Carrier}}\hat{\boldsymbol{z}},\end{aligned} \tag{2.43}$$

where the corresponding indices are shown in Table 2.2.

Table 2.2：Longitudinal Wave Beating Component Mapping

Wave Beating	**VVB**
Amplitude	
$\lvert\tilde{A}\rvert \pm \lvert\tilde{A}'\rvert$	$G^{\pm}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \equiv \left\lvert -\frac{\sin\theta_q}{\sqrt{2}}\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\lvert\ell\rvert}(q_\parallel)\right\rvert \pm \left\lvert -\frac{\sin\theta_q}{\sqrt{2}}\sin\left(\frac{\beta}{2}\right)\tilde{F}_{\lvert\ell'\rvert}(q_\parallel)\right\rvert$
Wavenumber	
$\Delta k \equiv k'-k, \quad \bar{k}\equiv\frac{k'+k}{2}$	$\Delta J \equiv J'-J, \quad \bar{J}\equiv\frac{J'+J}{2}$
Relative phase	
$\alpha' \equiv \alpha + \mathrm{Arg}[\tilde{A}'] - \mathrm{Arg}[\tilde{A}]$	$\alpha_{\lvert\ell\rvert,\lvert\ell'\rvert} \equiv \alpha + \mathrm{Arg}[\tilde{F}_{\lvert\ell'\rvert}(q_\parallel)] + \mathrm{Arg}[\tilde{F}_{\lvert\ell\rvert}(q_\parallel)] = \alpha - \frac{\pi}{2}(\lvert\ell'\rvert+\lvert\ell\rvert)$

For the purpose of simulation, we retain the previously neglected term $G^{-}_{|\ell|,|\ell'|}(q_\parallel;\beta)$ and obtain the exact longitudinal component as

$$\begin{aligned}\tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel,t;\alpha,\beta) = \Bigg[&G^{+}_{|\ell|,|\ell'|}(q_\parallel;\beta)\cos\left(\tfrac{\Delta J}{2}\phi_{\boldsymbol{q}_\parallel}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right) \\ &- i\underbrace{G^{-}_{|\ell|,|\ell'|}(q_\parallel;\beta)}_{\text{Previously neglected}}\sin\left(\tfrac{\Delta J}{2}\phi_{\boldsymbol{q}_\parallel}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\Bigg] \times e^{i\left(\bar{J}\phi_{\boldsymbol{q}_\parallel}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}\hat{\boldsymbol{z}}.\end{aligned} \tag{2.44}$$

The time-evolving field of $\mathrm{Re}\left[\tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel,t;\alpha,\beta)\right]$ for different VVB cases can be accessed via the QR code in Figure 2.12, where Figure 2.13 shows only the instantaneous field at $\omega t = 0$. In the animation, the field behaves as:

$$\begin{aligned}\text{rotating field}: \bar{J} \neq 0,\\ \text{standing field}: \bar{J} = 0.\end{aligned}$$

Similar to the beating wave, this behavior is depicted by the $\phi_{\boldsymbol{q}_\parallel}$-dependent phase factor in the carrier wave in Eq. (2.43).

Figure 2.12：Link to the time evolution animation for Figure 2.13.

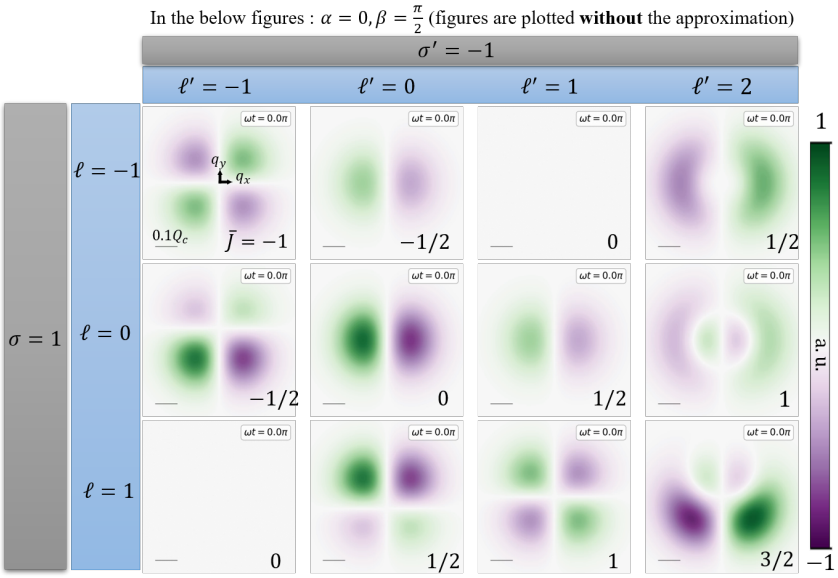


Figure 2.13 : Based on Eq. (2.44), distribution of $\mathrm{Re}\big[\tilde{\boldsymbol{\mathcal{A}}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel}, t; \alpha = 0, \beta = \frac{\pi}{2})\big]$ at $\omega t = 0$, with $\bar{J}$ indicated at the bottom right. Animation accessible via Figure 2.12.

On the other hand, the intensity, given by the squared absolute value, is depicted by the envelope function

$$\left|\tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel}; \alpha, \beta)\right|^2 \approx \Big|\underbrace{G^{+}_{|\ell|,|\ell'|}(q_{\parallel}; \beta)\cos\left(\tfrac{\Delta J}{2}\phi_{\boldsymbol{q}_{\parallel}} + \tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)}_{\text{Envelope}}\Big|^2. \tag{2.45}$$

Thus, the node structure of intensity shown in Figure 2.14 satisfies :

$$\text{number of nodes} = |\Delta J|.$$

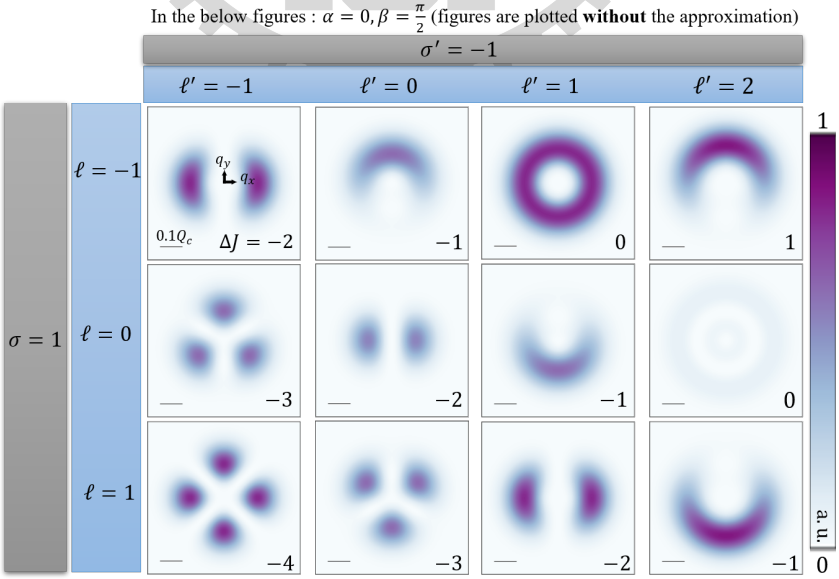


Figure 2.14 : Based on Eq. (2.44), distribution of $\left|\tilde{\boldsymbol{\mathcal{A}}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel}; \alpha = 0, \beta = \frac{\pi}{2})\right|^2$, with ΔJ indicated at the bottom right.

This structure is described by the envelope function in Eq. (2.43). Note that when the approximation, $G^{+}_{|\ell|,|\ell'|}(q_{\parallel}) \gg G^{-}_{|\ell|,|\ell'|}(q_{\parallel}) \approx 0$, is not perfectly satisfied, the nodes of the envelope will not be exactly zero, which is described by Eq. (2.44).

Finally, by rewriting the Eq (2.45) as

$$\left|\tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta)\right|^2 \approx \left|G^{+}_{|\ell|,|\ell'|}(q_{\parallel};\beta)\cos\left[\tfrac{\Delta J}{2}\left(\phi_{\boldsymbol{q}_{\parallel}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{\Delta J}\right)\right]\right|^2, \tag{2.46}$$

and focusing on the variation with α, we can know how the intensity changes with the relative phase α within $\alpha_{|\ell|,|\ell'|} = \alpha - \frac{\pi}{2}(|\ell'|+|\ell|)$ as

$$\text{the node structure is rotated (counter-clockwise) by } -\frac{\alpha}{\Delta J}.$$

Figure 2.15 depicts this change as the four cases shown in the upper row are the same as the upper-row cases in Figure 2.14. The different rows depict the intensity distributions with different α values and show that the node structure is rotated (counter-clockwise) by $-\frac{\alpha_{|\ell|,|\ell'|}}{\Delta J}$.

However, when $\Delta J = 0$, the two twisted light fields have the same TAM, and thus they are coherent and exhibit constructive/destructive interference across the whole $\boldsymbol{Q}$ plane. This behavior can be seen in Eq. (2.45), where the envelope is independent of $\phi_{\boldsymbol{q}_{\parallel}}$ and the relative phase α tunes the overall magnitude.

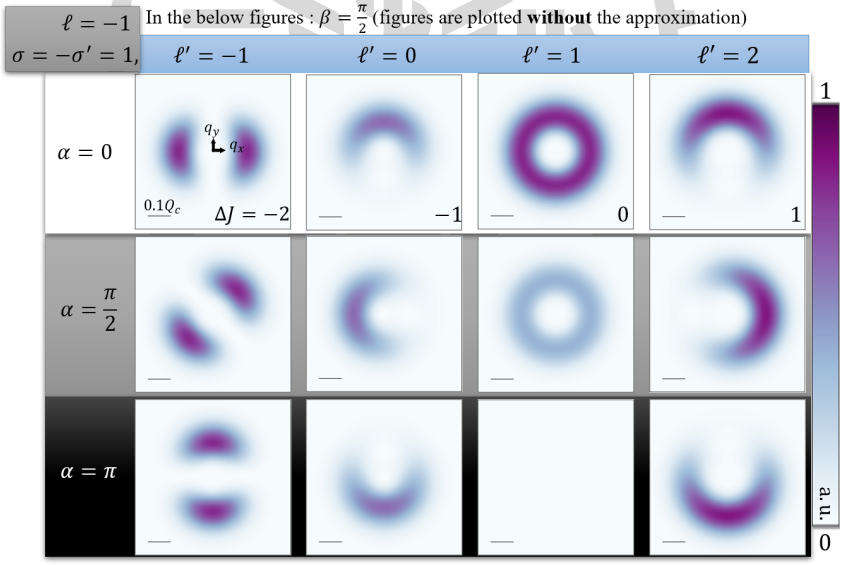


Figure 2.15 : Based on Eq. (2.44), distribution of $\left|\tilde{\boldsymbol{\mathcal{A}}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel};\alpha=0,\beta=\frac{\pi}{2})\right|^2$, with ΔJ indicated at the bottom right.

2.2.2 Analogy between 2D Wave Beating and the Transverse Component

To describe the transverse component of VVBs, we need to consider the additional polarization components of light. When superposing two twisted light fields, we can rewrite the different polarizations as

$$\hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}} \longrightarrow (\hat{\boldsymbol{x}} + \bar{\sigma}\hat{\boldsymbol{y}}) - i\tfrac{\Delta\sigma}{2}\hat{\boldsymbol{y}}$$
$$\hat{\boldsymbol{x}} + i\sigma'\hat{\boldsymbol{y}} \longrightarrow (\hat{\boldsymbol{x}} + \bar{\sigma}\hat{\boldsymbol{y}}) + i\tfrac{\Delta\sigma}{2}\hat{\boldsymbol{y}},$$

where the difference and average SAM are

$$\Delta\sigma \equiv \sigma' - \sigma, \text{ and } \bar{\sigma} \equiv \tfrac{\sigma'+\sigma}{2}.$$

Since we are interested in manipulating the polarization, it is natural to consider light with opposite SAM. In this case, we can observe that for different SAM

$$\Delta\sigma = \pm 2, \bar{\sigma} = 0,$$

the $\hat{\boldsymbol{x}}$ components of the polarizations remain the same, as in 1D wave beating (additive mixing), while the $\hat{\boldsymbol{y}}$ components differ in sign (subtractive mixing). The wave beating of the two components below is shown under the approximation : $|\tilde{A}| + |\tilde{A}'| \gg |\tilde{A}| - |\tilde{A}'|$.

Along the $\hat{\boldsymbol{x}}$ axis (additive mixing):

$$\mathrm{Re}\big[\tilde{A}e^{i(k\phi-\omega t)} + \tilde{A}'e^{i(k'\phi-\omega t+\alpha)}\big]\hat{\boldsymbol{x}}$$
$$\approx \mathrm{Re}\left\{\left(|\tilde{A}| + |\tilde{A}'|\right)\cos\left(\tfrac{\Delta k}{2}\phi + \tfrac{\alpha}{2}\right)e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}+\frac{1}{2}(\mathrm{Arg}[\tilde{A}']+\mathrm{Arg}[\tilde{A}])\right)}\right\}\hat{\boldsymbol{x}}$$

Along the $\hat{\boldsymbol{y}}$ axis (subtractive mixing):

$$\mathrm{Re}\big[-i\tfrac{\Delta\sigma}{2}\tilde{A}e^{i(k\phi-\omega t)} + i\tfrac{\Delta\sigma}{2}\tilde{A}'e^{i(k'\phi-\omega t+\alpha)}\big]\hat{\boldsymbol{y}}$$
$$\approx \mathrm{Re}\left\{-\tfrac{\Delta\sigma}{2}\left(|\tilde{A}| + |\tilde{A}'|\right)\sin\left(\tfrac{\Delta k}{2}\phi + \tfrac{\alpha}{2}\right)e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}+\frac{1}{2}(\mathrm{Arg}[\tilde{A}']+\mathrm{Arg}[\tilde{A}])\right)}\right\}\hat{\boldsymbol{y}}$$

In 2D:

$$\mathrm{Re}\left[\tilde{A}e^{i(k\phi-\omega t)}\left(\hat{\boldsymbol{x}} - i\tfrac{\Delta\sigma}{2}\hat{\boldsymbol{y}}\right) + \tilde{A}'e^{i(k'\phi-\omega t+\alpha)}\left(\hat{\boldsymbol{x}} + i\tfrac{\Delta\sigma}{2}\hat{\boldsymbol{y}}\right)\right]$$
$$\approx \mathrm{Re}\Bigg\{\underbrace{\left(|\tilde{A}| + |\tilde{A}'|\right)\left[\cos\left(\tfrac{\Delta k}{2}\phi + \tfrac{\alpha}{2}\right)\hat{\boldsymbol{x}} - \tfrac{\Delta\sigma}{2}\sin\left(\tfrac{\Delta k}{2}\phi + \tfrac{\alpha}{2}\right)\hat{\boldsymbol{y}}\right]}_{\text{Envelope in 2D (polarization)}}\underbrace{e^{i\left(\bar{k}\phi-\omega t+\frac{\alpha}{2}+\frac{1}{2}(\mathrm{Arg}[\tilde{A}']+\mathrm{Arg}[\tilde{A}])\right)}}_{\text{Carrier}}\Bigg\}$$

By analogy with VVBs, under the approximation $H^{+}_{|\ell|,|\ell'|}(q_\parallel;\beta) \gg H^{-}_{|\ell|,|\ell'|}(q_\parallel;\beta) \approx 0$, the transverse component can be expressed as

$$\begin{aligned}
&\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel, t;\alpha,\beta)\\
&= \tfrac{1}{\sqrt{2}}\cos\left(\tfrac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel)e^{i(\ell\phi_{\boldsymbol{q}_\parallel}-\omega t)}\left(\hat{\boldsymbol{x}} - i\tfrac{\Delta\sigma}{2}\hat{\boldsymbol{y}}\right) + \tfrac{1}{\sqrt{2}}\sin\left(\tfrac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)e^{i(\ell'\phi_{\boldsymbol{q}_\parallel}-\omega t+\alpha)}\left(\hat{\boldsymbol{x}} + i\tfrac{\Delta\sigma}{2}\hat{\boldsymbol{y}}\right)\\
&\approx \underbrace{H^{+}_{|\ell|,|\ell'|}(q_\parallel;\beta)\left[\cos\left(\tfrac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel} + \tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\hat{\boldsymbol{x}} - \tfrac{\Delta\sigma}{2}\sin\left(\tfrac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel} + \tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\hat{\boldsymbol{y}}\right]}_{\text{Polarization}} \times \underbrace{e^{i\left(\bar{\ell}\phi_{\boldsymbol{q}_\parallel}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}}_{\text{Carrier}}
\end{aligned} \tag{2.47}$$

where the corresponding indices are shown in Table 2.3.

Table 2.3：Transverse Wave Beating Component Mapping

Wave Beating	**VVB**
Amplitude	
$\lvert\tilde{A}\rvert \pm \lvert\tilde{A}'\rvert$	$H^{\pm}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \equiv \left\lvert \frac{1}{\sqrt{2}} \cos\left(\frac{\beta}{2}\right) \tilde{F}_{\lvert\ell\rvert}(q_\parallel) \right\rvert \pm \left\lvert \frac{1}{\sqrt{2}} \sin\left(\frac{\beta}{2}\right) \tilde{F}_{\lvert\ell'\rvert}(q_\parallel) \right\rvert$
Wavenumber	
$\Delta k \equiv k' - k, \quad \bar{k} \equiv \frac{k'+k}{2}$	$\Delta\ell \equiv \ell' - \ell, \quad \bar{\ell} \equiv \frac{\ell'+\ell}{2}$

For the purpose of simulation, we retain the previously neglected term $H^{-}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta)$ and obtain the exact transverse component as

$$\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel,t;\alpha,\beta) = \Bigg\{ H^{+}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \left[\cos\left(\tfrac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel} + \tfrac{\alpha_{\lvert\ell\rvert,\lvert\ell'\rvert}}{2}\right)\hat{\boldsymbol{x}} - \tfrac{\Delta\sigma}{2}\sin\left(\tfrac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel} + \tfrac{\alpha_{\lvert\ell\rvert,\lvert\ell'\rvert}}{2}\right)\hat{\boldsymbol{y}}\right]$$
$$- i \underbrace{H^{-}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta)}_{\text{Previously neglected}} \left[\sin\left(\tfrac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel} + \tfrac{\alpha_{\lvert\ell\rvert,\lvert\ell'\rvert}}{2}\right)\hat{\boldsymbol{x}} + \tfrac{\Delta\sigma}{2}\cos\left(\tfrac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel} + \tfrac{\alpha_{\lvert\ell\rvert,\lvert\ell'\rvert}}{2}\right)\hat{\boldsymbol{y}}\right] \Bigg\}$$
$$\times e^{i\left(\bar{\ell}\phi_{\boldsymbol{q}_\parallel} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(\lvert\ell'\rvert+\lvert\ell\rvert)\right)} \tag{2.48}$$

When the approximation is valid but not perfectly satisfied, the linear polarization described by Eq. (2.47) becomes elliptical, as described by Eq. (2.48). Both approximations,

$$H^{+}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \gg H^{-}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \approx 0 \text{ and } G^{+}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \gg G^{-}_{\lvert\ell\rvert,\lvert\ell'\rvert}(q_\parallel;\beta) \approx 0,$$

ultimately representing the same idea, namely

$$\lvert\cos\left(\tfrac{\beta}{2}\right) F_{\lvert\ell\rvert}(q_\parallel)\rvert \approx \lvert\sin\left(\tfrac{\beta}{2}\right) F_{\lvert\ell'\rvert}(q_\parallel)\rvert.$$

The time-evolving field of $\mathrm{Re}\left[\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel,t;\alpha,\beta)\right]$ for different VVB cases can be accessed via the QR code in Figure 2.16, where Figure 2.17 shows only the field at $\omega t = 0$. In the animation, the transverse field (magnitude of the instantaneous polarization, orange arrow) behaves as:

$$\text{rotating field} : \bar{\ell} \neq 0,$$
$$\text{standing field} : \bar{\ell} = 0.$$

In other words, the polarization remains stationary across the whole $\phi_{\boldsymbol{q}_\parallel}$ angle when the average OAM index $\bar{\ell}$ is zero. Similar to the beating wave, this behavior is depicted by the $\phi_{\boldsymbol{q}_\parallel}$ dependent phase factor in the carrier wave in Eq. (2.47).

Figure 2.16：Link to the time evolution animation for Figure 2.17.

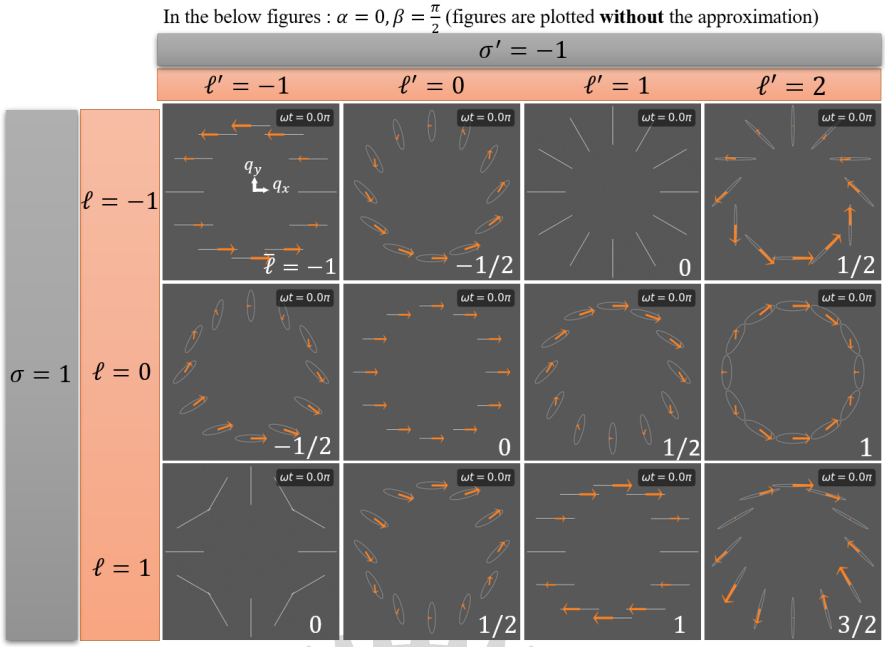


Figure 2.17 : Based on Eq. (2.48), polarization and instantaneous polarization distributions of $\mathrm{Re}\left[\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel}, t; \alpha = 0, \beta = \frac{\pi}{2})\right]$ at $\omega t = 0$, with $\bar{\ell}$ indicated at the bottom right. Animation accessible via Figure 2.16.

On the other hand, the polarization (white trajectory) is depicted by the envelope function, where

$$\text{Winding of light} = \Delta\ell.$$

Figure 2.19 shows that the winding of the light is determined by $\Delta\ell$.

Figure 2.18 : Link to the time evolution animation for Figure 2.19.

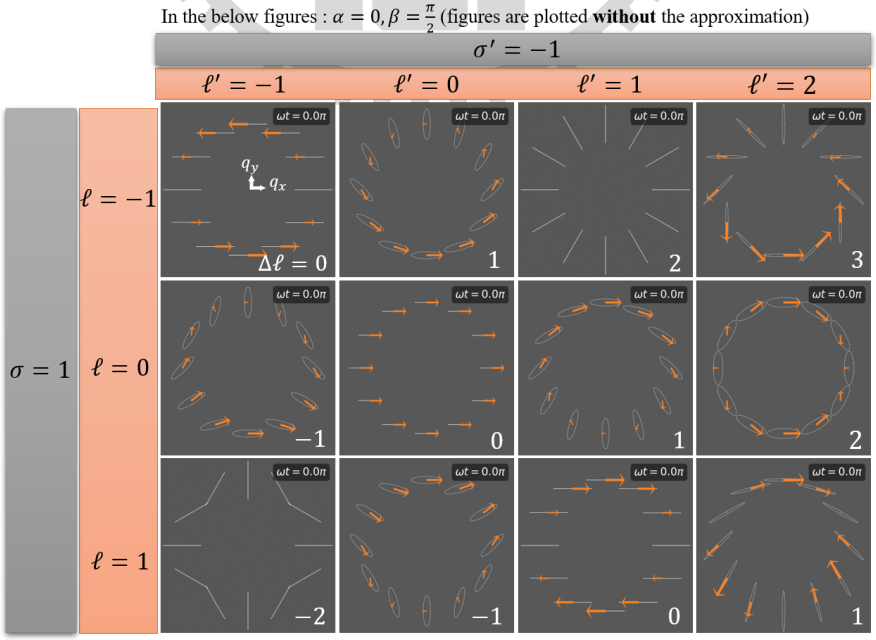


Figure 2.19 : Same as Figure 2.17, with $\Delta\ell$ indicated at the bottom right. Animation accessible via Figure 2.18

Finally, by changing α within $\alpha_{|\ell|,|\ell'|}$, each polarization

$$\text{rotated by (counter-clockwise) } \frac{\alpha}{2}.$$

This can be seen in Figure 2.21, where the four VVBs in the upper row are the same as those in the upper row of Figure 2.17. The different rows depict the polarization distributions with different α values.

Figure 2.20：Link to the time evolution animation for Figure 2.21.

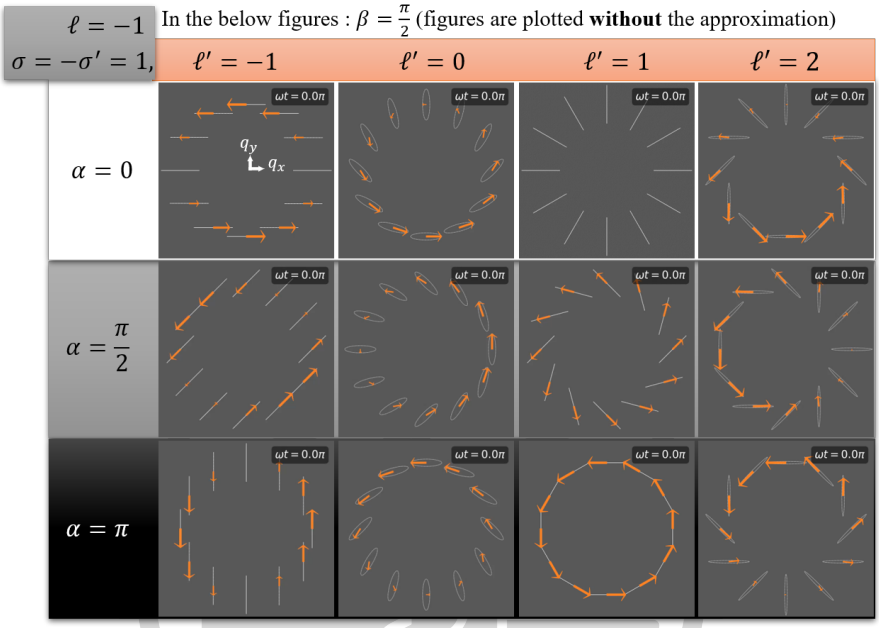


Figure 2.21：Based on Eq. (2.48), polarization and instantaneous polarization distributions of $\mathrm{Re}\left[\tilde{\mathcal{A}}^{\parallel}_{1,-1,-1,\ell'}(\boldsymbol{q}_{\parallel}, t; \alpha, \beta = \pi/2)\right]$ at $\omega t = 0$ for various α. Animation accessible via Figure 2.20

In conclusion, the superposition of two twisted light beams is analogous to wave beating, consisting of an envelope function and a time-dependent carrier wave, whose features are described by the difference and average of the wavenumbers, Δk and $\bar{k}$. Whether the VVB exhibits a rotating or standing field is determined by the average angular momentum $\bar{J}$ or $\bar{\ell}$, depending on the component. On the other hand, the number of nodes in the intensity distribution of the longitudinal component is determined by the difference in TAM ΔJ. For the transverse component, the difference in OAM $\Delta\ell$ determines the winding of light. Both of these conclusions hold when the approximation $|\cos\left(\frac{\beta}{2}\right) F_{|\ell|}(q_{\parallel})| \approx |\sin\left(\frac{\beta}{2}\right) F_{|\ell'|}(q_{\parallel})|$ is well satisfied.

In this sense, we can select specific OAM and SAM such that their beating effect determines both the winding of the light and the node structure in the intensity distribution of the longitudinal component. In the next chapter, we take advantage of this beating effect to selectively excite excitons with finite center-of-mass momentum.

Chapter 3 Light-Matter Interaction

3.1 Theory of Light–Matter Interaction

3.1.1 Fermi's golden rule

In this section, we examine the interaction between light and matter using **Fermi's golden rule**. The transition rate for an optical excitation from the ground state $|GS\rangle$, with fully occupied valence bands, to an exciton state $|\Psi^{S,\boldsymbol{Q}}\rangle$ with band index S and center-of-mass (COM) momentum $\boldsymbol{Q}$ is given by

$$\Gamma = \frac{2\pi}{\hbar}\left|\langle\Psi^{S,\boldsymbol{Q}}|\hat{H}_I|GS\rangle\right|^2 \delta(E^{S,\boldsymbol{Q}} - \hbar\omega). \tag{3.1}$$

The delta function enforces energy conservation between the exciton energy $E^{S,\boldsymbol{Q}}$ and the photon energy $\hbar\omega$. Within the pseudospin model [1, 7, 41], the exciton state is written as

$$|\Psi^{S,\boldsymbol{Q}}\rangle = \frac{1}{\sqrt{\Omega}}\sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})\hat{c}^{\dagger}_{c,\boldsymbol{k}+\boldsymbol{Q}}\hat{h}^{\dagger}_{v,-\boldsymbol{k}}|GS\rangle. \tag{3.2}$$

Here, the particle operators $\hat{c}^{\dagger}_{c,\boldsymbol{k}}$ and $\hat{h}^{\dagger}_{v,-\boldsymbol{k}}$ create an electron in the conduction band with crystal momentum $\boldsymbol{k}$ and a hole in the valence band with crystal momentum $-\boldsymbol{k}$, respectively. The coefficient $\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ represents the amplitude of the electron–hole configuration for exciton band S, and Ω is the area of the TMD monolayer. The matrix element in Eq. (3.1) can be expressed, within the **rotating wave approximation** [42], as

$$\langle\Psi^{S,\boldsymbol{Q}}|\hat{H}_I|GS\rangle = \frac{1}{\sqrt{\Omega}}\sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}*}(vc\boldsymbol{k})\langle\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}|H_I|\psi_{v,\boldsymbol{k}}\rangle, \tag{3.3}$$

where, for weak fields, the time-independent part of light–matter interaction Hamiltonian is given by [1, 7]

$$H_I = \frac{|e|}{2m_e}\boldsymbol{A}(\boldsymbol{r})\cdot\boldsymbol{p},$$

with $\boldsymbol{A}(\boldsymbol{r})$ the vector potential in the **Coulomb gauge**, $\boldsymbol{p}$ the momentum operator, and m_e (e) the electron mass (charge). If the vector potential is expanded in a **plane-wave basis** as in

Eq. (2.31), the matrix element in Eq. (3.3) becomes

$$\begin{aligned}\langle\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}|H_I|\psi_{v,\boldsymbol{k}}\rangle &= \frac{|e|}{2m_e}\sum_{\boldsymbol{q}_\parallel}\left[\boldsymbol{\mathcal{A}}(\boldsymbol{q}_\parallel)\cdot\langle\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}|e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}}\hat{\boldsymbol{p}}|\psi_{v,\boldsymbol{k}}\rangle\right] \\ &\approx \frac{|e|}{2m_e}\sum_{\boldsymbol{q}_\parallel}\left[\boldsymbol{\mathcal{A}}(\boldsymbol{q}_\parallel)\cdot\delta_{\boldsymbol{q}_\parallel,\boldsymbol{Q}}\langle\psi_{c,\boldsymbol{k}}|\hat{\boldsymbol{p}}|\psi_{v,\boldsymbol{k}}\rangle\right],\end{aligned} \tag{3.4}$$

where the approximation follows from the **electric dipole approximation**. In this approximation, the Bloch state wave functions are written as $\psi_{n,\boldsymbol{k}}(\boldsymbol{r}) = u_{n,\boldsymbol{k}}(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, and the plane-wave components combine to yield the Kronecker delta $\delta_{\boldsymbol{q}_\parallel,\boldsymbol{Q}}$, enforcing the equality between the in-plane wavevector and the exciton COM momentum.

Substituting this result back, the matrix element in Eq. (3.3) [1, 7] becomes

$$\langle\Psi^{S,\boldsymbol{Q}}|H_I|GS\rangle = \frac{E_g}{2i\hbar}\left(\boldsymbol{\mathcal{A}}(\boldsymbol{Q})\cdot\boldsymbol{D}^{S,\boldsymbol{Q}*}\right), \tag{3.5}$$

where superscript * denotes complex conjugation, E_g is the band gap at the K/K' valleys and $\boldsymbol{D}^{S,\boldsymbol{Q}}$ is the exciton transition dipole moment , which is defined as

$$\boldsymbol{D}^{S,\boldsymbol{Q}} = \frac{e\hbar}{im_0E_g\sqrt{\Omega}}\sum_{vc\boldsymbol{k}}\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})\langle\psi_{v,\boldsymbol{k}}|\boldsymbol{p}|\psi_{c,\boldsymbol{k}}\rangle \tag{3.6}$$

The transition dipole moment obtained from density functional theory combined with the Bethe–Salpeter equation, is given in the small-$\boldsymbol{Q}$ limit by Ref [1, 7]:

Table 3.1 ： Table of transition dipole moments with corresponding exciton types.

Exciton Type	**Transition Dipole Moment**
Valley excitons (bright)	$\boldsymbol{D}^{K,\boldsymbol{Q}} = \frac{D^B}{\sqrt{2}}(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}),$ $\boldsymbol{D}^{K',\boldsymbol{Q}} = \frac{D^B}{\sqrt{2}}(\hat{\boldsymbol{x}} - i\hat{\boldsymbol{y}})$
Valley-mixed excitons (bright)	$\boldsymbol{D}^{B+,\boldsymbol{Q}} = D^B\left(\cos\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{x}} + \sin\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{y}}\right),$ $\boldsymbol{D}^{B-,\boldsymbol{Q}} = iD^B\left(-\sin\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{x}} + \cos\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{y}}\right)$
Gray excitons	$\boldsymbol{D}^{G,\boldsymbol{Q}} = D^G\,\hat{\boldsymbol{z}}.$

Here, the subscript symbols K/K', $B+/B-$, and G denote the K/K'valley bright, upper/ lower band bright, and gray exciton, respectively. The magnitude of the dipole moment of BXs (GX) is D^B (D^G) [1, 7].

3.2 Manifestation of Excitonic Optical Transitions with Beating in VVBs

Under the excitation of VVBs, the corresponding Fermi's golden rule can then be written as [1]

$$\Gamma^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 \rho(\hbar\omega = E^{S,\boldsymbol{Q}}), \tag{3.7}$$

where the optical matrix element is defined as

$$\begin{aligned}\tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} &\equiv \frac{1}{\sqrt{\Omega}}\sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}*}(vc\boldsymbol{k})\left\langle \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}\left|\frac{|e|}{2m_e}\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\cdot\boldsymbol{p}\right|\psi_{v,\boldsymbol{k}}\right\rangle \\ &\approx \frac{E_g}{2i\hbar}\left(\tilde{\boldsymbol{\mathcal{A}}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\cdot\boldsymbol{D}^{S,\boldsymbol{Q}*}\right).\end{aligned} \tag{3.8}$$

Here we can see that the transition is strongly determined by the inner product of the vector potential and the transition dipole moment, which ultimately involves the beating of the two twisted light beams that determines the weighting of different plane-wave components and thus the structure of the VVB in the $\boldsymbol{Q}$ plane.

3.2.1 Optical Matrix Element and Transition Rate for Gray Exciton

Let us start from the simple case, the GX, whose transition dipole moment is approximately isotropic across the $\boldsymbol{Q}$ plane as shown in Table 3.1, which means the optical matrix element will reflect the pattern of the light. With Eq. (3.8), we can derive the time dependent optical matrix elements for GX as

$$\begin{aligned}\tilde{M}^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta) &= \frac{E_g}{2i\hbar}\left(\tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta)\cdot\boldsymbol{D}^{G,\boldsymbol{Q}*}\right) \\ &\approx \underbrace{\frac{E_g}{2i\hbar}G^{+}_{|\ell|,|\ell'|}(Q;\beta)\cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)}_{\text{Envelope}}\times\underbrace{e^{i\left(\bar{J}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}}_{\text{Carrier}}\underbrace{(\hat{\boldsymbol{z}}\cdot D^{G}\hat{\boldsymbol{z}})}_{\text{Constant}}\end{aligned} \tag{3.9}$$

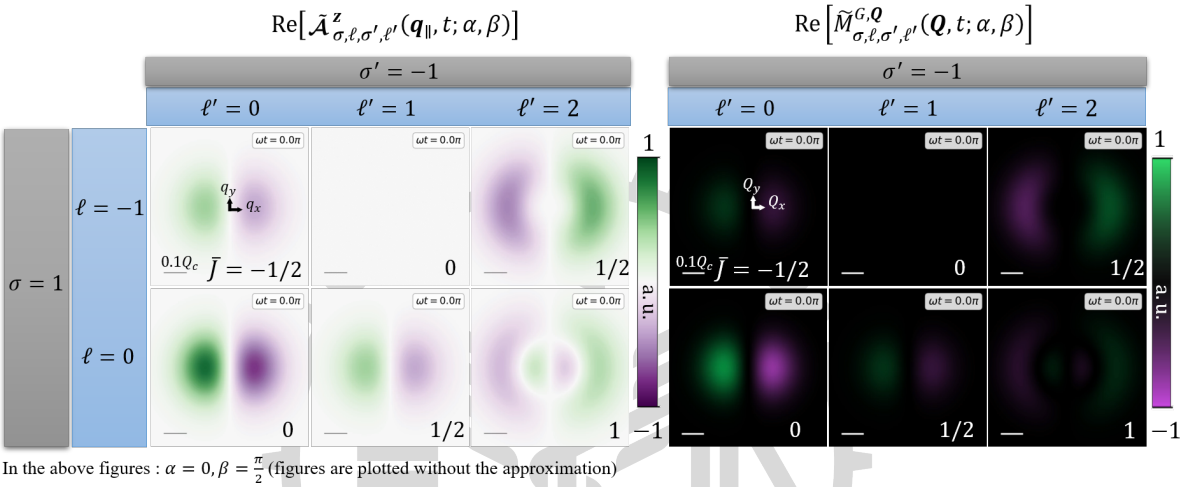
For the purpose of simulation, the exact optical matrix element for GX without the approximation, $G^{+}_{|\ell|,|\ell'|}(q_{\parallel};\beta) \gg G^{-}_{|\ell|,|\ell'|}(q_{\parallel};\beta) \approx 0$ is obtained as

$$\begin{aligned}\tilde{M}^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta) = \frac{E_g D^G}{2i\hbar}\Bigg[&G^{+}_{|\ell|,|\ell'|}(Q;\beta)\cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \\ &-\underbrace{iG^{-}_{|\ell|,|\ell'|}(Q;\beta)}_{\text{Previously neglected}}\sin\left(\frac{\Delta J}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)\Bigg]e^{i\left(\bar{J}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}\end{aligned} \tag{3.10}$$

As shown in Eq. (3.9), the optical matrix element for GX reflects the pattern of VVB's longitudinal component. In comparison, the real part of the field and optical matrix element is shown in Figure 3.2 and the animation can be accessed via the QR code in Figure 3.1 Whether the pattern rotates is controlled by the average TAM $\bar{J}$ in the carrier of the light in Eq. (2.43), which is carried over to the carrier in Eq. (3.9). Therefore,

Figure 3.1 : Link to the time evolution animation for Figure 3.2.

$$\text{rotating field} : \bar{J} \neq 0,$$
$$\text{standing field} : \bar{J} = 0.$$



In the above figures : $\alpha = 0, \beta = \frac{\pi}{2}$ (figures are plotted without the approximation)

Figure 3.2 : Based on Eq. (2.44) and Eq. (3.10), distribution of $\mathrm{Re}\left[\tilde{\mathcal{A}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel,t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (left panel) and $\mathrm{Re}\left[\tilde{M}^{G,\boldsymbol{Q}}_{1,\ell,-1,\ell'}(\boldsymbol{Q},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (right panel) at $\omega t = 0$, with $\bar{J}$ indicated at the bottom right. Animation accessible via Figure 3.1.

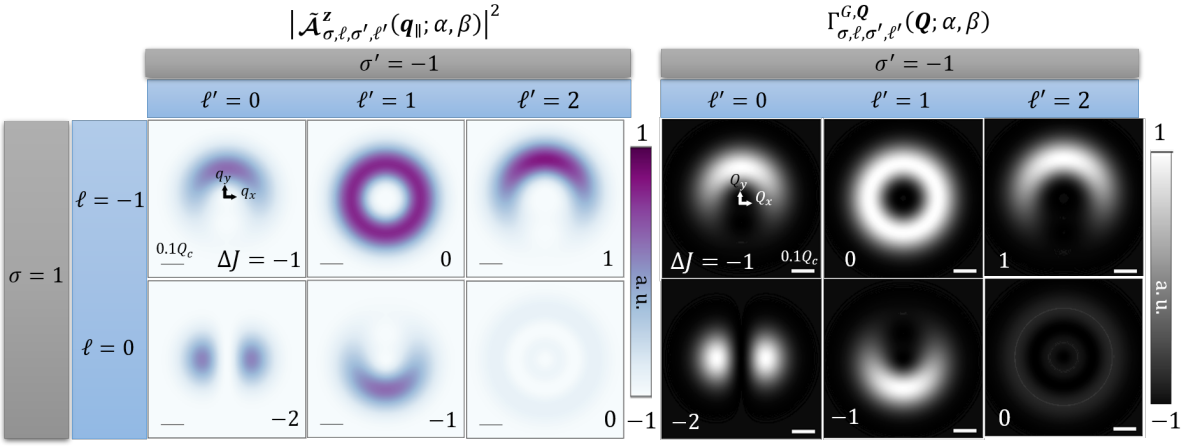
On the other hand, transition rate (squared absolute value) is determined by the envelope function:

$$\Gamma^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta) \propto \left|\tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\right|^2 \approx \underbrace{\left|\frac{E_g D^G}{2i\hbar} G^{+}_{|\ell|,|\ell'|}(Q;\beta)\cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{q}_\parallel} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right|^2}_{\text{Envelope}} \tag{3.11}$$

Thus, the node structure of transition rate shown in Figure 3.3 follows the light's pattern :

$$\text{number of nodes} = |\Delta J|.$$

Interestingly, when number of nodes $\Delta J = \pm 1$, we can create a asymmetry pattern in the $\boldsymbol{Q}$ plane thus selectively excite GXs with specific COM momentum. Eventually create GXs that may driven in specific directions.



In the above figures : $\alpha = 0, \beta = \frac{\pi}{2}$ (figures are plotted without the approximation)

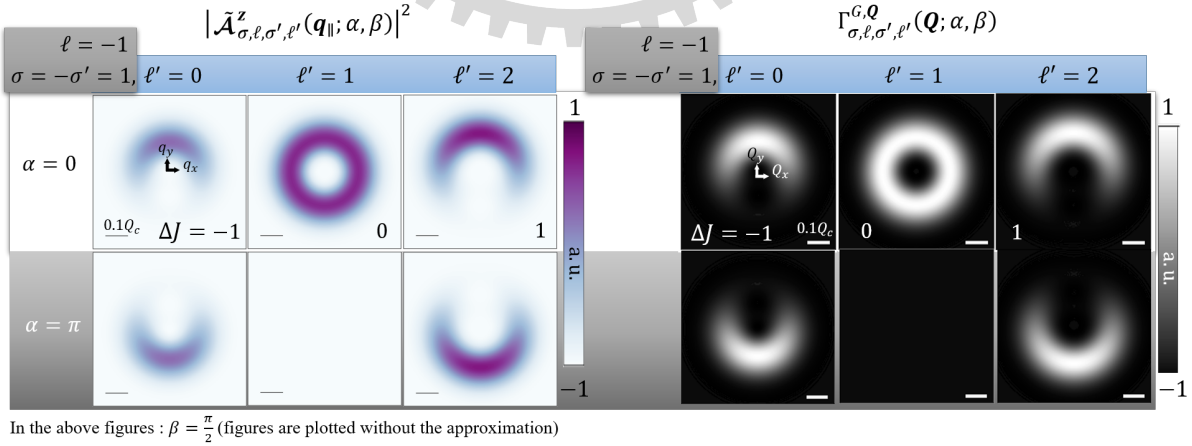
Figure 3.3 : Based on Eq. (2.44) and Eq. (3.10), distribution of $\left|\tilde{\mathcal{A}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel, t; \alpha = 0, \beta = \frac{\pi}{2})\right|^2$(left panel) and $\Gamma^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)$(right panel), with ΔJ indicated at the bottom right.

Similar to the longitudinal component of VVBs, the node structure of transition rate is rotated by the relative phase α of the light within $\alpha_{|\ell|,|\ell'|}$. This is depicted in Figure 3.4, where the three cases shown in the upper row (both light and transition rate) are the same as those in the upper row in Figure 3.3. Focusing on the variation with α, the node structure is

$$\text{rotated (counter-clockwise) by } -\frac{\alpha}{\Delta J}.$$

For $\Delta J = 0$, two coherent twisted light fields interfere constructively/destructively across the whole $\boldsymbol{Q}$ plane with different α values, and so does the transition rate. In this sense,

$$\Delta J = 0 : \alpha \text{ switch the transition of gray exciton}$$



In the above figures : $\beta = \frac{\pi}{2}$ (figures are plotted without the approximation)

Figure 3.4 : Based on Eq. (2.44) and Eq. (3.10), distribution of $\left|\tilde{\mathcal{A}}^{z}_{1,-1,-1,\ell'}(\boldsymbol{q}_\parallel, t; \alpha, \beta = \frac{\pi}{2})\right|^2$(left panel) and $\Gamma^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)$(right panel), with $-\frac{\alpha}{\Delta J}$ indicated at the bottom right.

3.2.2 Optical Matrix Element and Transition Rate for Bright Exciton

Bright excitons, whose transition dipole moments are oriented in the in-plane direction will be excited by the transverse component of the VVBs. With Eq. (3.8), the alignment between polarization and transition dipole moment is depicted by their inner product within the optical matrix elements as shown below

$$\tilde{M}^{B\pm,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta)=\frac{E_g}{2i\hbar}\left(\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta)\cdot\boldsymbol{D}^{B\pm,\boldsymbol{Q}*}\right)$$

$$\approx\frac{E_g}{2i\hbar}H^{+}_{|\ell|,|\ell'|}(Q;\beta)\times\underbrace{\left\{\left[\cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)\widehat{\boldsymbol{x}}-\frac{\Delta\sigma}{2}\sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)\widehat{\boldsymbol{y}}\right]\cdot\boldsymbol{D}^{B\pm,\boldsymbol{Q}*}\right\}}_{\text{Product of polarization and }\boldsymbol{D}^{B\pm,\boldsymbol{Q}*}}$$

$$\times e^{i\left(\bar{\ell}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)} \tag{3.12}$$

By explicitly evaluating the dot product, the time-dependent optical matrix elements are

$$\tilde{M}^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta)$$

$$\approx\underbrace{\frac{E_gD^B}{4i\hbar}H^{+}_{|\ell|,|\ell'|}(Q;\beta)\left[\left(1-\tfrac{\Delta\sigma}{2}\right)\cos\left(\tfrac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)+\left(1+\tfrac{\Delta\sigma}{2}\right)\cos\left(\tfrac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]}_{\text{Envelope}}$$

$$\times\underbrace{e^{i\left(\bar{\ell}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}}_{\text{Carrier}} \tag{3.13}$$

$$\tilde{M}^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta)$$

$$\approx\underbrace{\frac{-iE_gD^B}{4i\hbar}H^{+}_{|\ell|,|\ell'|}(Q;\beta)\left[\left(1-\tfrac{\Delta\sigma}{2}\right)\sin\left(\tfrac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)-\left(1+\tfrac{\Delta\sigma}{2}\right)\sin\left(\tfrac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]}_{\text{Envelope}}$$

$$\times\underbrace{e^{i\left(\bar{\ell}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)}}_{\text{Carrier}} \tag{3.14}$$

Here, we can already see that the SAM difference, $\Delta\sigma$, determines the pattern of the envelope function for both BXs' optical matrix elements and further leads to the interaction between the winding of light, $\Delta\ell$, and the winding number of the exciton, ±2, resulting in the effective angular dependence $\Delta\ell\pm2$, which affects the pattern of the transition rate.

For the purpose of simulation, the exact optical matrix elements for the upper and lower band

BXs without the approximation, $H^{+}_{|\ell|,|\ell'|}(q_{\parallel};\beta) \gg H^{-}_{|\ell|,|\ell'|}(q_{\parallel};\beta) \approx 0$ are

$$\tilde{M}^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta) = \frac{E_g D^B}{4i\hbar}\times$$
$$\Big\{H^{+}_{|\ell|,|\ell'|}(Q;\beta)\left[\left(1-\tfrac{\Delta\sigma}{2}\right)\cos\left(\tfrac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)+\left(1+\tfrac{\Delta\sigma}{2}\right)\cos\left(\tfrac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]$$
$$-i\underbrace{H^{-}_{|\ell|,|\ell'|}(Q;\beta)}_{\text{Previously neglected}}\left[\left(1-\tfrac{\Delta\sigma}{2}\right)\sin\left(\tfrac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)+\left(1+\tfrac{\Delta\sigma}{2}\right)\sin\left(\tfrac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]\Big\}$$
$$\times\, e^{i\left(\bar{\ell}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)} \tag{3.15}$$

$$\tilde{M}^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta) = \frac{-iE_g D^B}{4i\hbar}\times$$
$$\Big\{H^{+}_{|\ell|,|\ell'|}(Q;\beta)\left[\left(1-\tfrac{\Delta\sigma}{2}\right)\sin\left(\tfrac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)-\left(1+\tfrac{\Delta\sigma}{2}\right)\sin\left(\tfrac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]$$
$$-i\underbrace{H^{-}_{|\ell|,|\ell'|}(Q;\beta)}_{\text{Previously neglected}}\left[-\left(1-\tfrac{\Delta\sigma}{2}\right)\cos\left(\tfrac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)+\left(1+\tfrac{\Delta\sigma}{2}\right)\cos\left(\tfrac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\tfrac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]\Big\}$$
$$\times\, e^{i\left(\bar{\ell}\phi_{\boldsymbol{Q}}-\omega t+\frac{\alpha}{2}-\frac{\pi}{4}(|\ell'|+|\ell|)\right)} \tag{3.16}$$

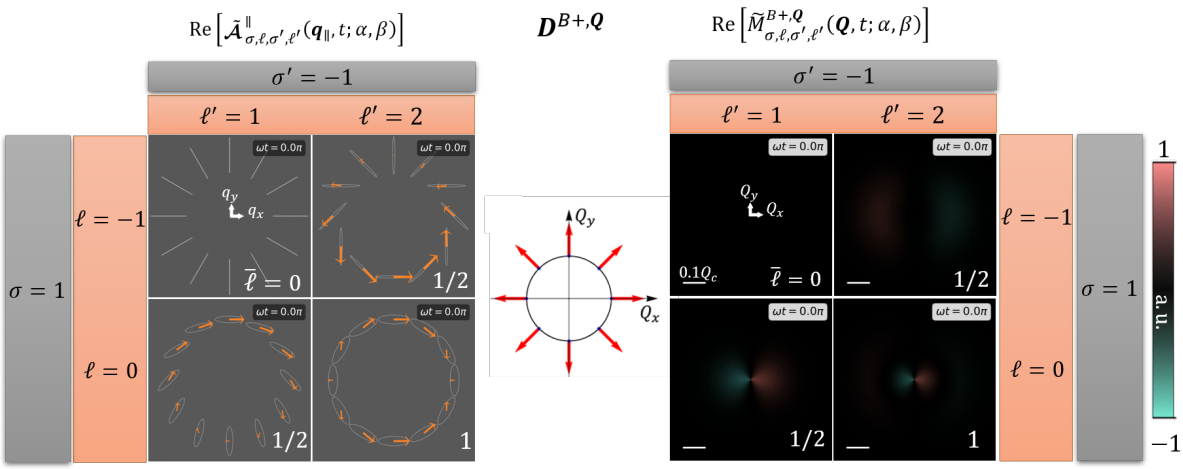
The time-evolution of $\mathrm{Re}\left[\tilde{M}^{B\pm,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q},t;\alpha,\beta)\right]$ for different VVB cases can be accessed via the QR code in Figure 3.5 and Figure 3.6 , where Figure 3.7 and Figure 3.8 show only the instantaneous results ($\omega t = 0$). In the animation, the optical matrix element behaves as:

$$\bar{\ell} \neq 0, \text{ rotating,}$$
$$\bar{\ell} = 0, \text{ standing ,}$$

which reflects the light's rotating feature. The perfect match of the polarization and transition dipole moment can be seen with the VVB with $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 1 (\Delta\sigma = -2, \Delta\ell - 2 = 0)$ shown in the top left in both Figure 3.7 and Figure 3.8, where the radial pattern of the polarization matches the "longitudinal" pattern of $\boldsymbol{D}^{B+,\boldsymbol{Q}}$ and is perpendicular to $\boldsymbol{D}^{B-,\boldsymbol{Q}}$. Though the transverse component vanished at the specific time, the time evolution can be observed via the animation, showing that only upper-band BXs can be excited, while the lower BXs cannot. This selective excitation can be observed more clearly in the pattern of the transition rate.

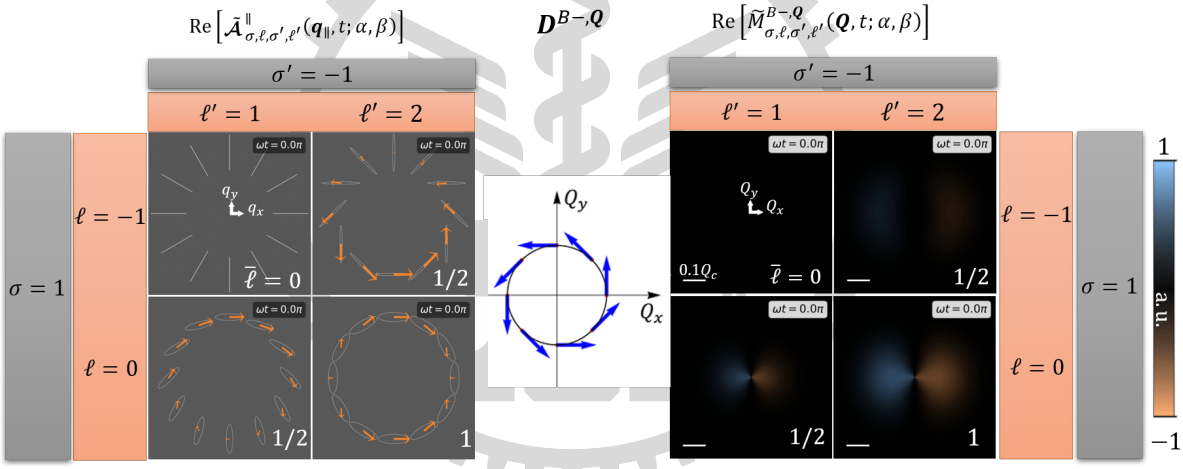
Figure 3.5 : Link to the time evolution animation for Figure 3.7.

Figure 3.6 : Link to the time evolution animation for Figure 3.8.



In the above figures : $\alpha=0,\beta=\frac{\pi}{2},\Delta\sigma=-2$ (figures are plotted **without** the approximation)

Figure 3.7：Based on Eq. (2.48) and Eq. (3.15), distribution of $\mathrm{Re}\left[\tilde{\mathcal{A}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (left panel) and $\mathrm{Re}\left[\tilde{M}^{B+,\boldsymbol{Q}}_{1,\ell,-1,\ell'}(\boldsymbol{Q},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (right panel) at $\omega t=0$. The transition dipole moment for upper band BX is plotted in the middle. Animation accessible via Figure 3.5.



In the above figures : $\alpha=0,\beta=\frac{\pi}{2},\Delta\sigma=-2$ (figures are plotted **without** the approximation)

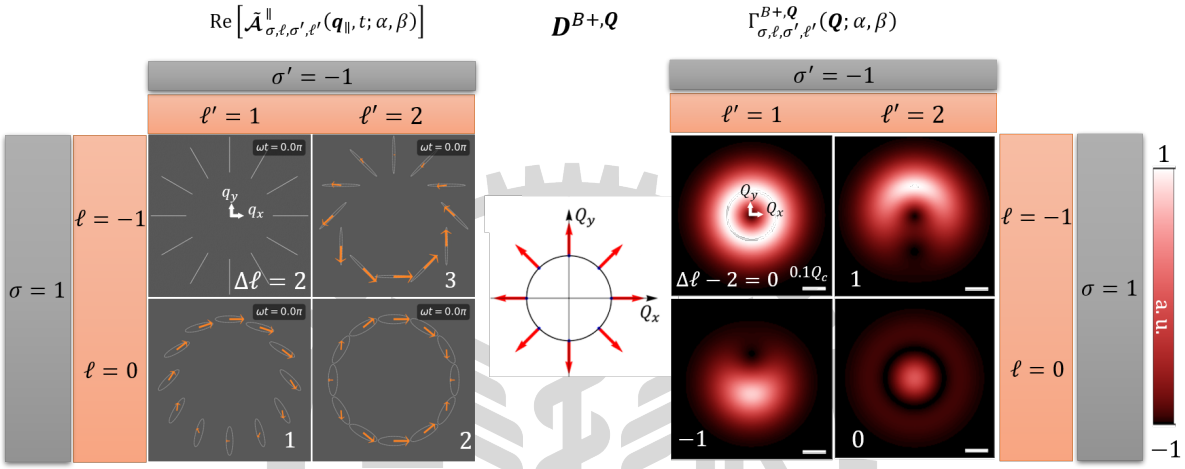
Figure 3.8：Based on Eq. (2.48) and Eq. (3.16), distribution of $\mathrm{Re}\left[\tilde{\mathcal{A}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (left panel) and $\mathrm{Re}\left[\tilde{M}^{B-,\boldsymbol{Q}}_{1,\ell,-1,\ell'}(\boldsymbol{Q},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (right panel) at $\omega t=0$. The transition dipole moment for lower band BX is plotted in the middle. Animation accessible via Figure 3.6.

The transition rates are obtained as

$$\Gamma^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta) \propto \left|\widetilde{M}^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\right|^2$$
$$\approx \left|\frac{E_g D^B}{4i\hbar} H^{+}_{|\ell|,|\ell'|}(Q;\beta)\left[\left(1-\frac{\Delta\sigma}{2}\right)\cos\left(\frac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)+\left(1+\frac{\Delta\sigma}{2}\right)\cos\left(\frac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]\right|^2 \tag{3.17}$$

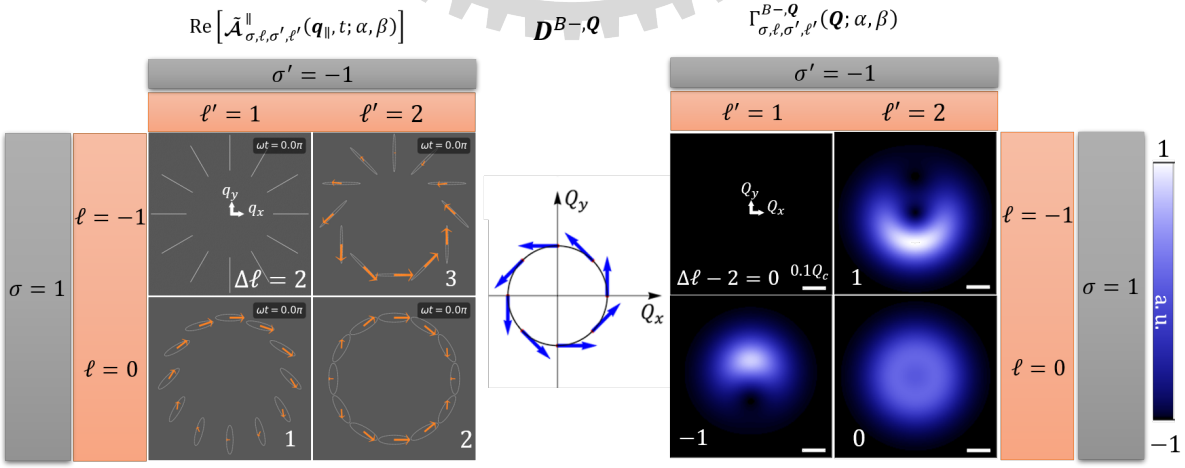
$$\Gamma^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta) \propto \left|\widetilde{M}^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\right|^2$$
$$\approx \left|\frac{-iE_g D^B}{4i\hbar} H^{+}_{|\ell|,|\ell'|}(Q;\beta)\left[\left(1-\frac{\Delta\sigma}{2}\right)\sin\left(\frac{\Delta\ell-2}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)-\left(1+\frac{\Delta\sigma}{2}\right)\sin\left(\frac{\Delta\ell+2}{2}\phi_{\boldsymbol{Q}}+\frac{\alpha_{|\ell|,|\ell'|}}{2}\right)\right]\right|^2, \tag{3.18}$$

which correspond to the time-averaged results.



In the above figures : $\alpha=0,\beta=\frac{\pi}{2},\Delta\sigma=-2$ (figures are plotted **without** the approximation)

Figure 3.9：Based on Eq. (2.48) and Eq. (3.15), distribution of $\mathrm{Re}\left[\tilde{\mathcal{A}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (left panel) and $\Gamma^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha=0,\beta=\frac{\pi}{2})$ (right panel), with $\Delta\ell$ and $\Delta\ell-2$ are indicated at the bottom. The transition dipole moment for upper band BX is plotted in the middle.



In the above figures : $\alpha=0,\beta=\frac{\pi}{2},\Delta\sigma=-2$ (figures are plotted **without** the approximation)

Figure 3.10：Based on Eq. (2.48) and Eq. (3.16), distribution of $\mathrm{Re}\left[\tilde{\mathcal{A}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel},t;\alpha=0,\beta=\frac{\pi}{2})\right]$ (left panel) and $\Gamma^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha=0,\beta=\frac{\pi}{2})$ (right panel), with $\Delta\ell$ and $\Delta\ell-2$ are indicated at the bottom. The transition dipole moment for lower band BX is plotted in the middle.

The VVB with $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 1 (\Delta\sigma = -2, \Delta\ell - 2 = 0)$ shown in the top left in both Figure 3.7 and Figure 3.8 shows more clearly that only the upper-band bright exciton is excited while the lower band is not. Besides this feature, the pattern of the transition rate also exhibits a node structure, where

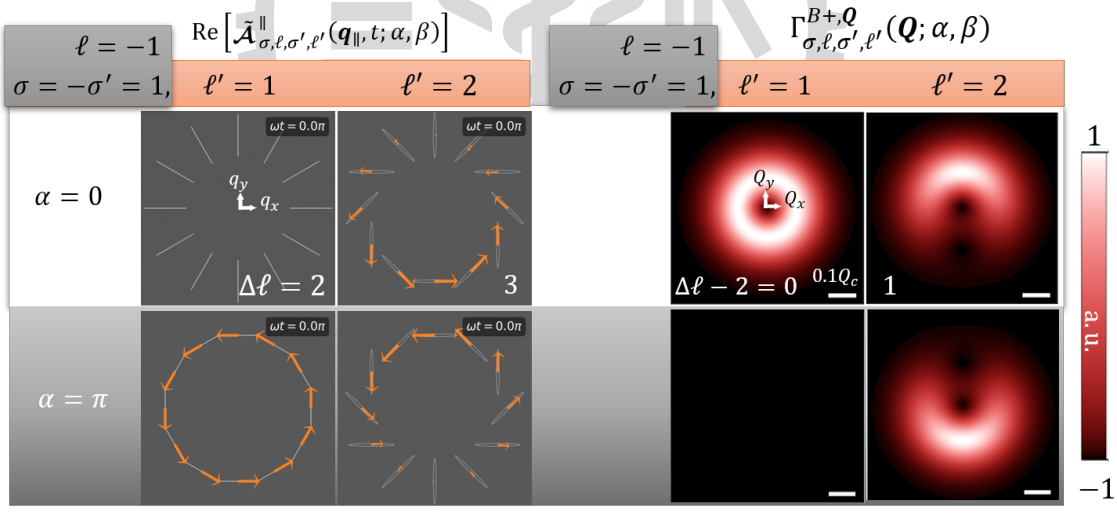
$$\text{with } \Delta\sigma = -2, \text{ number of nodes} = |\Delta\ell - 2|$$

Note that the minus sign here follows the sign of the SAM difference ($\Delta\sigma = \pm 2$) in Eq (3.17) and Eq (3.18), which determines the $\Delta\ell \pm 2$ structure in the $\phi_{\boldsymbol{Q}_\parallel}$ angle node pattern (which is determined by the trigonometric functions).

Again, when the number of nodes $\Delta\ell - 2 = \pm 1$, we can create a asymmetry pattern in the $\boldsymbol{Q}$ plane thus selectively excite BXs with specific COM momentum. Eventually create BXs that may driven in specific directions.

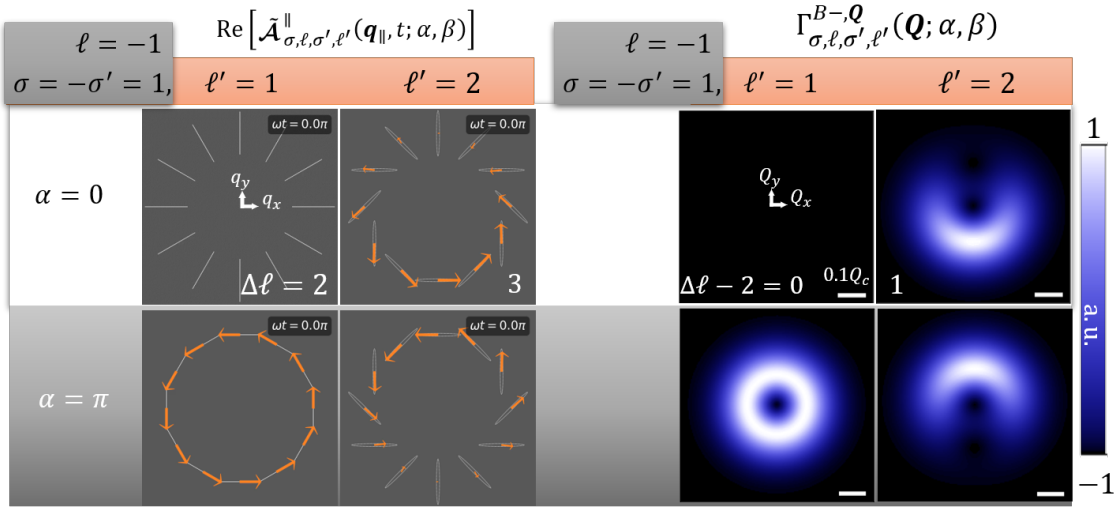
The node structure of the transition rate is rotated by the relative phase α within $\alpha_{|\ell|,|\ell'|}$. This is depicted in Figure 3.11 and Figure 3.12, where the cases shown in the upper row (both the light and transition rate) are the same as those in the upper row of Figure 3.9 and Figure 3.10. Focusing on the variation with α, the node structure is

$$\text{rotated (counter-clockwise) by } -\frac{\alpha}{\Delta\ell - 2}.$$



In the above figures : $\beta = \frac{\pi}{2}$ (figures are plotted without the approximation)

Figure 3.11 : Based on Eq. (2.48) and Eq. (3.15), distribution of Re$\left[\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel, t; \alpha, \beta = \pi/2)\right]$ (left panel) and $\Gamma^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q}; \alpha, \beta = \pi/2)$ (right panel), with different α. The transition dipole moment for upper band BX is plotted in the middle.



In the above figures : $\beta = \frac{\pi}{2}$ (figures are plotted without the approximation)

Figure 3.12 : Based on Eq. (2.48) and Eq. (3.16), distribution of $\mathrm{Re}\left[\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel},t;\alpha,\beta=\pi/2)\right]$ (left panel) and $\Gamma^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta=\pi/2)$ (right panel), with different α. The transition dipole moment for lower band BX is plotted in the middle.

Again, the minus sign in $\Delta\ell-2$ implicitly contains the sign of the SAM difference $\Delta\sigma$. Specifically, choosing the VVB configuration $\sigma,\ell,\sigma',\ell' = 1,-1,-1,1$ satisfies the condition:

$$\text{with } \Delta\sigma=-2, \quad \Delta\ell-2=0: \quad \alpha \text{ switches the transition of BXs.}$$

To further observe the relation between SAM difference, $\Delta\sigma$, and the winding of light and the exciton, we discuss the total transition rate for the exciton in the next subsection.

3.2.3 Relevant Energy Scales for Experimental Observation

To observe the selective transition in experiment, one may observe the blue shift associated with the splitting of valley-mixed exciton branches when transitioning from the conventional VVB excitation case ($\ell = \ell' = 0$) to the finite-COM-momentum excitation case with $\ell, \ell' \neq 0$ (see Figure 3.13). To provide an energy-scale reference for observing this effect and to distinguish it from possible contributions related to dark excitons, we summarize the relevant exciton energy scales for representative monolayer TMD materials (see Table 3.2).

These energy scales provide a reference for selecting suitable TMD materials and interpreting the spectral shift associated with momentum-selective exciton excitation induced by VVB excitation. Although Tungsten-based TMD materials exhibit larger dark–bright exciton splittings, the valley-mixed exciton splitting energy scale has not been systematically reported for these materials to our knowledge.

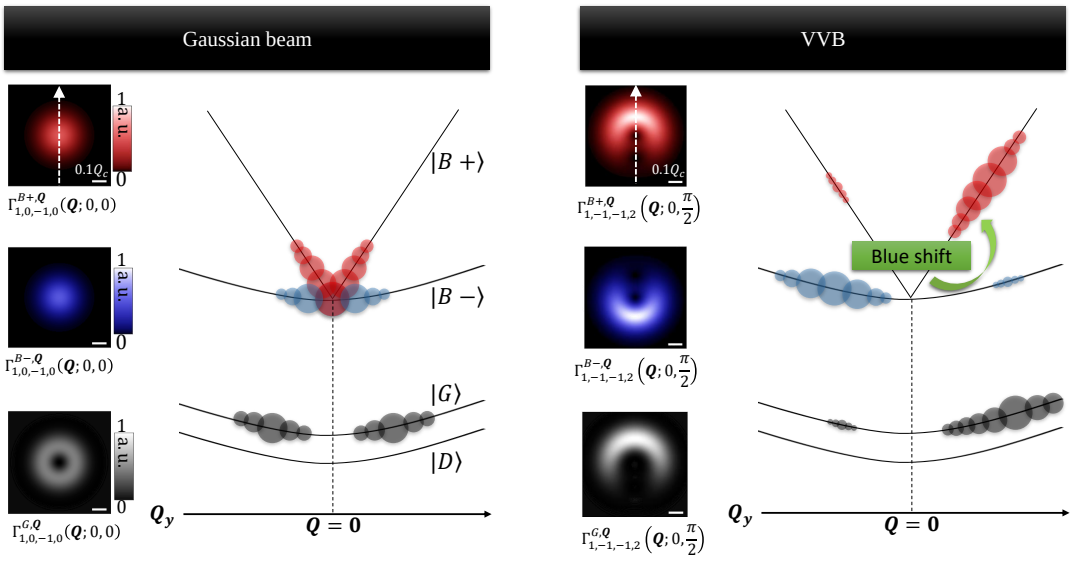


Figure 3.13 : Schematic of exciton population distribution along the Q_y axis for a VVB state ($\sigma, \ell, \sigma', \ell'$=1, 0, −1, 0; $\alpha = 0, \beta = 0$, Gaussian beam) and a VVB state ($\sigma, \ell, \sigma', \ell'$=1, −1, −1, 2; $\alpha = 0, \beta = \pi/2$).

Material	Valley-mixed exciton splitting energy scale	Dark–bright exciton splitting
MoS_2	1–2 meV [7, 43]	0, −5 meV [44]; −14 meV [45]
$MoSe_2$	–	+16, +11 meV [44]; +1.4 meV [45]
WS_2	–	−11 meV [44]; −55 meV [46]
WSe_2	–	−16 meV [44]; −40 meV [46, 47]

Table 3.2 : Relevant exciton energy scales in monolayer TMDs. For the dark–bright exciton splitting, negative (positive) values indicate that the DX lies below (above) BX.

3.2.4 Selective Excitation of Excitons

By summing the contributions in momentum space excited by the VVBs, we obtain the total transition rate, which characterizes the overall absorption strength of the material. Under the paraxial approximation, where $\theta_{\boldsymbol{Q}} = \frac{Q}{\sqrt{Q^2+q_0^2}} \approx \frac{Q}{q_0}$ is taken to simplify the integral, the total transition rate can be evaluated as:

$$\Gamma^{S}_{\sigma,\ell,\sigma',\ell'} \equiv \sum_{\boldsymbol{Q}} \Gamma^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} = \frac{\Omega}{(2\pi)^2} \int_0^{\infty} \int_0^{2\pi} \Gamma^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\, Q\, \mathrm{d}\phi_{\boldsymbol{Q}}\, \mathrm{d}Q.$$

The total transition rates for GX and BXs are obtained as

$$\Gamma^{S}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) = \frac{\Omega}{(2\pi)^2} \int_0^{\infty} \int_0^{2\pi} \frac{2\pi}{\hbar} \left| \widetilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q}) \right|^2 \mathrm{d}\phi_{\boldsymbol{Q}} Q \mathrm{d}Q. \tag{3.19}$$

With the exact formulation of the optical matrix element (Eq (3.10), Eq (3.15), and Eq (3.16)) the total transition rate is obtained as

$$\Gamma^{\boldsymbol{G}}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) = \frac{\pi\Phi}{2\hbar\Omega} \left[\frac{E_g D^B}{a_0 q_0 W_0 \hbar} \right]^2 \times \left\{ \left[1 + \cos^2\left(\tfrac{\beta}{2}\right)|\ell| + \sin^2\left(\tfrac{\beta}{2}\right)|\ell'| \right] + \delta_{\Delta J,0} \cos\alpha_{|\ell|,|\ell'|} \sin\beta \frac{(\frac{|\ell|+|\ell'|}{2}+1)!}{\sqrt{|\ell|!\,|\ell'|!}} \right\} \tag{3.20}$$

$$\Gamma^{\boldsymbol{B}\pm}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) = \frac{\pi\Phi}{4\hbar\Omega} \left[\frac{E_g D^B}{a_0 \hbar} \right]^2 \times \left[1 \pm (\delta_{\Delta\ell,2}\delta_{\Delta\sigma,-2} + \delta_{\Delta\ell,-2}\delta_{\Delta\sigma,2}) \cos\alpha_{|\ell|,|\ell'|} \sin\beta \frac{(\frac{|\ell|+|\ell'|}{2})!}{\sqrt{|\ell|!\,|\ell'|!}} \right] \tag{3.21}$$

where the constant radiant power is $\Phi = (a_0 A_0^{\mathrm{LG}} W_0)^2$, and $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$.

In addition, considering that the two BXs may not be spectrally distinguishable, the total transition rate for both BXs under the condition can be similarly written as

$$\Gamma^{\boldsymbol{B}}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) = \sum_{S=B\pm} \Gamma^{\boldsymbol{S}}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) = \frac{\pi\Phi}{2\hbar\Omega} \left[\frac{E_g D^B}{a_0 \hbar} \right]^2, \tag{3.22}$$

which is independent of α, resulting in a constant total transition rate.

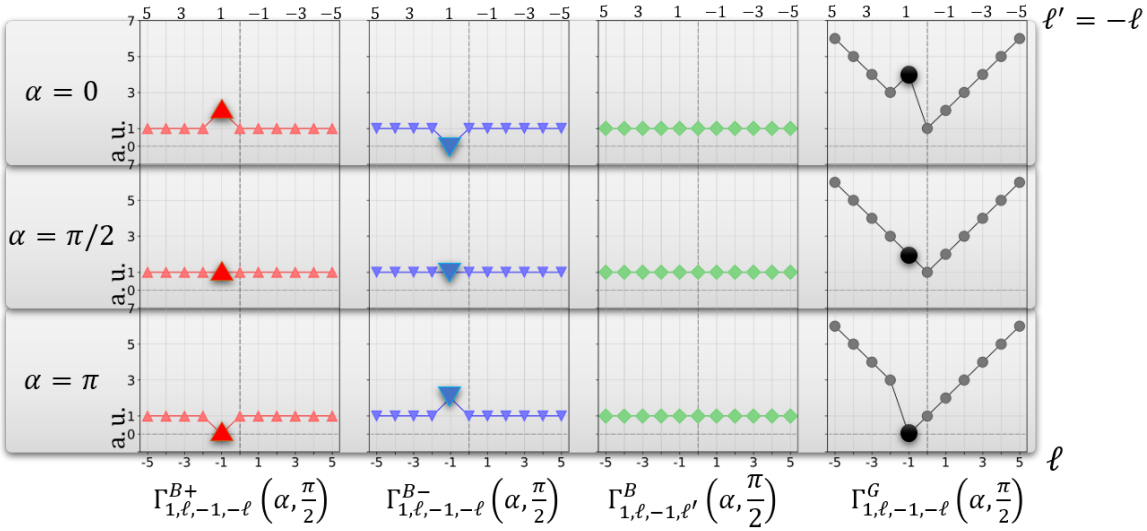


Figure 3.14 : Normalized total transition rates of BXs, total BXs, and GX excited by a VVB with $\sigma, \ell, \sigma', \ell' = 1, \ell, -1, -\ell$, with $\beta = \pi/2$ and varying ℓ and α.

Eq. (3.20), Eq. (3.21), and Eq. (3.22) explicitly show the selection rules for different BXs and GX, where the condition $\Delta\sigma = -2, \Delta\ell - 2 = 0$ and $\Delta J = 0$ will be satisfied by the same VVB with $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 1$. Only under this specific VVB's excitation, the excitation can be switched on and off with the relative phase between the composite LG beams, which can be seen in Figure 3.14.

For GX total transition rate, the $\ell(\ell')$-dependent terms arise in the square brackets and the following factorial term. The former results from the enhancement of longitudinal component, where the light becomes stronger as $\ell(\ell')$ increases. This effect was reported by Oscar Javier Gomez Sanchez *et al.* in Ref. [1] and is represented by the $\sin\theta_q$ dependence in Eq (2.36).

On the other hand, the condition for coherence is shown as a selection rule with $\delta_{\Delta J,0}$. When matched, α switches the transition of gray exciton as we have seen in Figure 3.4. This selection rule originates from the SOI contribution of the longitudinal component and therefore appears explicitly in the GX transition through the longitudinal field.

As for the constant in the front, the beam waist W_0, which appears in the denominator of the prefactor for the GX total transition rate, indicates that smaller W_0 enhances the longitudinal field component and thus increases the excitation efficiency of GXs.

For BXs total transition rate, the $\ell(\ell')$-dependent terms arise only through the combined selection rules. In the example for $\Delta\ell - 2 = 0$, the SAM difference, $\Delta\sigma$, the winding of light, ℓ, and the winding number of the exciton, 2, are specifically chosen. This effect is not shown in the VVB formulation (Eq (2.48)), which has no explicit SOI effect. It is not until the formulation of the optical matrix elements (Eq (3.13) and Eq (3.14)) that we see evidence of these selection rules.

Note that Figure 3.14 is simulated under the condition $\Delta\sigma = -2$. If we consider $\Delta\sigma = 2$, the selection rule will then select the VVB with the opposite OAM configuration, whose $\sigma, \ell, \sigma', \ell' = -1, 1, 1, -1$ and the conditions, $\Delta\sigma = 2, \Delta\ell + 2 = 0$ and $\Delta J = 0$ will be fulfilled.

This shows that changing the sign of $\Delta\sigma$ does not change the physical result, since simultaneously switching the OAM and SAM between the composite LG beams gives the corresponding VVB with the same physical selection behavior.

Focus on the VVB which fulfill the selection rule with $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 1$, Figure 3.15 shows the excitation is smoothly switched by α.

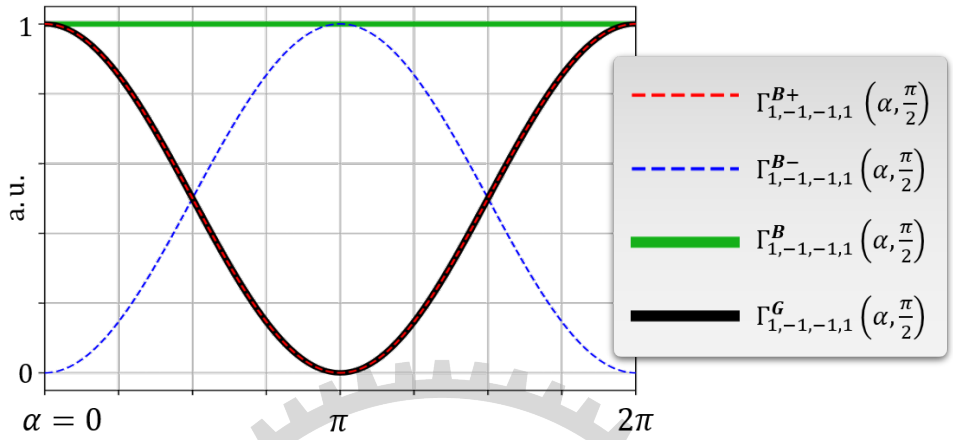


Figure 3.15：Normalized total transition rates of BXs, total BXs, and GXs excited by a VVB with $1, -1, -1, 1; \alpha, \beta = \pi/2$ for azimuthal angles $\alpha = 0$ to 2π.

These observations originate from SOI associated with the longitudinal component of the LG beam, together with matching between SAM difference $\Delta\sigma$, winding of light $\Delta\ell$, and winding number of exciton. The former arises from the gauge transformation, specifically the second term in Eq. (2.28). The latter originates from the correspondence between the winding texture of BXs' transition dipole moment (winding number) and the winding of polarization textures in VVBs. In both cases, Fermi's golden rule provides a theoretical framework for interpreting these results and establishes a basis for further studies involving SOI in the transverse component and their interaction with excitons, or for going beyond the approximation $|\cos\left(\frac{\beta}{2}\right) F_{|\ell|}(q_{\parallel})| \approx |\sin\left(\frac{\beta}{2}\right) F_{|\ell'|}(q_{\parallel})|$.

Chapter 4 Wave Packets

4.1 Internal Structure of Excitons

4.1.1 Global Phase in Valley-Mixed Exciton

In this section, we establish the theoretical foundation for further studies of exciton wave packets. A wave packet of the particle refers to a superposition of momentum eigenstates, where the carrier wave (ripples) possesses a characteristic wavelength determined by the dominant momentum component, while the envelope describes the finite spatial extent of the particle. The velocity associated with the envelope, known as the group velocity, corresponds to the particle velocity in classical physics. From this perspective, the envelope of a wave packet provides a good intuition of the particle's position and motion [42, 48].

Using time-dependent perturbation theory [42], the time-dependent exciton state under weak photoexcitation can be expressed as a superposition of finite-momentum exciton (SFME) states [7],

$$|\Psi(t)\rangle = |GS\rangle + \sum_{S=B\pm}\sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t) e^{-i\omega^{S,\boldsymbol{Q}}t}|\Psi^{S,\boldsymbol{Q}}\rangle, \tag{4.1}$$

which represents a linear combination of the ground state $|GS\rangle$ and exciton states $|\Psi^{S,\boldsymbol{Q}}\rangle$ with center-of-mass (COM) momentum $\boldsymbol{Q}$. Other notations follow Eq. (3.2). Within first-order perturbation theory, the amplitude coefficients are given by

$$\tilde{c}^{S,\boldsymbol{Q}(1)}(t) = \tilde{\gamma}^{S,\boldsymbol{Q}}(t)\left(\frac{\tilde{M}^{S,\boldsymbol{Q}}}{\hbar\omega^{S,\boldsymbol{Q}}}\right), \tag{4.2}$$

where time–dependent coefficient $\tilde{\gamma}^{S,\boldsymbol{Q}}(t) \approx \tilde{\gamma}^{\boldsymbol{0}}(t)$, and the natural frequency is approximated as $\omega^{S,\boldsymbol{Q}} = E^{S,\boldsymbol{Q}}/\hbar \approx E^{S,\boldsymbol{0}}/\hbar = \omega^{\boldsymbol{0}}$. The amplitude coefficients can thus be separated into time–dependent and time–independent parts, where the latter contains the optical matrix element introduced in Eq. (3.8). For simplicity, we consider excitation by a single LG beam with radial mode index $p = 0$, such that the optical matrix element takes the form

$$\tilde{M}^{S,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \approx \frac{E_g}{2i\hbar}\left(\boldsymbol{\mathcal{A}}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{Q})\cdot\boldsymbol{D}^{S,\boldsymbol{Q}*}\right). \tag{4.3}$$

Here, derivations analogous to those in the previous chapter—including the rotating-wave approximation and electric dipole approximation—are omitted for brevity. By adopting the polarization unit vector $\hat{\boldsymbol{\varepsilon}}$ instead of photon helicity σ, we relax the restriction to circular polarization. The vector potential is denoted by $\boldsymbol{\mathcal{A}}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{q}_{\parallel})$, which corresponds to the complex amplitude introduced in Eq. (2.32) but with polarization unit vector not limited to circularly polarized.

In the work of Guan-Hao Peng *et al.* [7], the envelope function is defined as

$$F^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,t) \equiv \frac{1}{\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \tag{4.4}$$

To explicitly isolate the static contribution, we use the relation $F^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,t) \approx \tilde{\gamma}_{\boldsymbol{0}}(t)\, f^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c)$, which defines the static part of envelope function as

$$f^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{S,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \tag{4.5}$$

where the natural frequency of the exciton state $\omega^{S,\boldsymbol{Q}} = E^{S,\boldsymbol{Q}} \approx E^{S,\boldsymbol{0}} = \omega^{\boldsymbol{0}}$ is assumed to be approximately constant. However, the valley-mixed exciton state obtained from the pseudo-spin model [7],

$$|\Psi^{B\pm,\boldsymbol{Q}}\rangle = e^{i\delta(\boldsymbol{Q})} \left(e^{-i\phi_{\boldsymbol{Q}}} |\Psi^{K,\boldsymbol{Q}}\rangle \pm e^{i\phi_{\boldsymbol{Q}}} |\Psi^{K',\boldsymbol{Q}}\rangle \right) / \sqrt{2}, \tag{4.6}$$

naturally carries a $\boldsymbol{Q}$-dependent global phase $\delta(\boldsymbol{Q})$ arising from its internal structure. Consequently, the optical matrix element acquires the same phase factor,

$$\tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} = e^{i\delta(\boldsymbol{Q})} \left(e^{i\phi_{\boldsymbol{Q}}} \tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{-i\phi_{\boldsymbol{Q}}} \tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \right) / \sqrt{2}. \tag{4.7}$$

As a result, this $\boldsymbol{Q}$-dependent phase can modify the resulting static part of envelope function across momentum space, provided that the internal phase factors ($e^{\pm i\phi_{\boldsymbol{Q}}}$) partially compensate or reshape the global phase. This observation motivates a more careful examination of the internal structure of the exciton wave function and the explicit extraction of common factors.

4.2 Wave functions of finite-momentum excitons

To address this issue without resorting to a numerically expensive construction of the exciton wave function, we instead extract its $\boldsymbol{Q}$–dependent phase structure analytically. We begin by considering the internal structure of the exciton wave function [49, 50],

$$\Psi^{S,\boldsymbol{Q}}(\boldsymbol{r}_e,\boldsymbol{r}_h) = \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{r}_e) \psi^{*}_{v,\boldsymbol{k}}(\boldsymbol{r}_h), \tag{4.8}$$

where $\psi_{c,\boldsymbol{k}}(\boldsymbol{r}_e)$ and $\psi_{v,\boldsymbol{k}}(\boldsymbol{r}_h)$ denote the single-particle Bloch wave functions for an electron in the conduction band and a hole in the valence band, respectively. Here, $\boldsymbol{r}_e$ and $\boldsymbol{r}_h$ are the corresponding electron and hole position vectors. The coefficient $\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ represents the amplitude of the electron–hole configuration for exciton band S.

For the purpose of extracting the COM momentum $\boldsymbol{Q}$-dependent phase term, it is convenient to rewrite the exciton wave function in terms of a weighted-average position vector $\boldsymbol{R}_c$[1]

[1] Although this weighted-average position vector is not strictly identical to the conventional COM coordinate in classical mechanics, we retain the subscript "c" to avoid confusion with the Bravais lattice vector $\boldsymbol{R}$.

and the relative coordinate $\boldsymbol{r}$:

$$\begin{aligned} \boldsymbol{R}_c = a\boldsymbol{r}_e + b\boldsymbol{r}_h, \qquad \boldsymbol{r} = \boldsymbol{r}_e - \boldsymbol{r}_h, \qquad a+b=1, \\ \boldsymbol{r}_e = \boldsymbol{R}_c + b\boldsymbol{r}, \qquad \boldsymbol{r}_h = \boldsymbol{R}_c - a\boldsymbol{r}. \end{aligned} \tag{4.9}$$

Using these relations, the exciton wave function can be rewritten as

$$\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) \psi^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - a\boldsymbol{r}). \tag{4.10}$$

In addition, Bloch's theorem states that, in a material with a periodic lattice structure—such as a two-dimensional TMD characterized by Bravais lattice vectors $\boldsymbol{R}$—the single-particle wave function satisfies [42, 50]

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}') e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{4.11}$$

where the wave function is expressed as the product of a cell-periodic function, $u_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}' + \boldsymbol{R})$, and a plane-wave component $e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}$ [2]. Using this property, the exciton wave function can be rewritten as

$$\begin{aligned} &\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) \\ &= \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q}) \psi_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) \psi^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) \\ &= \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q}) \left[u_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) e^{i(\boldsymbol{k}+a\boldsymbol{Q})\cdot(\boldsymbol{R}_c+b\boldsymbol{r})} u^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) e^{-i(\boldsymbol{k}-b\boldsymbol{Q})\cdot(\boldsymbol{R}_c-a\boldsymbol{r})} \right] \\ &= \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} \left[u_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) u^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) \right]. \\ &= e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) \end{aligned} \tag{4.12}$$

The first arrow is obtained by shifting crystal momentum by $-b\boldsymbol{Q}$, which allows the $\boldsymbol{Q}$-dependent phase originating from the plane-wave components to be written in the compact form $e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}$. Finally, we extract $U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})$. Once we eliminate the $\boldsymbol{Q}$-dependence of $U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})$, it should be easier to construct the internal structure of the exciton wave function

Finally, we adopt the convention

$$a = 1, \qquad b = 0,$$

then

$$U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \left[u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - \boldsymbol{r}) \right]. \tag{4.13}$$

Further simplification can be achieved by performing a Taylor expansion around $\boldsymbol{Q} = \boldsymbol{0}$

[2] The vector $\boldsymbol{r}'$ denotes a general position vector and may represent the weighted-average, relative, electron, or hole position vector.

for the implicitly $\boldsymbol{Q}$-dependent quantities, namely $\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ and $u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)$. Explicitly,

$$\begin{aligned}
\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) &= \sum_{m=0}^{\infty} \frac{1}{m!} \left(\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\right)^m \Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k}) \quad &= \Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k}) + \boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k}) + \cdots, \\
u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) &= \sum_{m=0}^{\infty} \frac{1}{m!} \left(\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\right)^m u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) \quad &= u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) + \boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) + \cdots.
\end{aligned} \tag{4.14}$$

Their product can therefore be expanded as

$$\begin{aligned}
\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) = &\underbrace{\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)}_{\text{zeroth order}} \\
&+ \underbrace{\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)\right] + \left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})\right]u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)}_{\text{first order}} \\
&+ \underbrace{\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})\right]\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)\right]}_{\text{second order}} \\
&+ \text{higher-order terms}.
\end{aligned} \tag{4.15}$$

4.3 Zeroth-order approximation

Substituting this expansion into Eq. (4.13), the exciton wave function can be decomposed into contributions from different orders of $\boldsymbol{Q}$,

$$\begin{aligned}
U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\left\{\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)\right\}u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}) \\
&= \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\left[\Psi^{S,\boldsymbol{Q}(0)}(\boldsymbol{R}_c,\boldsymbol{r}) + \Psi^{S,\boldsymbol{Q}(1)}(\boldsymbol{R}_c,\boldsymbol{r}) + \Psi^{S,\boldsymbol{Q}(2)}(\boldsymbol{R}_c,\boldsymbol{r}) + \cdots\right] \\
&\approx \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}) \quad \text{———} \quad \text{zeroth–order},
\end{aligned} \tag{4.16}$$

where the final approximation refers to the zeroth-order approximation [3] in the small-$\boldsymbol{Q}$ limit. Consequently, the $\boldsymbol{Q}$-dependent phase factor can be extracted as [43]

$$\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}U^{S,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \tag{4.17}$$

In addition, the valley exciton wave functions with zero COM momentum satisfy

$$U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) = U^{K',\boldsymbol{0}*}(\boldsymbol{R}_c,\boldsymbol{r}) \tag{4.18}$$

[3] A similar expansion scheme is commonly employed in evaluating the electron–hole exchange interaction (EHEI) within the Bethe–Salpeter equation, where the first-order contribution is typically retained. Although this may appear inconsistent with the zeroth-order approximation adopted here, the zeroth-order contribution to the EHEI vanishes to leading order, thereby necessitating the inclusion of higher-order terms.

as derived in appendix A.3. Therefore, each wave function can be expressed in polar form as

$$\begin{aligned} U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \left|U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| \exp\Big[i\,\mathrm{Arg}\big(U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\big)\Big], \\ U^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \left|U^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| \exp\Big[i\,\mathrm{Arg}\big(U^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\big)\Big]. \end{aligned} \tag{4.19}$$

From the complex-conjugation relation above, the two wave functions have identical amplitudes while their phases are opposite in sign, i.e.,

$$\left|\Psi^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| = \left|\Psi^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right|, \qquad \mathrm{Arg}\Big[\Psi^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\Big] = -\mathrm{Arg}\Big[\Psi^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\Big]. \tag{4.20}$$

4.4 Exciton Wave Packet for Excitons

4.4.1 General Effect from Exciton Wave Function

In the work of Guan-Hao Peng *et al.* [7], the envelope function is defined by

$$F^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,t) \equiv \frac{1}{\sqrt{\Omega}}\sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \tag{4.4}$$

and its static part is

$$f^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}}\sum_{\boldsymbol{Q}} \tilde{M}^{S,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \tag{4.5}$$

In this work, we include the internal exciton wave function and define the envelope function more generally as

$$\begin{aligned} \Phi^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r},t) &\equiv \frac{1}{\sqrt{\Omega}}\sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t)e^{-i\omega^{S,\boldsymbol{Q}}t}\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}), \\ &= \frac{1}{\sqrt{\Omega}}\sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t)e^{-i\omega^{S,\boldsymbol{Q}}t}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \end{aligned} \tag{4.21}$$

The second equality follows from Eq (4.12) and the corresponding static part is

$$\Phi^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) = \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}}\sum_{\boldsymbol{Q}} \tilde{M}^{S,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.22}$$

$U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r})$, the internal structure of the exciton wave function, is the additional factor that distinguishes Eq. (4.5) from Eq. (4.22). In addition, it is both $\boldsymbol{Q}$–dependent and complex-valued. Below, we discuss how this internal structure generally influences the wave packet and the possible sources of its phase contributions.

First, for simplicity we neglect the electron-hole exchange interaction and consider exciton states with well-defined valley degrees of freedom, K or K', to discuss the possible sources of

phase contributions. With Eq (4.13), the valley exciton wave function's internal structure is

$$U^{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda^{K/K',\boldsymbol{Q}}(vc\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \left[u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - \boldsymbol{r})\right]. \tag{4.23}$$

The internal structure contains three main contributions :

$$\begin{aligned} \text{Amplitude of the e–h configuration : } & \Lambda^{K/K',\boldsymbol{Q}}(vc\boldsymbol{k}) \\ \text{Phase associated with electron–hole separation: } & e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \\ \text{Cell-periodic functions of electron and hole : } & u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - \boldsymbol{r}) \end{aligned} \tag{4.24}$$

Amplitude of the e–h configuration : $\Lambda^{K/K',\boldsymbol{Q}}(vc\boldsymbol{k})$ is generally complex-valued and depends on both $\boldsymbol{Q}$ and $\boldsymbol{k}$. Although one can assume it is $\boldsymbol{Q}$-independent as in [43] or apply the zeroth-order approximation used in the previous section, such approximations are only valid in the small-$\boldsymbol{Q}$ limit. This is not generally sufficient for the excitation of excitons by twisted light, which produces a donut-shaped distribution in $\boldsymbol{Q}$ space when carrying non-zero OAM. In other words, the excitation necessarily involves exciton states with finite $\boldsymbol{Q}$.

Phase associated with electron–hole separation: The phase factor, $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, indicates that the relative position between the electron and hole introduces an additional phase to the wave packet. Interestingly, this means that the exciton may exhibit different wave packet shapes for different electron-hole separations, with two characteristic cases given by

$$\begin{aligned} \text{radiative recombination : } & |\boldsymbol{r}| = 0 \\ \text{characteristic exciton radius : } & |\boldsymbol{r}| \sim a_{\mathrm{B}} \end{aligned} \tag{4.25}$$

This connection indicates that the wave packet may reflect the dynamics of the exciton.

Cell-periodic functions of electron and hole : The internal structure of the electron and hole's wave function will affect the wave packet envelope. Therefore, obtaining the full wave packet in this case requires knowledge of the corresponding electron and hole wave functions.

Other possible phase contributions (valley-mixed exciton) : For a valley-mixed exciton, the wave function is composed of exciton states from both valleys, with a $\phi_{\boldsymbol{Q}}$-dependent relative phase between them. This relative phase can therefore influence the spatial structure of the wave packet. More details are discussed in the following subsection.

Without explicitly calculating the wavefunction, it is still possible to simulate the wavepacket if we neglect these additional phase contributions. In the following subsections, we provide physical considerations for evaluating the validity of this approximation and simulate the wave packet to investigate how the light field influences the exciton wave packet.

4.4.2 Valley Excitons

In the zeroth-order approximation, Eq. (4.17) suggests

$$\Psi^{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.26}$$

Accordingly, considering the internal structure of the exciton states, the static part of the envelope function is constructed through Eq (4.22) as

$$\Phi^{K/K'}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) \approx \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}} U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \sum_{\boldsymbol{Q}} \tilde{M}^{K/K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \tag{4.27}$$

where the internal structure of the zero COM exciton wave function is given by Eq. (4.13) as

$$U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) = \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \Lambda^{K/K',\mathbf{0}}(vc\boldsymbol{k}) u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - \boldsymbol{r}). \tag{4.28}$$

In the following, we introduce physically motivated approximations under which the additional contributions from $U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ can be neglected, thereby avoiding the need to explicitly calculate the exact exciton wavefunction.

Amplitude of the e–h configuration : The momentum dependence of the coefficient $\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k})$ can be estimated using the two-dimensional hydrogen model [51]. For the $1s$ exciton state, it is given by

$$\Lambda^{1s,\mathbf{0}}(vc\boldsymbol{k}) = \frac{\sqrt{2\pi}a^*_B}{\left[1+\left(\frac{a^*_B\boldsymbol{k}}{2}\right)^2\right]^{3/2}}, \tag{4.29}$$

where the effective Bohr radius is $a^*_B \approx 1\,\text{nm}$. Since $\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k})$ is a purely real function, it does not introduce any additional phase to the wave packet.

Cell-periodic functions of electron and hole : We can consider the case where the electron and hole spatially overlap, i.e., $\boldsymbol{r} = \mathbf{0}$. Under this condition, $e^{i\boldsymbol{k}\cdot\boldsymbol{r}} = 1$, and therefore this phase factor introduces no additional phase contribution. Beyond simplifying the analysis, this condition corresponds to the configuration required for radiative recombination of excitons.

Cell-periodic functions of electron and hole : Finally, the cell-periodic functions satisfy

$$u_{n,\boldsymbol{k}}(\boldsymbol{R}_c) = u_{n,\boldsymbol{k}}(\boldsymbol{R}_c + \boldsymbol{R}). \tag{4.30}$$

Since the lattice constant (on the order of Å) is much smaller than the optical beam waist ($W_0 \sim 1.5\,\mu$m), the spatial variation of the cell-periodic functions is confined within a unit cell and does not significantly affect the overall envelope profile. Equivalently, neglecting the cell-periodic functions corresponds to describing the envelope function on the scale of at discrete lattice sites, which does not qualitatively alter the resulting wave-packet structure.

Simulations

Based on the considerations above, we neglect the additional phase by assuming

$$U^{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = 1,$$

such that the static part of envelope function, Eq (4.22), becomes

$$\Phi^{K/K'}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{K/K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \tag{4.31}$$

By adopting Born's statistical interpretation [42], the exciton wave packet can be characterized by the probability density $|\Phi^{K/K'}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c, \boldsymbol{r} = \mathbf{0})|^2$.

Before the simulation, we recall that the transverse component of the light field contributes to the optical matrix element, yielding

$$\tilde{M}^{S,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \approx \frac{E_g}{2i\hbar} \left(\boldsymbol{\mathcal{A}}^{\parallel}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{Q}) \cdot \boldsymbol{D}^{S,\boldsymbol{Q}*} \right). \tag{4.32}$$

The transverse component of the vector potential in the Coulomb gauge is given in Eq. (2.32). Accordingly, it can be written as

$$\boldsymbol{\mathcal{A}}^{\parallel}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{Q}) = \mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})\hat{\boldsymbol{\varepsilon}} \approx \tilde{F}_{|\ell|}(\boldsymbol{Q}) e^{i\ell\phi_{\boldsymbol{Q}}} \hat{\boldsymbol{\varepsilon}}, \tag{4.33}$$

where the approximation follows from the paraxial condition adopted in Eq. (2.36).

For topological charge $\ell = 0$ and ± 1, the light's distribution (see Figure 4.1 and Figure 4.3) is preserved in both the optical matrix element and the resulting probability distribution (see Figure 4.2 and Figure 4.4),

Wave packet shape reflects light pattern

This behavior directly reflects the spatial structure of the LG beam in angular spectrum space and produces corresponding spatial profiles in real space (see Figure 2.3). In addition, K and K' valley excitons are selectively excited by circularly polarized light with $\hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{x}} - i\hat{\boldsymbol{y}}$, respectively, under the transition dipole convention adopted in Table 3.1.

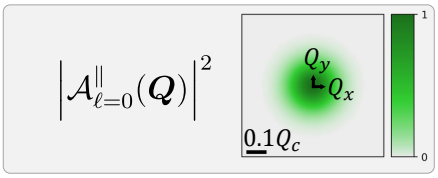


Figure 4.1 : For $\ell = 0$, normalized $|\mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})|^2$. The scale bar indicates 0.1 light cone range.

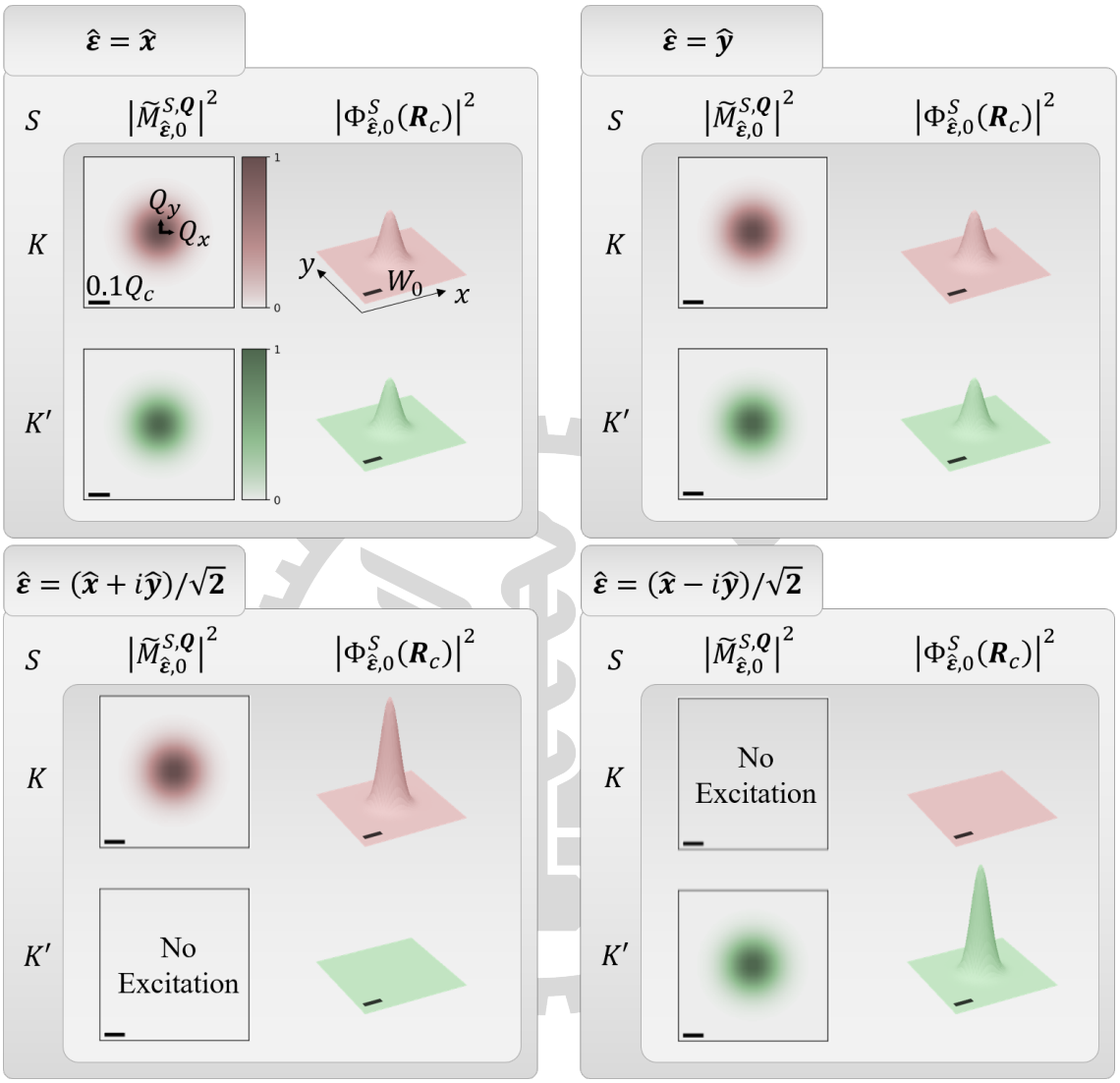


Figure 4.2 : For $\ell = 0$, normalized $|\tilde{M}^{K/K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}|^2$ and $|\Phi^{K/K'}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c)|^2$ with different polarization unit vectors $\hat{\boldsymbol{\varepsilon}}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$.

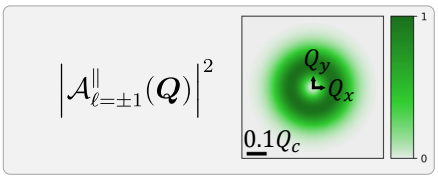


Figure 4.3：For $\ell = \pm 1$, normalized $|\mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})|^2$. The scale bar indicates 0.1 light cone range.

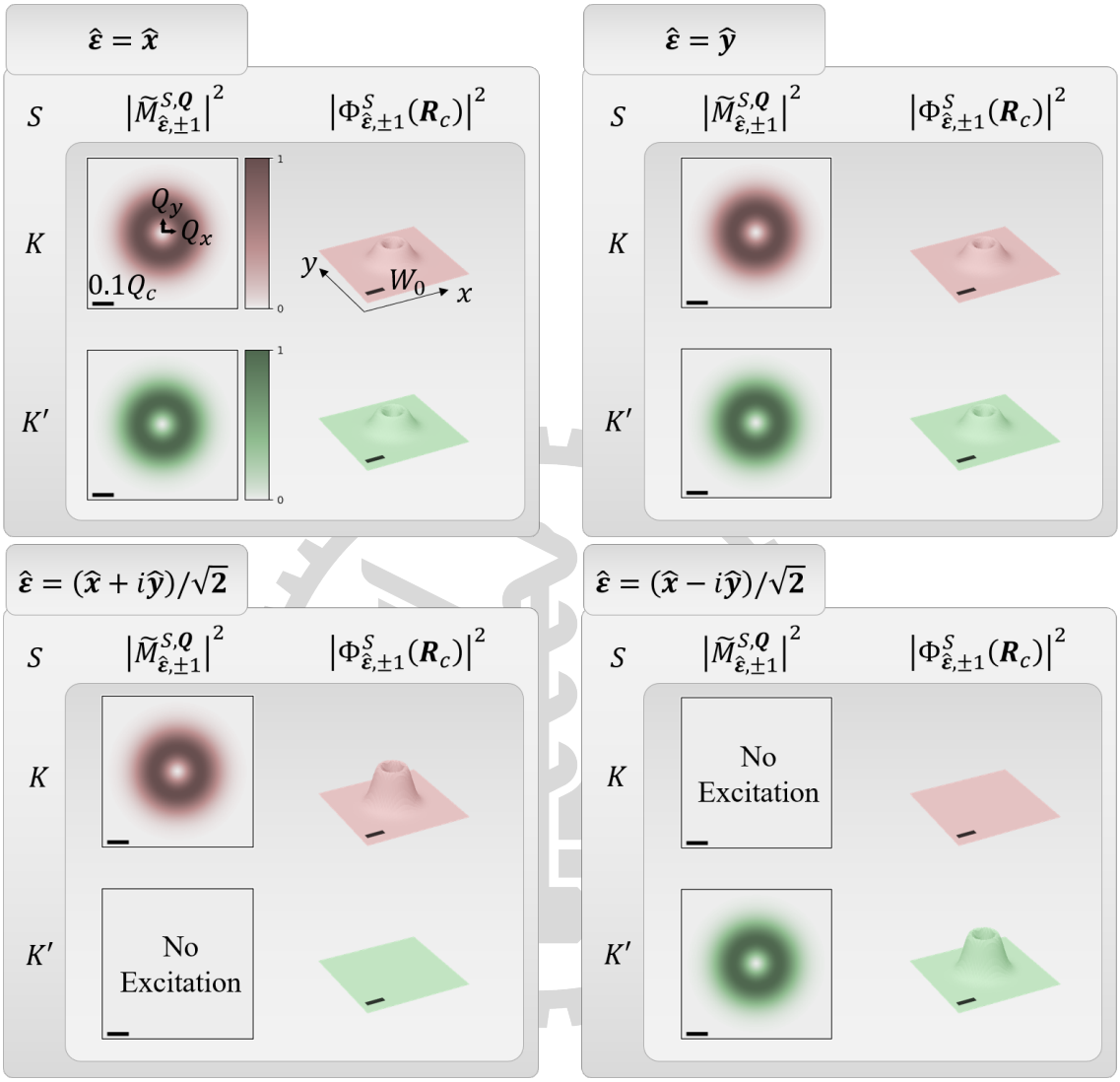


Figure 4.4：For $\ell = \pm 1$, normalized $|\tilde{M}^{K/K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}|^2$ and $|\Phi^{K/K'}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c)|^2$ with different polarization unit vectors $\hat{\boldsymbol{\varepsilon}}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$.

4.4.3 Valley-mixed Excitons

For valley–mixed excitons, the corresponding wave functions can be constructed within the pseudo–spin model [7] from the valley exciton wave functions. Notably, zeroth-order approximation (Eq. (4.17)) remains applicable in this case. One obtains

$$\Psi^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{B\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.34}$$

Following Eq. (4.22), the static part of envelope function is written as

$$\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) \approx \frac{1}{\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{B\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.35}$$

Here, $U^{B\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ is retained inside the summation because the $\phi_{\boldsymbol{Q}}$ dependence remains embedded in its internal structure. Using both Eq. (4.6) and Eq. (4.7), the $\boldsymbol{Q}$–dependent terms can be written more explicitly by rewriting the static part of envelope function in terms of the valley indices K and K':

$$\begin{aligned}
&\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) \\
&= \frac{1}{2\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \left(e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} \left[e^{-i\phi_{\boldsymbol{Q}}}U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \pm e^{i\phi_{\boldsymbol{Q}}}U^{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right] \\
&= \frac{1}{2\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}} \Bigg[U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \sum_{\boldsymbol{Q}} \left(e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} e^{-i\phi_{\boldsymbol{Q}}} \\
&\qquad \pm U^{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \sum_{\boldsymbol{Q}} \left(e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} e^{i\phi_{\boldsymbol{Q}}} \Bigg].
\end{aligned} \tag{4.36}$$

In this form, the phase of $U^{K/K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})$, $e^{+/-i\mathrm{Arg}\left[U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]}$, can no longer be neglected. As a result, a complete construction of the exciton wave function appears, at first sight, unavoidable.

Simulations

Nevertheless, we can first obtain the wave packet neglecting the innerstructure by assuming

$$U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) = 1,$$

such that the static part of envelope function, Eq (4.22), becomes

$$\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \tag{4.37}$$

In addition, this formulation is the same as Eq (4.5) in Guan-Hao Peng *et al*'s paper [7],

$$f^{S}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{S,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \tag{4.5}$$

thus we will have the same reults as in the paper.

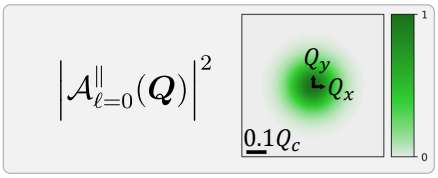


Figure 4.1 : (Repeated) For $\ell = 0$, normalized $|\mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})|^2$. The scale bar indicates 0.1 light cone range.

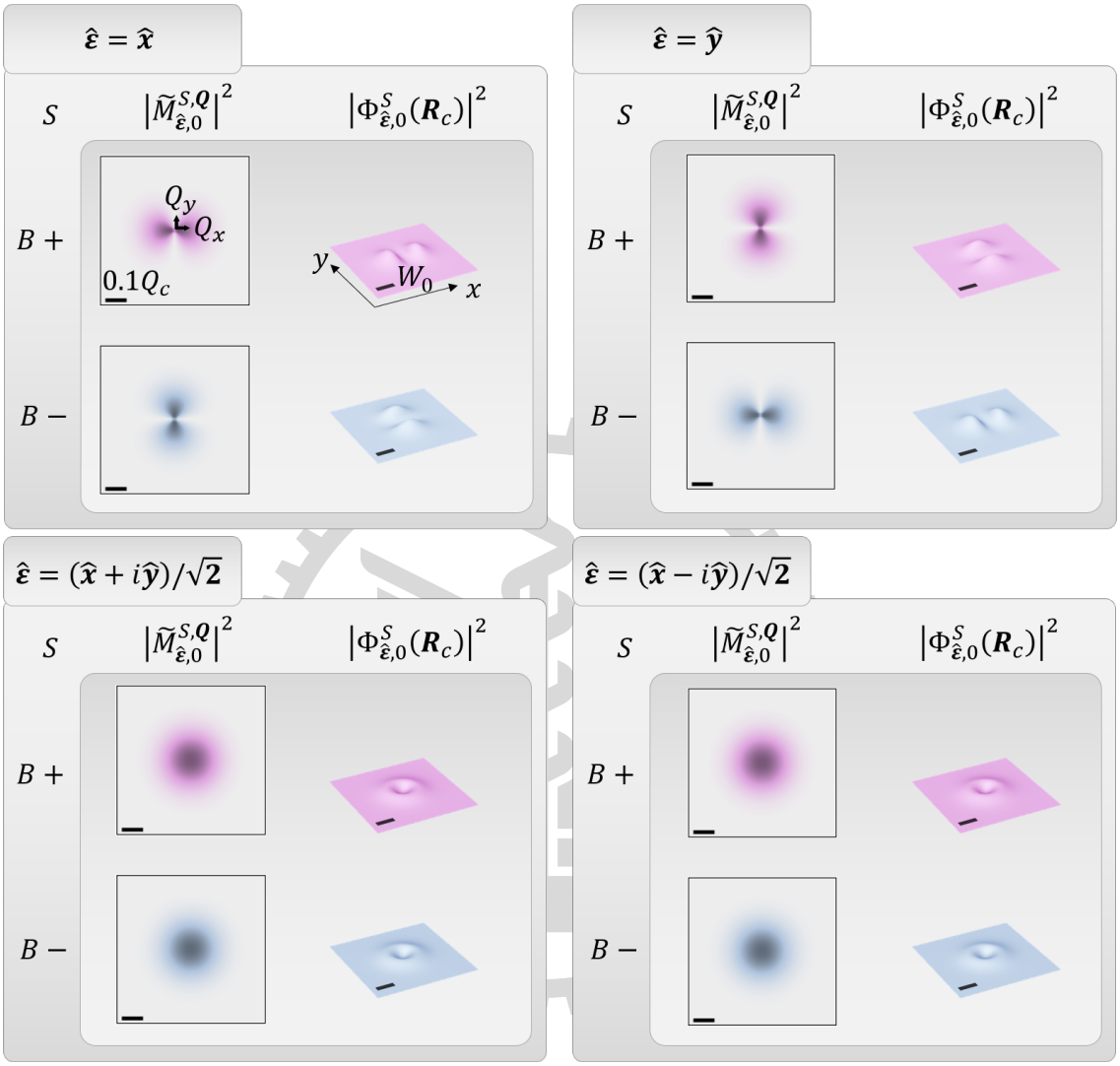


Figure 4.5 : For $\ell = 0$, normalized $|\tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}|^2$ and $|\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c)|^2$ for different polarization unit vectors $\hat{\boldsymbol{\varepsilon}}$. The scale bar indicates 0.1 light cone range.

The central peak feature can be understood by evaluating

$$\begin{aligned}\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c = \boldsymbol{0}) &= \frac{1}{\hbar\omega^0\sqrt{\Omega}}\sum_{\boldsymbol{Q}}\tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \\ &\propto \sum_{\phi_{\boldsymbol{Q}}} e^{i(\ell+1)\phi_{\boldsymbol{Q}}}\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{K,\boldsymbol{Q}*}\right) \pm e^{i(\ell-1)\phi_{\boldsymbol{Q}}}\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{K',\boldsymbol{Q}*}\right) \\ &\propto \delta_{\ell,-1} \pm \delta_{\ell,1}\end{aligned} \tag{4.38}$$

This indicates that only LG beams carrying $\ell = \pm 1$ may yield a central peak, which conforms to Figure 4.5, where every case has a central depression since $\ell = 0$.

On the other hand, for the $\ell = 1$ cases in Figure 4.6, three of the cases have a central peak,

as expected from Eq. (4.38). The remaining case, corresponding to right-circularly polarized light, can be further analyzed with

$$\begin{aligned}
&\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}}=(\hat{\boldsymbol{\varepsilon}}+i\hat{\boldsymbol{y}})/\sqrt{2},\ell}(\boldsymbol{R}_c=\boldsymbol{0}) \\
&\propto \sum_{\phi_{\boldsymbol{Q}}} e^{i(\ell+1)\phi_{\boldsymbol{Q}}}\left[(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}})/\sqrt{2}\cdot \boldsymbol{D}^{K,\boldsymbol{Q}*}\right] \pm e^{i(\ell-1)\phi_{\boldsymbol{Q}}}\cancelto{0}{\left[(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}})/\sqrt{2}\cdot \boldsymbol{D}^{K',\boldsymbol{Q}*}\right]} \\
&\propto \delta_{\ell,-1},
\end{aligned} \tag{4.39}$$

where only K valley exciton is excited and carries the $e^{i(\ell+1)\phi_{\boldsymbol{Q}}}$ phase into the envelope function. Thus, the wave packet will have a central peak only at $\ell=-1$ when excited by right-circularly polarized light.

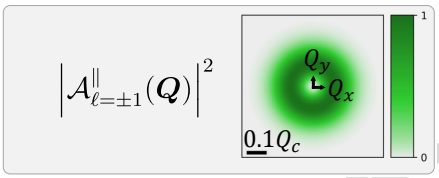


Figure 4.3 : (Repeated) For $\ell=\pm1$, normalized $|\mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})|^2$. The scale bar indicates 0.1 light cone range.

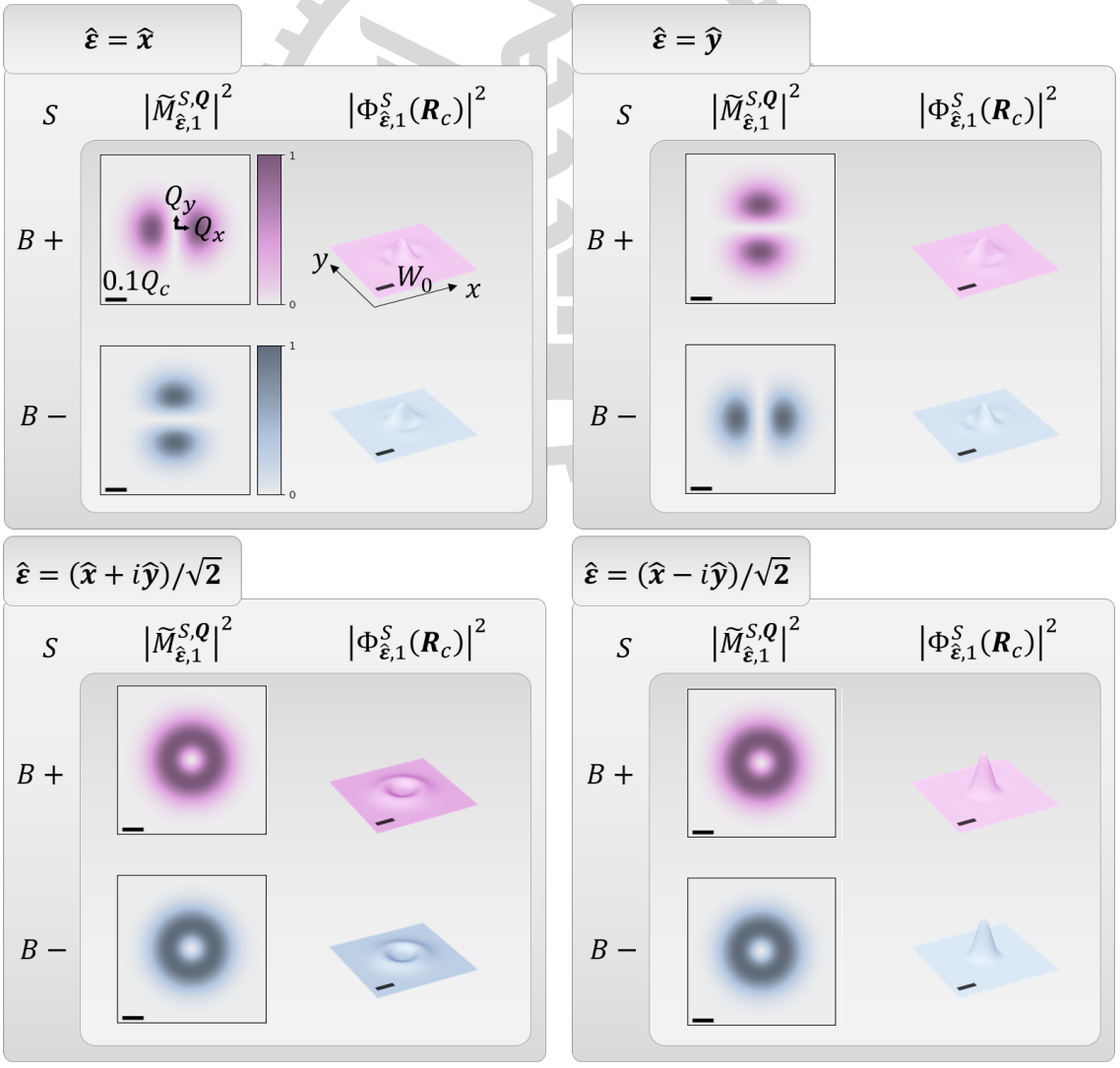


Figure 4.6 : For $\ell=1$, normalized $|\tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}|^2$ and $|\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell=1}(\boldsymbol{R}_c)|^2$ for different polarization unit vectors $\hat{\boldsymbol{\varepsilon}}$. The scale bar indicates 0.1 light cone range.

For valley-mixed excitons' cases, the light structures may not preserved in the resulting probability distribution. One may potentialy anology the central peak value between light and resulting probability distribution to topological feature :

	Light Field	**Exciton Wavepacket**
$\ell = 0:$	Central Peak	Donut Shape
$\ell = \pm 1:$	Donut Shape	Central Peak

(4.40)

This qualitative change highlights the crucial role of the phase factors in Eq. (4.6), as additional phase contribute to the results.

4.4.4 Phase in the Internal Exciton Wave Function

Throughout the above subsections, we assume $U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = 1$ to simplify the simulations. However, the phase of $U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})$ may significantly affect the spatial structure of the wave packet. Thus, in the following simulation we investigate the effect of this phase by rewriting Eq (4.22) as

$$\Phi^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c, \boldsymbol{r}) = \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} \left|U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})\right| e^{i\,\mathrm{Arg}\left[U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r})\right]} \tag{4.41}$$

Simulations

For a valley-mixed exciton, the internal wave function contains $\boldsymbol{Q}$-dependent phase factors of the form $e^{\pm i\phi_{\boldsymbol{Q}}}$, as shown in Eq (4.36). To investigate how such a phase contribution affects the wave packet, we consider the simplified case

$$\left|U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})\right| = 1, \qquad \text{and} \quad e^{i\,\mathrm{Arg}\left[U^{B\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r})\right]} = e^{i\phi_{\boldsymbol{Q}}}. \tag{4.42}$$

to investigate whether the phase contribution from the internal wave function affects the wave packet. As a result, Eq (4.41) becomes

$$\Phi'^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} e^{i\phi_{\boldsymbol{Q}}} \tag{4.43}$$

Here we denote a prime to indicate that the resulting envelope function differs from Eq (4.37).

The following simulations demonstrate that

Internal phase alters the central-peak feature of the exciton wave packet.

The central peak is determined by the interplay between the topological charge ℓ and the $\phi_{\boldsymbol{Q}}$-dependent phases of the optical matrix element and the internal exciton wave function. This highlights the importance of carefully accounting for the phase of the internal exciton wave function, as it can alter key features of the exciton wave packet, particularly the topological charge ℓ at which the central peak occurs.

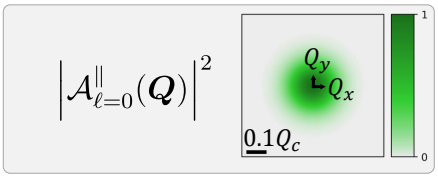


Figure 4.1 : (Repeated) For $\ell = 0$, normalized $|\mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})|^2$. The scale bar indicates 0.1 light cone range.

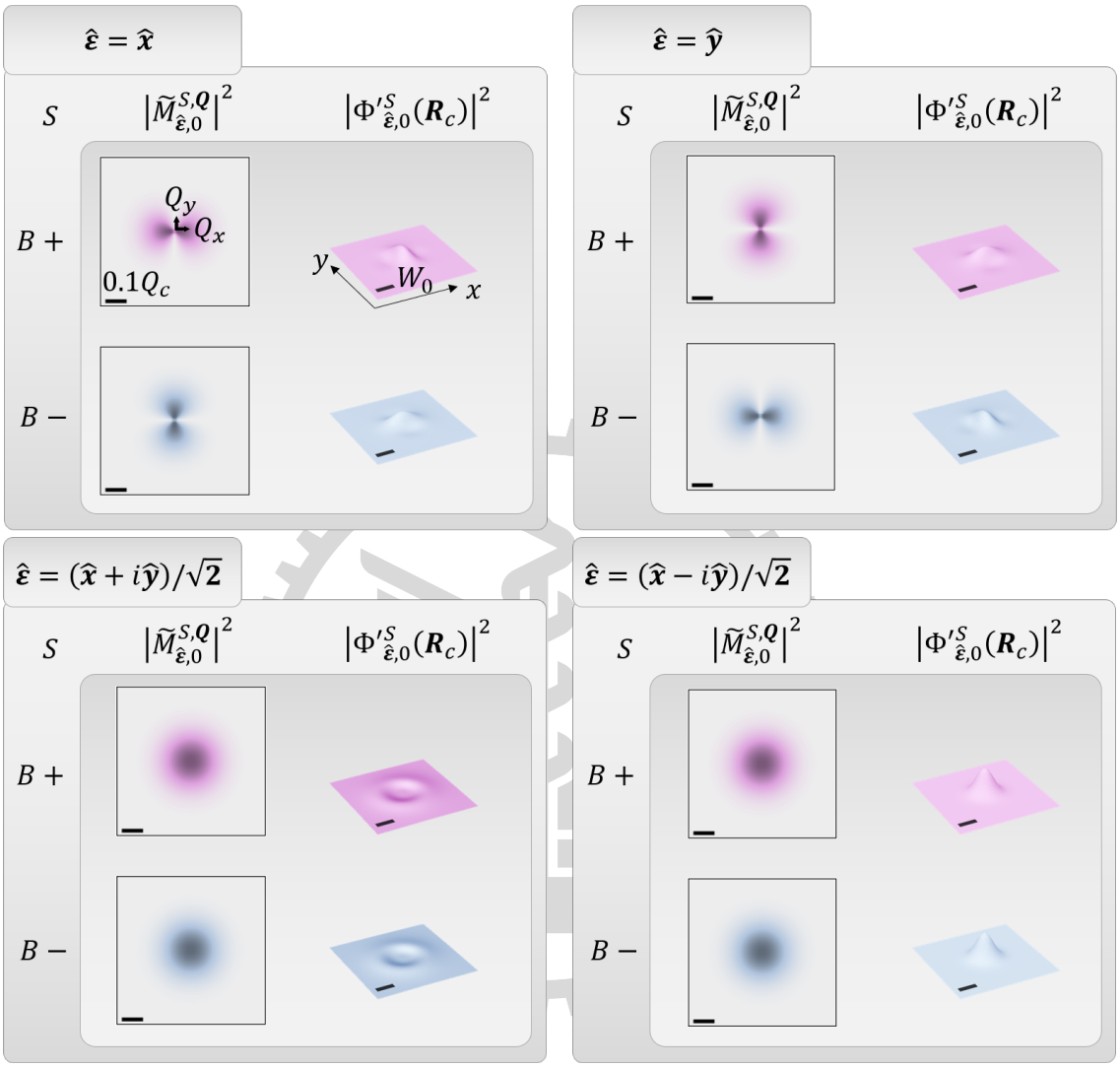


Figure 4.7 : For $\ell = 0$, normalized $|\tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}|^2$ and $|\Phi'^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c)|^2$ for different polarization unit vectors $\hat{\boldsymbol{\varepsilon}}$. The scale bar indicates 0.1 light cone range.

We can again utilize expect whether the central peak feature appear for what value ℓ is with

$$\begin{aligned}\Phi'^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c = \boldsymbol{0}) &= \frac{1}{\hbar\omega^{\boldsymbol{0}}\sqrt{\Omega}}\sum_{\boldsymbol{Q}} \tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} e^{i\phi_{\boldsymbol{Q}}} \\ &\propto \sum_{\phi_{\boldsymbol{Q}}} e^{i(\ell+2)\phi_{\boldsymbol{Q}}}\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{K,\boldsymbol{Q}*}\right) \pm e^{i(\ell-0)\phi_{\boldsymbol{Q}}}\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{K',\boldsymbol{Q}*}\right) \\ &\propto \delta_{\ell,-2} \pm \delta_{\ell,0}\end{aligned} \tag{4.44}$$

This indicates that only LG beams carrying $\ell = -2$ or 0 may yield a central peak, which conforms to Figure 4.8, where every case has a central depression since $\ell = 1$.

On the other hand, for the $\ell = 0$ cases in Figure 4.8, three of the cases have a central peak.

The remaining case, corresponding to right-circularly polarized light, can be further analyzed using with

$$\begin{aligned}
&\Phi'^{B\pm}_{\hat{\boldsymbol{\varepsilon}}=(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}})/\sqrt{2},\ell}(\boldsymbol{R}_c=\boldsymbol{0}) \\
&\propto \sum_{\phi_{\boldsymbol{Q}}} e^{i(\ell+2)\phi_{\boldsymbol{Q}}}\left[(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}})/\sqrt{2}\cdot \boldsymbol{D}^{K,\boldsymbol{Q}*}\right] \pm e^{i(\ell-0)\phi_{\boldsymbol{Q}}}\left[(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}})/\sqrt{2}\cdot \boldsymbol{D}^{K',\boldsymbol{Q}*}\right] \qquad (4.45)\\
&\propto \delta_{\ell,-2},
\end{aligned}$$

where only K valley exciton is excited and carries the $e^{i(\ell+2)\phi_{\boldsymbol{Q}}}$ phase into the envelope function. Thus, the wave packet will have a central peak only at $\ell=-2$ when excited by right-circularly polarized light.

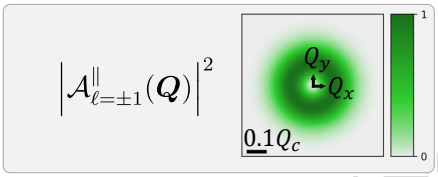


Figure 4.3 : (Repeated) For $\ell=\pm1$, normalized $|\mathcal{A}^{\parallel}_{\ell}(\boldsymbol{Q})|^2$. The scale bar indicates 0.1 light cone range.

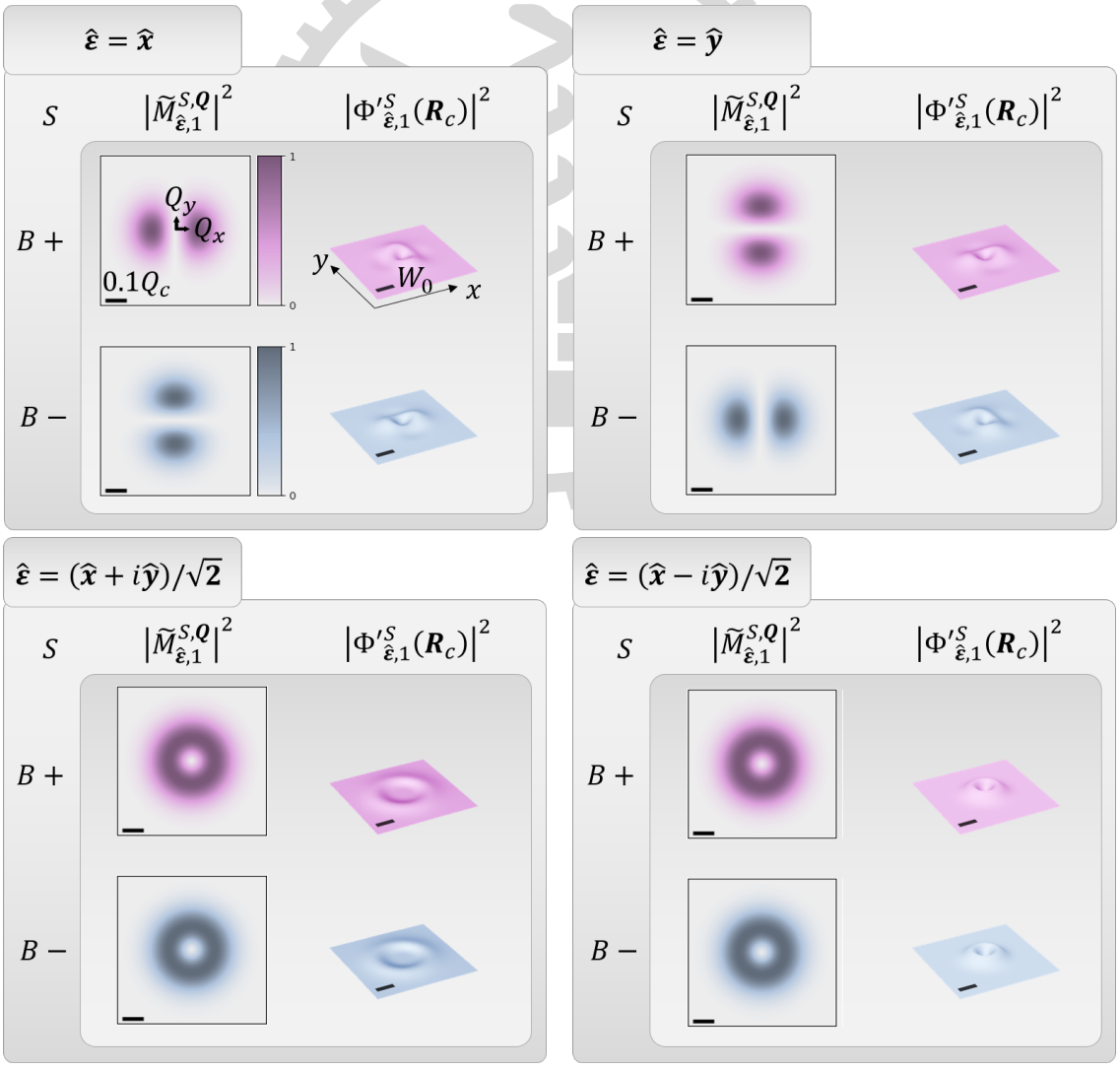


Figure 4.8 : For $\ell=1$, normalized $|\tilde{M}^{B\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}|^2$ and $|\Phi'^{B\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c)|^2$ for different polarization unit vectors $\hat{\boldsymbol{\varepsilon}}$. The scale bar indicates 0.1 light cone range.

Chapter 5 Summary and outlook

In this thesis, we present a theoretical investigation of excitonic transitions and wave-packet formation in two-dimensional transition-metal dichalcogenides under excitation by structured light, including scalar and vector vortex beams.

First, we investigate the angular-momentum structure of Laguerre–Gaussian (LG) beams and vector vortex beams (VVBs) in the angular spectrum representation. By transforming the vector potential into the Coulomb gauge, the spin–orbit interaction (SOI) naturally emerges in the longitudinal component of LG beams. For VVBs formed by the superposition of two LG beams, the resulting longitudinal angular geometry is governed by the total angular momentum difference $\Delta J = (\sigma' + \ell') - (\sigma + \ell)$, while the transverse polarization geometry is determined by the OAM difference. We demonstrate that these angular geometries are preserved beyond the equal-OAM condition, provided that sufficient spatial overlap exists between the constituent beams between the constituent beams.

Next, we investigate the interaction between VVBs and valley-mixed excitons using Fermi's golden rule. The exciton transition-rate distribution in center-of-mass momentum space is determined by the compatibility between the angular-momentum differences of the optical field and the winding number of the excitonic transition dipole moments. In particular, gray excitons can be interpreted as zero-winding excitonic states because their transition dipole moments are approximately isotropic, naturally coupling to longitudinal optical structures with zero TAM difference. Matching the optical and excitonic angular structures enables full-azimuthal momentum-selective excitation, whereas a mismatching leads to directional excitation.

Finally, we construct photoexcited exciton wave packets by considering the internal exciton structure through first-order time-dependent perturbation theory. A zeroth-order approximation is introduced to simplify the internal structure while avoiding the need to explicitly evaluate the full microscopic wave function. While the internal phase contribution can be neglected for valley excitons under appropriate conditions, it becomes a relative phase between valley components for valley-mixed excitons and plays a crucial role in determining their wave-packet structures. Further numerical treatment is therefore required to obtain a complete description of valley-mixed exciton wave packets.

This work establishes a connection between structured-light angular geometry and exciton momentum-space engineering, providing a framework for controlling and probing excitonic states using tailored optical fields.

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Appendix

A.1 Wave Equations from Maxwell's Equations in Free Space

In classical electrodynamics, Maxwell's equations in vacuum are given by

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}, t) = \frac{\rho}{\epsilon_0} = 0, \tag{A.1.1}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}, t) = 0, \tag{A.1.2}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}, t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r}, t)}{\partial t}, \tag{A.1.3}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \boldsymbol{E}(\boldsymbol{r}, t)}{\partial t} = \frac{1}{c^2}\frac{\partial \boldsymbol{E}(\boldsymbol{r}, t)}{\partial t}. \tag{A.1.4}$$

Here, $\boldsymbol{\nabla} \equiv \hat{\boldsymbol{x}}\frac{\partial}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial}{\partial z}$ denotes the spatial gradient operator in Cartesian coordinates. To reformulate the equations in terms of electromagnetic potentials, we first note that Eq. (A.1.2) implies that the magnetic field can be expressed as the curl of a vector potential $\boldsymbol{A}(\boldsymbol{r}, t)$,

$$\boldsymbol{B}(\boldsymbol{r}, t) = \boldsymbol{\nabla} \times \boldsymbol{A}(\boldsymbol{r}, t). \tag{A.1.5}$$

Substituting Eq. (A.1.5) into Faraday's law (Eq. (A.1.3)) gives a curl-free expression, which allows the introduction of a scalar potential $\phi(\boldsymbol{r}, t)$ such that

$$\boldsymbol{E}(\boldsymbol{r}, t) = -\boldsymbol{\nabla}\phi(\boldsymbol{r}, t) - \frac{\partial \boldsymbol{A}(\boldsymbol{r}, t)}{\partial t}. \tag{A.1.6}$$

This representation automatically satisfies Eq. (A.1.2) and Eq. (A.1.3). Finally, Gauss's law (Eq. (A.1.1)) and Ampère–Maxwell law (Eq. (A.1.4)) can be rewritten as coupled wave equations for the scalar and vector potentials:

$$\nabla^2\phi(\boldsymbol{r}, t) + \frac{\partial}{\partial t}\big(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t)\big) = 0, \tag{A.1.7}$$

$$\nabla^2 \boldsymbol{A}(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}(\boldsymbol{r}, t)}{\partial t^2} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t) + \frac{1}{c^2}\frac{\partial \phi(\boldsymbol{r}, t)}{\partial t}\right) = 0. \tag{A.1.8}$$

A.2 Scalar Helmholtz Equation

We begin in the Lorenz gauge and assume a monochromatic-wave ansatz of the form $\boldsymbol{A}^L(\boldsymbol{r}, t) = \mathrm{Re}\big[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\big]$. Substituting this expression into Eq. (2.8) yields the vector Helmholtz equation

$$\nabla^2 \boldsymbol{A}^L(\boldsymbol{r}) + q_0^2 \boldsymbol{A}^L(\boldsymbol{r}) = 0. \tag{A.2.1}$$

We then impose a transverse-field ansatz, $\boldsymbol{A}^L(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}\, A^L(\boldsymbol{r})$, where $\hat{\boldsymbol{\varepsilon}}$ is the polarization unit vector. Under this assumption, Eq. (A.2.1) reduces to a scalar equation for the amplitude $A^L(\boldsymbol{r})$,

$$\nabla^2 A^L(\boldsymbol{r}) + q_0^2 A^L(\boldsymbol{r}) = 0, \tag{A.2.2}$$

which is the scalar Helmholtz equation.

A.3 Valley Excitons with Zero Center-of-Mass Momentum

In this appendix, we derive the relation between valley exciton wave functions with zero center-of-mass momentum at the high-symmetry valleys K and K'. Using the Bloch theorem, the corresponding exciton wave functions can be written as

$$\begin{aligned} U^{K,\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) &\approx \psi_{c,\boldsymbol{K}}(\boldsymbol{R}_c)\psi^*_{v,\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r}) &&= u_{c,\boldsymbol{K}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r})e^{i\boldsymbol{K}\cdot\boldsymbol{r}}, \\ U^{K',\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) &\approx \psi_{c,-\boldsymbol{K}}(\boldsymbol{R}_c)\psi^*_{v,-\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r}) &&= u_{c,-\boldsymbol{K}}(\boldsymbol{R}_c)u^*_{v,-\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r})e^{-i\boldsymbol{K}\cdot\boldsymbol{r}}, \end{aligned} \tag{A.3.1}$$

Here, the vectors $\boldsymbol{K}$ and $\boldsymbol{K}'$ denote the momentum-space position vectors of the K and K' valleys in reciprocal space, respectively. The single-particle Bloch wave functions satisfy[1]

$$\psi^*_{n,-\boldsymbol{K}}(\boldsymbol{r}') = \psi_{n,\boldsymbol{K}}(\boldsymbol{r}'). \tag{A.3.2}$$

Using the Bloch form of the single-particle wave function,

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}')e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{A.3.3}$$

Eq. (A.3.2) gives

$$u^*_{n,-\boldsymbol{K}}(\boldsymbol{r}') = u_{n,\boldsymbol{K}}(\boldsymbol{r}'). \tag{A.3.4}$$

Substituting this relation into Eq. (A.3.1), we obtain

$$U^{K,\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) = U^{K',\mathbf{0}*}(\boldsymbol{R}_c, \boldsymbol{r}) \tag{A.3.5}$$

which establishes the complex-conjugation relation between the valley exciton wave functions at $\boldsymbol{K}$ and $\boldsymbol{K}'$.

[1] The vector $\boldsymbol{r}'$ denotes a general position vector and may represent the weighted-average, relative, electron, or hole coordinate.

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